

BOOK 4

FORCES, DENSITY & PRESSURE

4 Forces, density and pressure

4.1 Turning effects of forces

Candidates should be able to:

- 1 understand that the weight of an object may be taken as acting at a single point known as its centre of gravity
- 2 define and apply the moment of a force
- 3 understand that a couple is a pair of forces that acts to produce rotation only
- 4 define and apply the torque of a couple

4.2 Equilibrium of forces

Candidates should be able to:

- 1 state and apply the principle of moments
- 2 understand that, when there is no resultant force and no resultant torque, a system is in equilibrium
- 3 use a vector triangle to represent coplanar forces in equilibrium

4.3 Density and pressure

Candidates should be able to:

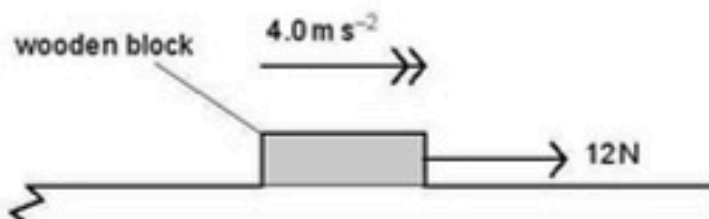
- 1 define and use density
- 2 define and use pressure
- 3 derive, from the definitions of pressure and density, the equation for hydrostatic pressure $\Delta p = \rho g \Delta h$
- 4 use the equation $\Delta p = \rho g \Delta h$
- 5 understand that the upthrust acting on an object in a fluid is due to a difference in hydrostatic pressure
- 6 calculate the upthrust acting on an object in a fluid using the equation $F = \rho g V$ (Archimedes' principle)

1 9702/1/M/J/02/Q10

A wooden block of mass 0.60 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at 4.0 m s^{-2} .

What is the magnitude of the frictional force acting on the block?

- A 2.4 N B 9.6 N
C 14 N D 16 N



2 9702/1/M/J/02/Q12

A submarine descends vertically at constant velocity. The three forces acting on the submarine are viscous drag, upthrust and weight.

Which relationship between their magnitudes is correct?

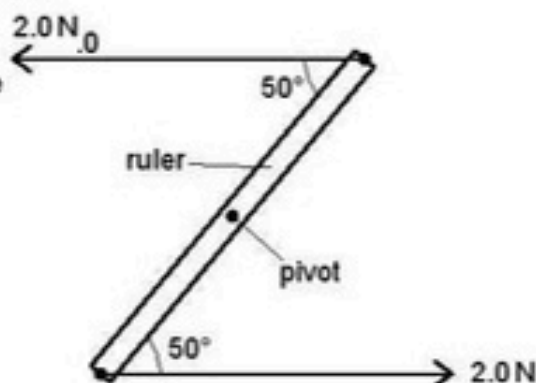
- A weight < drag B weight = drag C weight < upthrust D weight > upthrust

3 9702/11/O/N/11/Q14

A ruler of length 0.30 m is pivoted at its centre. Equal and opposite forces of magnitude 2.0 N are applied to the ends of the ruler, creating a couple as shown.

What is the magnitude of the torque of the couple on the ruler when it is in the position shown?

- A 0.23 Nm B 0.39 Nm
C 0.46 Nm D 0.60 Nm



4 9702/1/M/J/02/Q14

A cylindrical block of wood has a cross-sectional area A and weight W . It is totally immersed in water with its axis vertical. The block experiences pressures p_t and p_b at its top and bottom surfaces respectively.

Which of the following expressions is equal to the upthrust on the block?

- A $(p_b - p_t)A + W$ B $(p_b - p_t)$ C $(p_b - p_t)A$ D $(p_b - p_t)A - W$

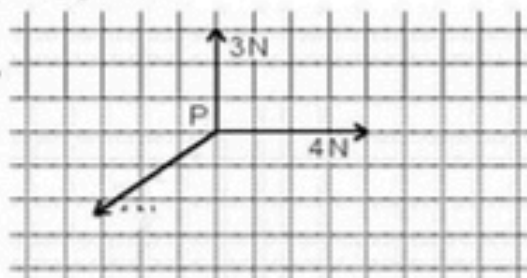
5 9702/1/M/J/02/Q15

The vector diagram shows three coplanar forces acting on an object at P.

The magnitude of the resultant of these three forces is 1 N .

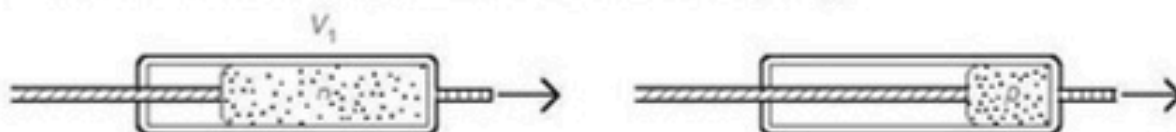
What is the direction of this resultant?

- A $\downarrow$ B $\searrow$
C $\swarrow$ D $\nearrow$



6 9702/1/M/J/02/Q17

Air in a bicycle pump is forced through a valve at a constant pressure p . In one stroke of the pump the volume of air in the pump chamber is reduced from V_1 to V_2 .



What is the work done on this air in one stroke of the pump?

- A $\frac{p(V_1 + V_2)}{2}$ B $p(V_1 + V_2)$ C $p(V_1 - V_2)$ D pV_1

7 9702/1/M/J/02/Q20

For a given liquid at atmospheric pressure, which process can occur at any temperature?

- A boiling B evaporation C melting D solidification

8 9702/1/M/J/02/Q21

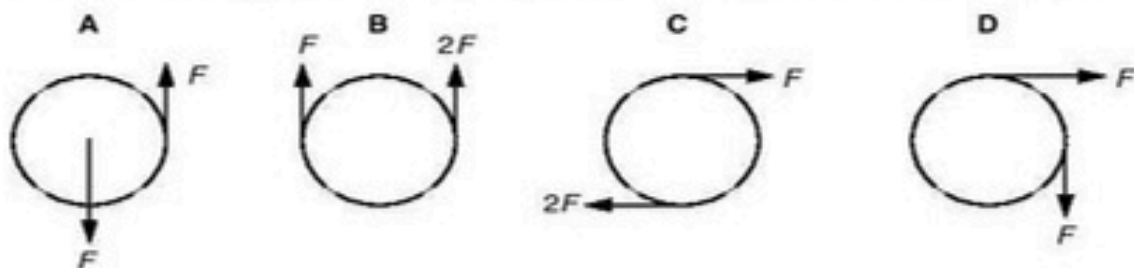
A mass of a liquid of density ρ is thoroughly mixed with an equal mass of another liquid of density 2ρ . No change of the total volume occurs.

What is the density of the liquid mixture?

- A $\frac{4}{3}\rho$ B $\frac{3}{2}\rho$ C $\frac{5}{3}\rho$ D 3ρ

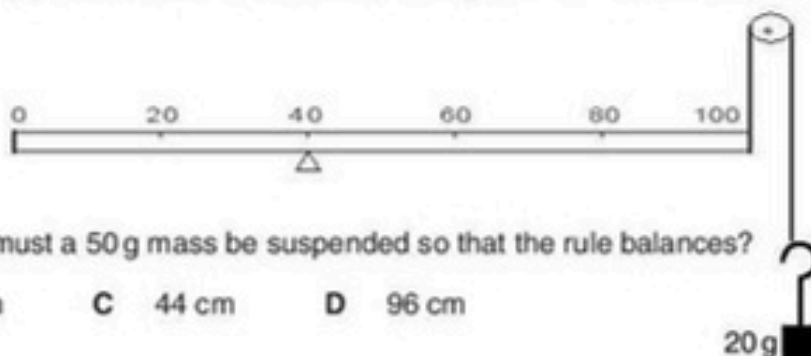
9 9702/1/O/N/02/Q13

Which of the following pairs of forces, acting on a circular object, constitutes a couple?



10 9702/1/O/N/02/Q14

A uniform metre rule of mass 100 g is supported by a knife-edge at the 40 cm mark and a string at the 100 cm mark. The string passes round a frictionless pulley and carries a mass of 20 g as shown in the diagram.



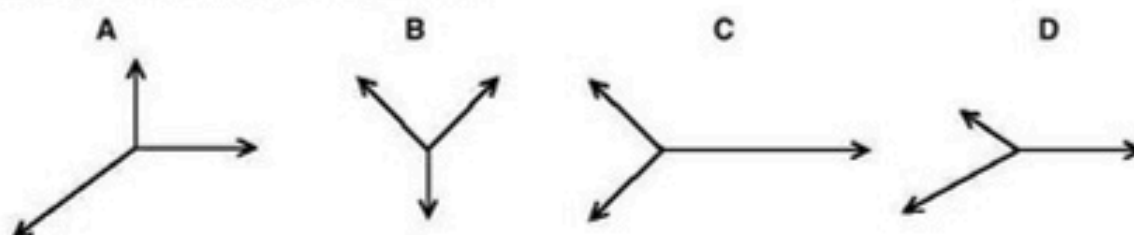
At which mark on the rule must a 50 g mass be suspended so that the rule balances?

- A 4 cm B 36 cm C 44 cm D 96 cm

11 9702/1/O/N/02/Q15

The diagrams represent systems of coplanar forces acting at a point. The lengths of the force vectors represent the magnitudes of the forces.

Which system of forces is in equilibrium?



12 9702/1/O/N/02/Q17

A car driver adjusts the pressure on a car's brakes so that the car travels at constant speed down a hill from P to Q.



The magnitude of the change in the car's kinetic energy is ΔE_k . The magnitude of the change in its gravitational potential energy is ΔE_p .

Which statement is correct?

- A $\Delta E_k > \Delta E_p$ B $\Delta E_k = \Delta E_p$ C $\Delta E_p > \Delta E_k > 0$ D $\Delta E_k = 0$

13 9702/1/O/N/02/Q20

Pollen grains are suspended in a liquid and are illuminated strongly. When observed under a microscope they are seen to be in continuous random motion.

What is the reason for this?

- A convection currents in the liquid
B evaporation of the liquid
C molecules of the liquid colliding with the pollen grains
D pollen grains colliding with each other

14 9702/1/O/N/02/Q21

At a depth of 20 cm in a liquid of density 1800 kg m^{-3} , the pressure due to the liquid is p . Another liquid has a density of 1200 kg m^{-3} .

What is the pressure due to this liquid at a depth of 60 cm?

- A $\frac{p}{2}$ B $\frac{3p}{2}$ C $2p$ D $3p$

15 9702/01/M/J/03/Q10

Two blocks X and Y, of masses m and $3m$ respectively, are accelerated along a smooth horizontal surface by a force F applied to block X as shown.

What is the magnitude of the force exerted by block X on block Y during this acceleration?



- A $\frac{F}{4}$ B $\frac{F}{3}$ C $\frac{F}{2}$ D $\frac{3F}{4}$

16 9702/1/O/N/02/Q22

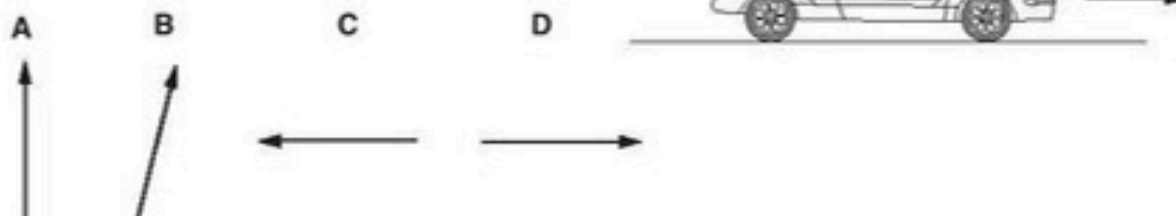
Which line in the table gives approximate ratios of density and molecular spacing for a substance in its solid, liquid and gas phases?

	density			molecular spacing		
	solid	liquid	gas	solid	liquid	gas
A	1000	1000	1	1	1	10
B	1000	100	1	1	10	1000
C	1000	1000	1	1	1	1000
D	1000	100	1	1	10	100

17 9702/01/M/J/03/Q11

A car with front-wheel drive accelerates in the direction shown.

Which diagram best shows the direction of the total force exerted by the road on the front wheels?

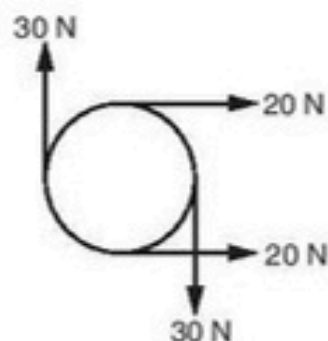


18 9702/01/M/J/03/Q13

The diagram shows four forces applied to a circular object.

Which of the following describes the resultant force and resultant torque on the object?

	resultant force	resultant torque
A	zero	zero
B	zero	non-zero
C	non-zero	zero
D	non-zero	non-zero

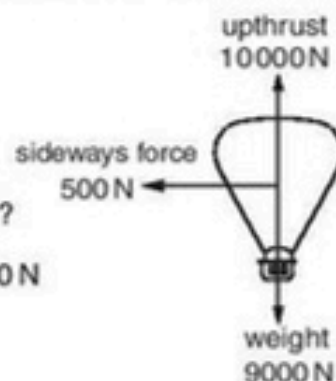


19 9702/01/M/J/03/Q14

A balloon is acted upon by three forces, weight, upthrust and sideways force due to the wind, as shown in the diagram.

What is the vertical component of the resultant force on the balloon?

- A 500 N B 1000 N C 10 000 N D 10 500 N

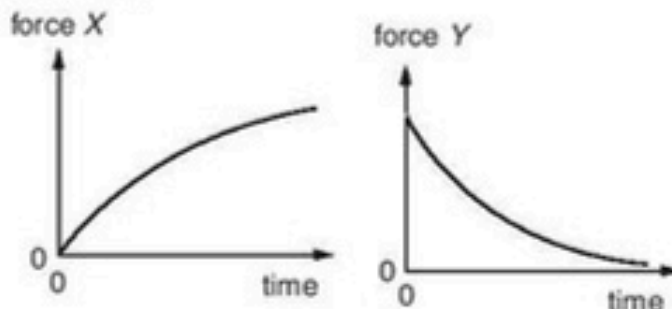


20 9702/01/M/J/03/Q15

A ball falls from rest through air and eventually reaches a constant velocity.
For this fall, forces X and Y vary with time as shown.

What are forces X and Y ?

	force X	force Y
A	air resistance	resultant force
B	air resistance	weight
C	upthrust	resultant force
D	upthrust	weight



21 9702/01/M/J/03/Q18

A motorist travelling at 10 m s^{-1} can bring his car to rest in a distance of 10 m.

If he had been travelling at 30 m s^{-1} , in what distance could he bring the car to rest using the same braking force?

- A 17 m B 30 m C 52 m D 90 m

22 9702/01/M/J/03/Q20

A child drinks a liquid of density ρ through a vertical straw.

Atmospheric pressure is p_0 and the child is capable of lowering the pressure at the top of the straw by 10%. The acceleration of free fall is g .

What is the maximum length of straw that would enable the child to drink the liquid?

- A $\frac{p_0}{10\rho g}$ B $\frac{9p_0}{10\rho g}$ C $\frac{p_0}{\rho g}$ D $\frac{10p_0}{\rho g}$

23 9702/01/O/N/03/Q10

A mass accelerates uniformly when the resultant force acting on it

- A is zero. C increases uniformly with respect to time.
B is constant but not zero. D is proportional to the displacement from a fixed point.

24 9702/01/O/N/03/Q15

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a .

Which quantities **must** be in the same direction?

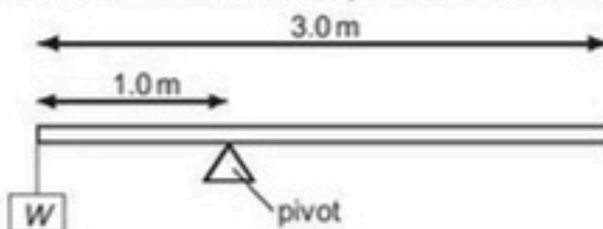
- A a and v only B a and F only C v and F only D v , F and a

25 9702/01/M/J/04/Q13

A uniform beam of weight 50 N is 3.0 m long and is supported on a pivot situated 1.0 m from one end. When a load of weight W is hung from that end, the beam is in equilibrium, as shown in the diagram.

What is the value of W ?

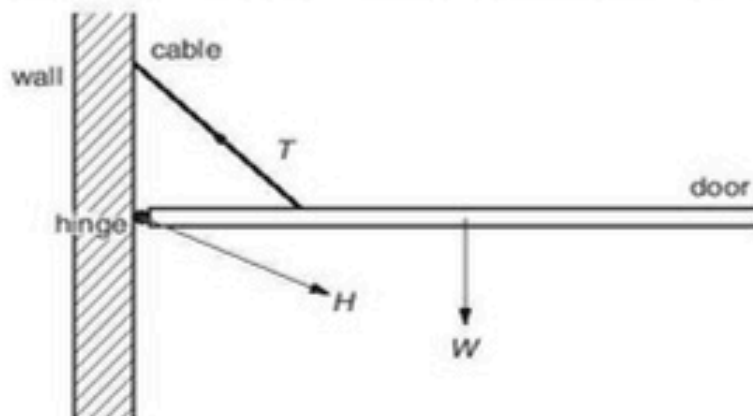
- A 25 N B 50 N
C 75 N D 100 N



26 9702/01/O/N/03/Q16

A hinged door is held closed in the horizontal position by a cable.

Three forces act on the door: the weight W of the door, the tension T in the cable, and the force H at the hinge.



Which list gives the three forces in increasing order of magnitude?

- A H, T, W B T, H, W
C W, H, T D W, T, H

27 9702/11/M/J/17/Q18

A mass is raised vertically. In time t , the increase in its gravitational potential energy is E_p and the increase in its kinetic energy is E_k .

What is the average power input to the mass?

- A $(E_p - E_k)t$ B $(E_p + E_k)t$ C $\frac{E_p - E_k}{t}$ D $\frac{E_p + E_k}{t}$

28 9702/01/M/J/04/Q12

An object, immersed in a liquid in a tank, experiences an upthrust.

What is the physical reason for this upthrust?

- A The density of the body differs from that of the liquid.
B The density of the liquid increases with depth.
C The pressure in the liquid increases with depth.
D The value of g in the liquid increases with depth.

29 9702/01/M/J/04/Q20

Particles of dust, suspended in water, are viewed through a microscope. The particles can be seen to move irregularly.

This movement is due to

- A convection currents in the water.
B evaporation of the water near the dust particles.
C gravitational forces acting on the particles of dust.
D water molecules hitting the dust particles in a random way.

30 9702/01/M/J/04/Q21

Two solid substances P and Q have atoms of mass M_P and M_Q respectively. They have N_P and N_Q atoms per unit volume.

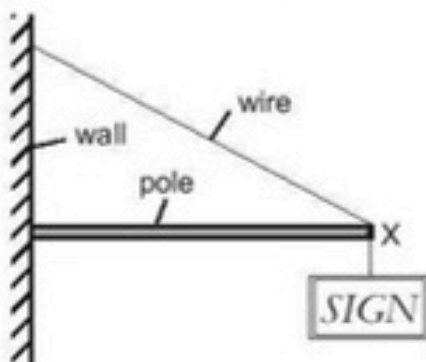
It is found by experiment that the density of P is greater than that of Q.

Which of the following deductions from this experiment must be correct?

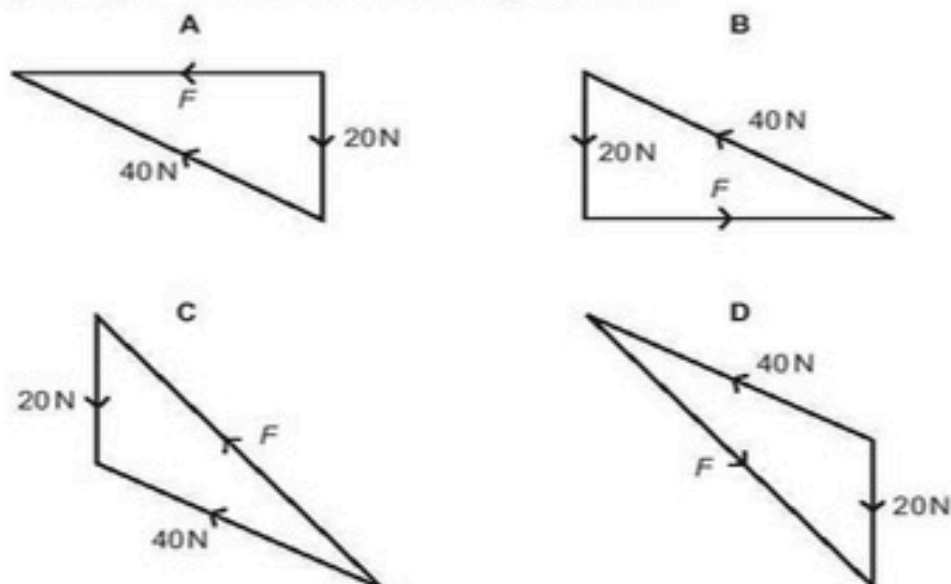
- A $M_P > M_Q$ B $N_P > N_Q$ C $M_P N_P > M_Q N_Q$ D $\frac{M_P}{N_P} > \frac{M_Q}{N_Q}$

31 9702/01/M/J/04/Q14

The diagram shows a sign of weight 20 N suspended from a pole, attached to a wall. The pole is kept in equilibrium by a wire attached at point X of the pole.



The force exerted by the pole at point X is F , and the tension in the wire is 40 N. Which diagram represents the three forces acting at point X?



32 9702/01/O/N/04/Q17

A cyclist is capable of generating an average power of 3.0 kW during a 4.0 km speed trial. His aerodynamic suit and position on the cycle reduce resistive forces to 180 N.

What is the approximate time achieved in the speed trial?

- A 140 s B 240 s C 1300 s D 2200 s

33 9702/01/O/N/04/Q19

Which statement applies to the boiling but **not** to the evaporation of a liquid?

- A All the bonds between molecules in the liquid are broken.
- B At normal atmospheric pressure, the process occurs at one temperature only.
- C Energy must be provided for the process to happen.
- D The separation of the molecules increases greatly.

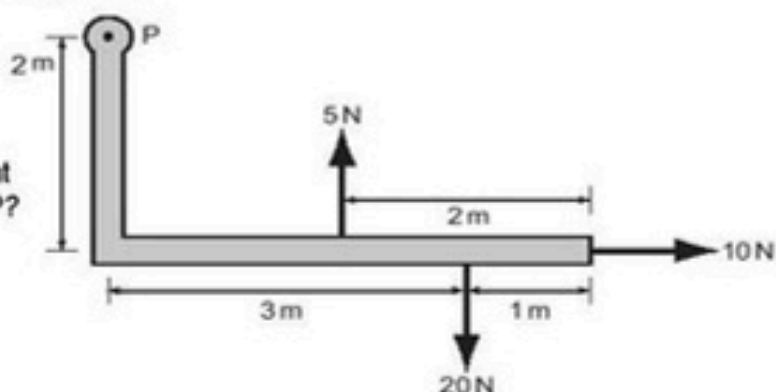
34 9702/01/M/J/05/Q13

An L-shaped rigid lever arm is pivoted at point P.

Three forces act on the lever arm, as shown in the diagram.

What is the magnitude of the resultant moment of these forces about point P?

- A 30 Nm B 35 Nm
C 50 Nm D 90 Nm



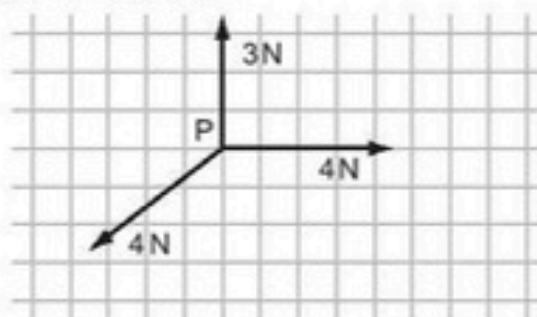
35 9702/01/M/J/05/Q14

The vector diagram shows three coplanar forces acting on an object at P.

The magnitude of the resultant of these three forces is 1 N.

What is the direction of this resultant?

- A ↓ B ↘ C ↙ D ↗



36 9702/01/M/J/05/Q18

The hydrostatic pressure p at a depth h in a liquid of density ρ is given by the formula $p = h\rho g$.

Which equation, or principle of physics, is used in the derivation of this formula?

- A density = mass ÷ volume C atmospheric pressure decreases with height
B potential energy = mgh D density increases with depth

37 9702/01/M/J/05/Q19

Pollen grains are suspended in a liquid and are illuminated strongly. When observed under a microscope they are seen to be in continuous random motion.

What is the reason for this?

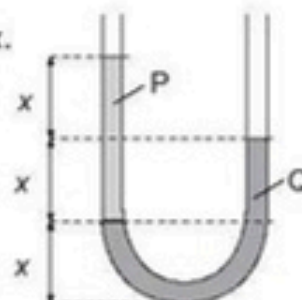
- A convection currents in the liquid
B evaporation of the liquid
C molecules of the liquid colliding with the pollen grains
D pollen grains colliding with each other

38 9702/13/O/N/14/Q22

The diagram shows two liquids, labelled P and Q, which do not mix. The liquids are in equilibrium in an open U-tube.

What is the ratio $\frac{\text{density of P}}{\text{density of Q}}$?

- A $\frac{1}{2}$ B $\frac{2}{3}$ C $\frac{3}{2}$ D 2

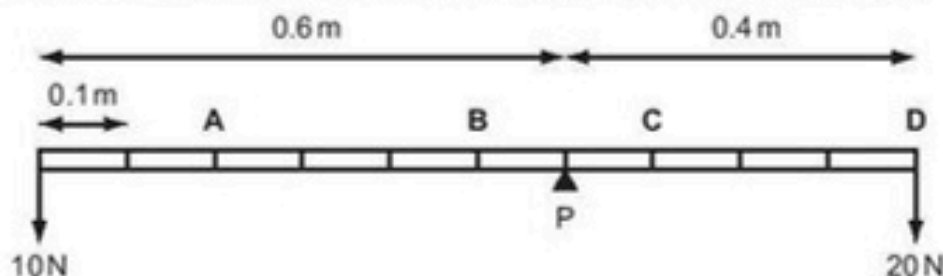


39 9702/01/O/N/05/Q12

A uniform beam of weight 100 N is pivoted at P as shown. Weights of 10 N and 20 N are attached to its ends.

The length of the beam is marked off at 0.1 m intervals.

At which point should a further weight of 20 N be attached to achieve equilibrium?



40 9702/01/O/N/05/Q17

Why does the pressure increase when a sealed container of gas is heated?

- A The gas molecules collide more often with each other.
- B The gas molecules expand when they are heated.
- C The gas molecules travel faster and hit the walls of the container more often.
- D There are more gas molecules present to collide with the walls of the container.

41 9702/01/O/N/05/Q18

Liquids X and Y are stored in large open tanks. Liquids X and Y have densities of 800 kg m^{-3} and 1200 kg m^{-3} respectively.

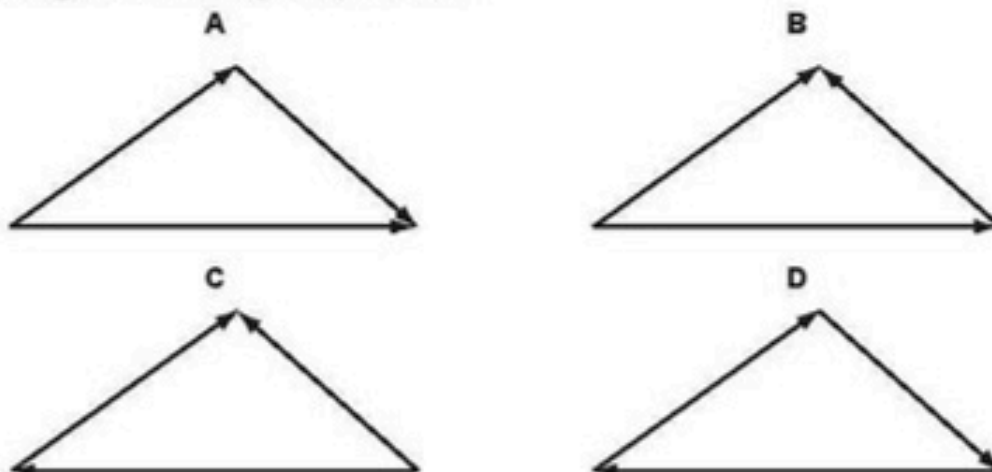
At what depths are the pressures equal?

	depth in liquid X	depth in liquid Y
A	8 m	12 m
B	10 m	10 m
C	15 m	10 m
D	18 m	8 m

42 9702/01/M/J/06/Q13

The diagrams show three forces acting on a body.

In which diagram is the body in equilibrium?

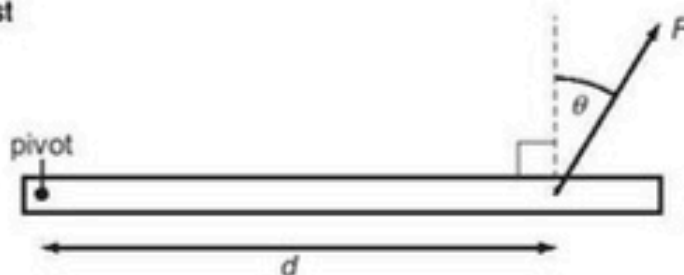


43 9702/01/M/J/06/Q14

A force F is applied to a beam at a distance d from a pivot. The force acts at an angle θ to a line perpendicular to the beam.

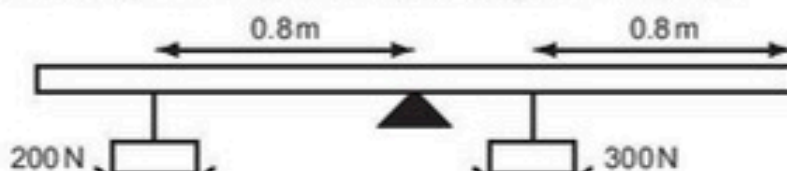
Which combination will cause the largest turning effect about the pivot?

	F	d	θ
A	large	large	large
B	large	large	small
C	small	small	large
D	small	large	small



44 9702/01/M/J/06/Q15

A rigid uniform bar of length 2.4 m is pivoted horizontally at its mid-point.



Weights are hung from two points of the bar as shown in the diagram. To maintain horizontal equilibrium, a couple is applied to the bar.

What is the torque and direction of this couple?

- A 40 N m clockwise B 40 N m anticlockwise C 80 N m clockwise D 80 N m anticlockwise

45 9702/01/M/J/06/Q19

Below are four short paragraphs describing the molecules in a beaker of water at 50°C .

Which paragraph correctly describes the molecules?

- A The molecules all travel at the same speed. This speed is not large enough for any of the molecules to leave the surface of the water. There are attractive forces between the molecules.
- B The molecules have a range of speeds. Some molecules travel sufficiently fast to leave the surface of the water. There are no forces between the molecules.
- C The molecules have a range of speeds. Some molecules travel sufficiently fast to leave the surface of the water. There are attractive forces between the molecules.
- D The molecules have a range of speeds. The fastest molecules are unable to leave the surface of the water. There are attractive forces between the molecules.

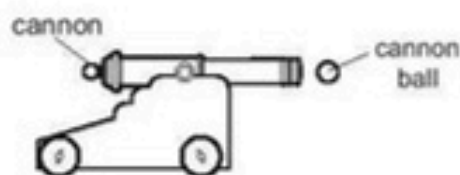
46 9702/01/O/N/06/Q12

The diagram shows a cannon ball fired from a cannon.

The mass of the cannon is 1000 kg and the mass of the cannon ball is 10 kg.

The recoil velocity of the cannon is 5 ms^{-1} horizontally.

What is the horizontal velocity of the cannon ball?



- A 200 ms^{-1} B 500 ms^{-1} C 2000 ms^{-1} D 5000 ms^{-1}

47 9702/01/O/N/06/Q13

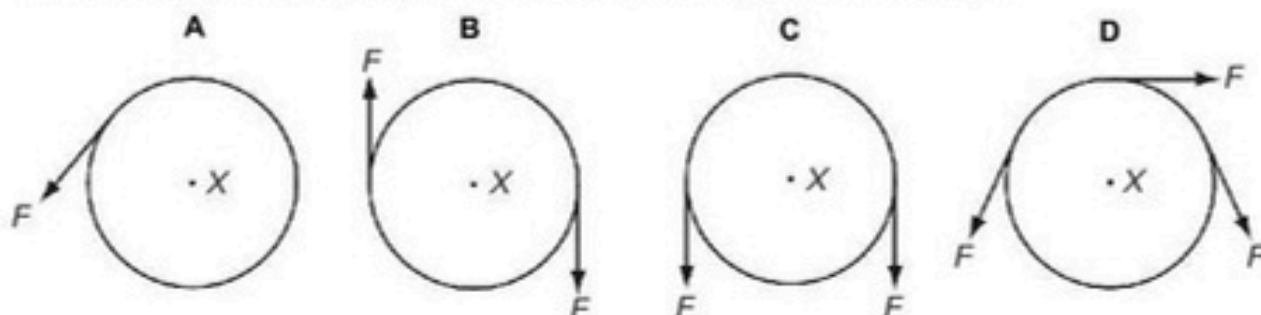
Which force is caused by a pressure difference?

- A friction B upthrust C viscous force D weight

48 9702/01/O/N/06/Q14

A rigid circular disc of radius r has its centre at X . A number of forces of equal magnitude F act at the edge of the disc. All the forces are in the plane of the disc.

Which arrangement of forces provides a moment of magnitude $2Fr$ about X ?

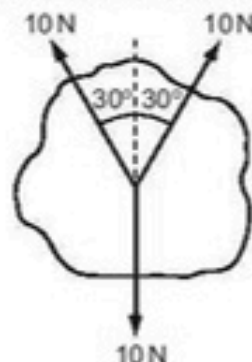


49 9702/01/O/N/06/Q15

Three coplanar forces, each of magnitude 10 N, act through the same point of a body in the directions shown.

What is the magnitude of the resultant force?

- A 0 N B 1.3 N
C 7.3 N D 10 N



50 9702/01/O/N/06/Q19

Which statement defines the density of a substance?

- A the force per unit area acting on the substance
B the increase in length per unit length of the substance
C the mass per unit volume of the substance
D the work done per unit time by the substance

51 9702/01/O/N/06/Q20

The table summarises some properties of evaporation.

Which row of the table is correct?

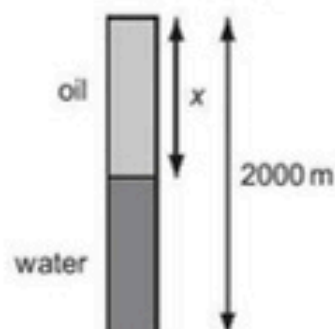
	involves a change in state from liquid to vapour	occurs at a fixed temperature	involves a reduction in the average kinetic energy of the remaining atoms
A	true	true	true
B	true	false	true
C	true	false	false
D	false	true	false

52 9702/01/O/N/06/Q21

A bore hole of depth 2000 m contains both oil and water as shown. The pressure at the bottom is 17.5 MPa. The density of the oil is 830 kg m^{-3} and the density of the water is 1000 kg m^{-3} .

What is the depth x of the oil?

- A 907 m B 1000 m
C 1090 m D 1270 m



53 9702/01/O/N/06/Q16

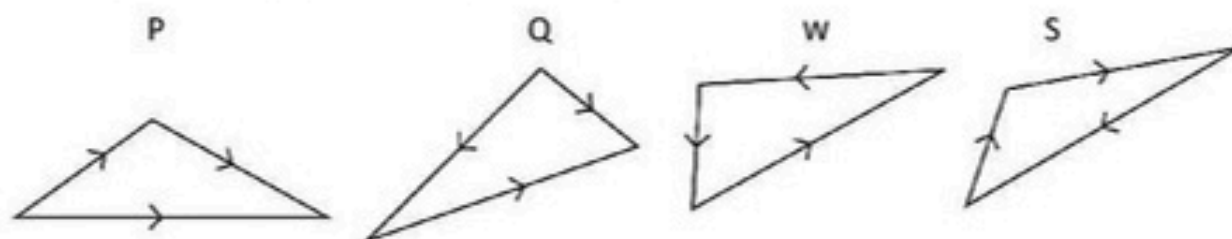
To get to his office from the entrance of the building, a man has to walk up six flights of stairs. The height of each flight is 2.5 m and the man has a mass of 80 kg.

What is the approximate gain in the man's gravitational potential energy during the climb?

- A 1200 J B 2000 J C 4800 J D 12 000 J

54 9702/01/O/N/07/Q13

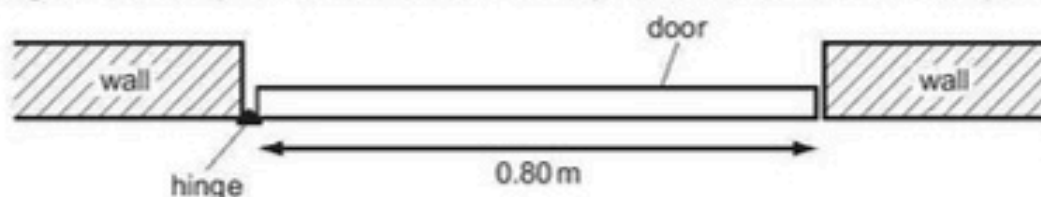
Which two vector diagrams represent forces in equilibrium?



- A P and Q B Q and R C R and S D S and P

55 9702/01/O/N/07/Q14

The diagram shows a plan view of a door which requires a moment of 12 N m to open it.



What is the minimum force that must be applied at the door's midpoint to ensure it opens?

- A 4.8 N B 9.6 N C 15 N D 30 N

56 9702/01/O/N/07/Q17

Two solid substances P and Q have atoms of mass M_P and M_Q respectively. There are n_P and n_Q atoms per unit volume respectively.

It is found by experiment that the density of P is greater than that of Q.

Which deduction from this experiment must be correct?

- A $M_P > M_Q$ B $n_P > n_Q$ C $M_P n_P > M_Q n_Q$ D $\frac{M_P}{n_P} > \frac{M_Q}{n_Q}$

57 9702/01/O/N/07/Q18

A submarine carries a pressure meter so that the crew can work out how far they are below the surface of the sea. At the surface, the meter indicates a pressure of 100 kPa. The density of seawater is 1030 kg m^{-3} .

What is the depth below the surface when the meter reads 450 kPa?

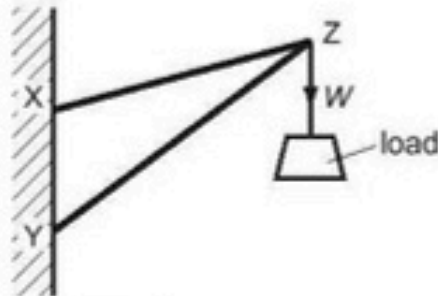
- A 34.6 m B 44.5 m C 340 m D 437 m

58 9702/01/M/J/08/Q13

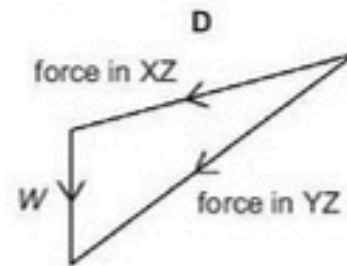
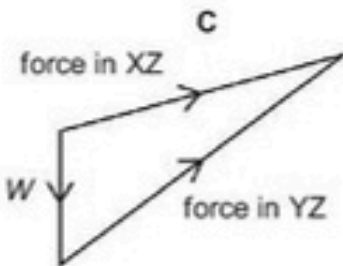
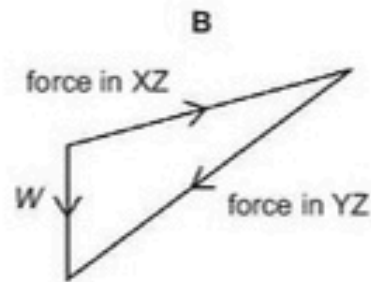
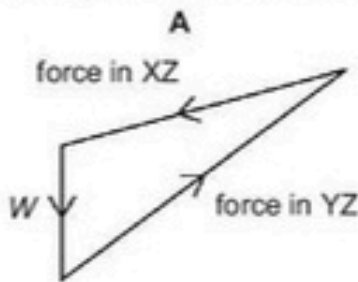
Two rigid rods, XZ and YZ, are fixed to a vertical wall at points X and Y.

A load of weight W is hung from point Z.

The load is not moving.



Which diagram shows the forces acting at point Z?



59 9702/01/M/J/08/Q20

Why does an ideal gas exert pressure on its container?

- A The molecules of the gas collide continually with each other.
 B The molecules of the gas collide continually with the walls of the container.
 C The molecules of the gas collide inelastically with the walls of the container.
 D The weight of the molecules exerts a force on the walls of the container.

60 9702/01/M/J/08/Q21

The formula for hydrostatic pressure is $p = \rho gh$.

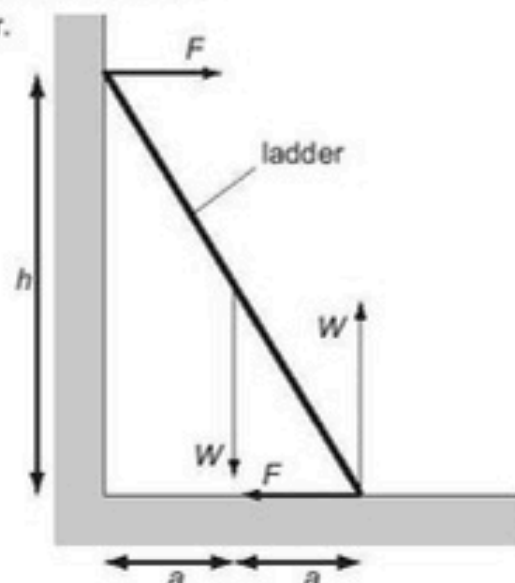
Which equation, or principle of physics, is used in the derivation of this formula?

- A density = mass ÷ volume C atmospheric pressure decreases with height
 B potential energy = mgh D density increases with depth

61 9702/11/O/N/14/Q12

A uniform ladder rests against a vertical wall where there is negligible friction. The bottom of the ladder rests on rough ground where there is friction. The top of the ladder is at a height h above the ground and the foot of the ladder is at a distance $2a$ from the wall.

The diagram shows the forces which act on the ladder.

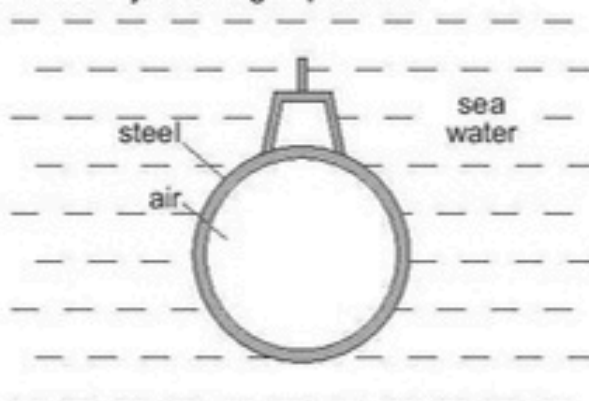


Which equation is formed by taking moments?

- A $Wa + Fh = 2Wa$
- B $Fa + Wa = Fh$
- C $Wa + 2Wa = Fh$
- D $Wa - 2Wa = 2Fh$

62 9702/12/M/J/11/Q15

A submarine is in equilibrium in a fully submerged position.



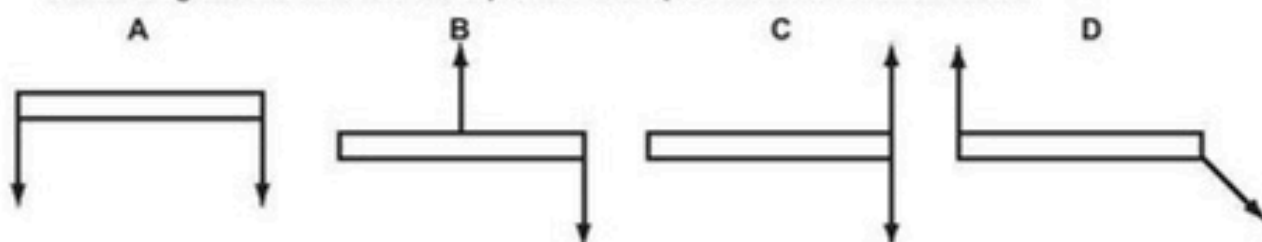
What causes the upthrust on the submarine?

- A The air in the submarine is less dense than sea water.
- B The sea water exerts a greater upward force on the submarine than the weight of the steel.
- C The submarine displaces its own volume of sea water.
- D There is a difference in water pressure acting on the top and bottom of the submarine.

63 9702/12/M/J/11/Q15

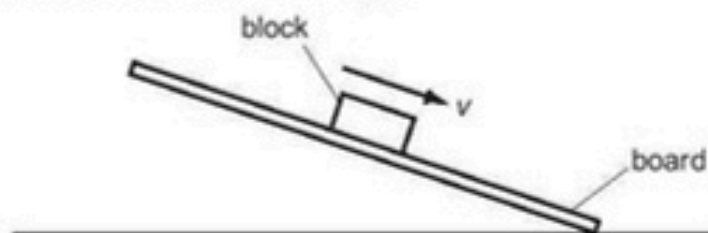
The diagrams all show a pair of equal forces acting on a metre rule.

Which diagram shows forces that provide a couple and zero resultant force?



64 9702/13/M/J/13/Q13

A wooden block rests on a rough board. The end of the board is then raised until the block slides down the plane of the board at constant velocity v .

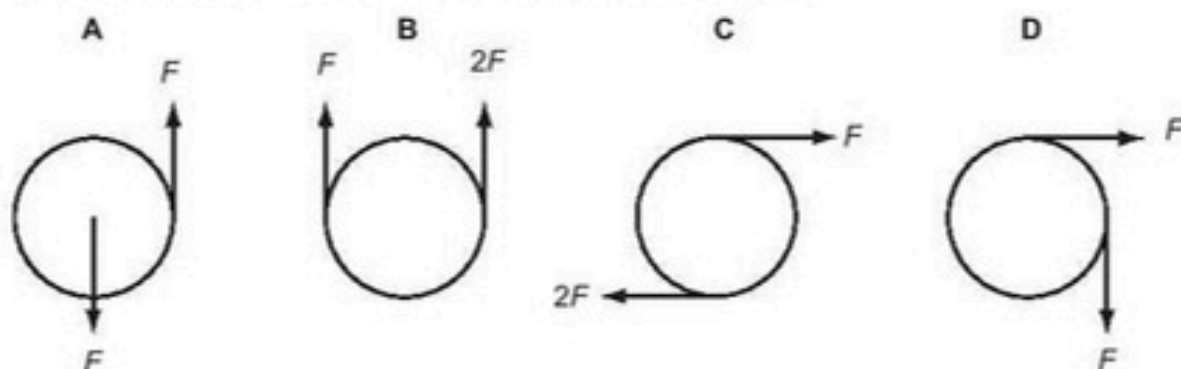


Which row describes the forces acting on the block when sliding with constant velocity?

	frictional force on block	resultant force on block
A	down the plane	down the plane
B	down the plane	zero
C	up the plane	down the plane
D	up the plane	zero

65 9702/01/O/N/08/Q14

Which pair of forces acts as a couple on the circular object?



66 9702/01/O/N/08/Q20

Why does the pressure of a gas increase when the gas is compressed at constant temperature?

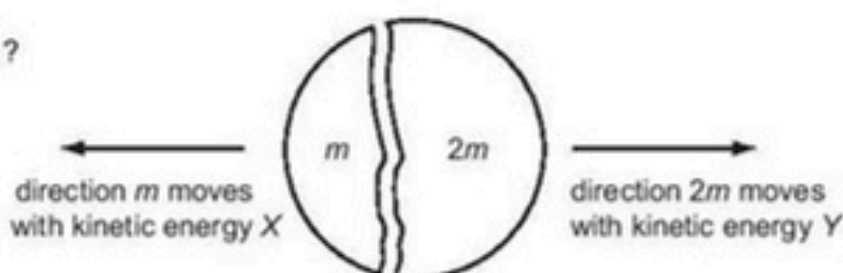
- A The gas molecules collide more often with each other.
- B The gas molecules expand under pressure.
- C The gas molecules hit the walls of the container more frequently.
- D The gas molecules travel faster.

67 9702/01/M/J/09/Q10

A stationary body explodes into two components of masses m and $2m$. The components gain kinetic energies X and Y respectively.

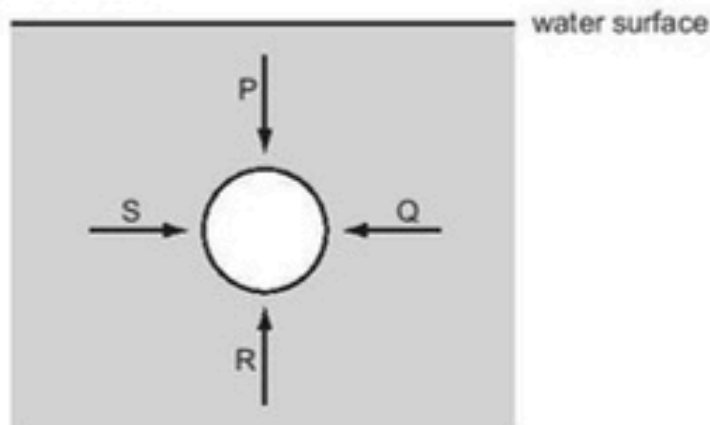
What is the value of the ratio $\frac{X}{Y}$?

- A $\frac{1}{4}$
- B $\frac{1}{2}$
- C $\frac{2}{1}$
- D $\frac{4}{1}$



68 9702/01/M/J/09/Q11

The diagram represents a sphere under water. P, Q, R, and S are forces acting on the sphere, due to the pressure of the water.



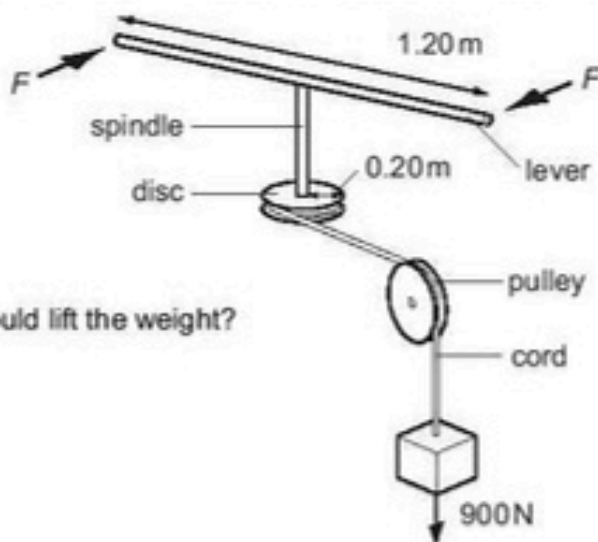
Each force acts perpendicularly to the sphere's surface. P and R act in opposite directions vertically. Q and S act in opposite directions horizontally.

Which information about the magnitudes of the forces is correct?

- A $P < R$; $S = Q$ B $P > R$; $S = Q$ C $P = R$; $S = Q$ D $P = R = S = Q$

69 9702/01/M/J/09/Q13

A spindle is attached at one end to the centre of a lever 1.20 m long and at its other end to the centre of a disc of radius 0.20 m. A cord is wrapped round the disc, passes over a pulley and is attached to a 900 N weight.



What is the minimum force F , applied to each end of the lever, that could lift the weight?

- A 75 N B 150 N
C 300 N D 950 N

70 9702/01/M/J/09/Q16

Which row best describes how the molecules move in solids, in liquids and in gases?

	solids	liquids	gases
A	fixed in position	only vibrate	move about freely
B	slowly in all directions	quickly in all directions	very quickly in all directions
C	vibrate about mean position	vibrate and move about	move about freely
D	vibrate in one direction only	vibrate in two directions	vibrate in all three directions

71 9702/01/M/J/09/Q17

Water can exist in three states: solid, liquid or vapour. Transitions between these states can involve melting, freezing, evaporation or boiling.

Under conditions of constant pressure, which transition can occur over a range of temperatures rather than at one fixed temperature?

- A boiling B evaporation C freezing D melting

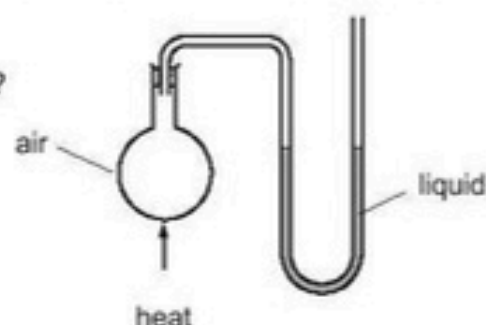
72 9702/01/M/J/09/Q18

The diagram shows a flask connected to a U-tube containing liquid. The flask contains air at atmospheric pressure.

The flask is now gently heated and the liquid level in the right-hand side of the U-tube rises through a distance h . The density of the liquid is ρ .

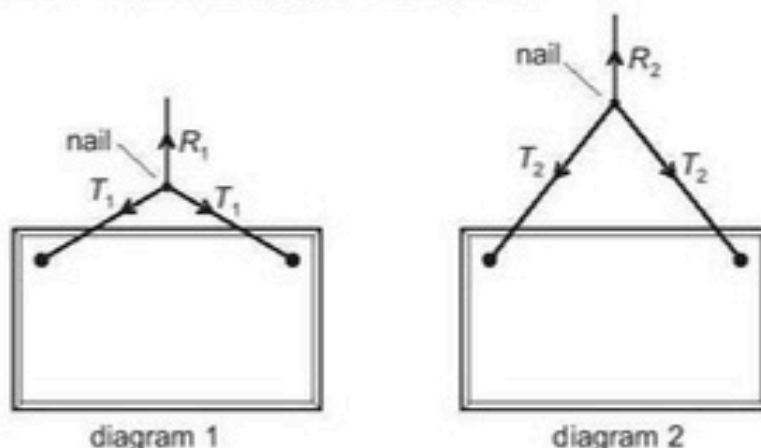
What is the increase in pressure of the heated air in the flask?

- A $h\rho$ B $\frac{1}{2}h\rho g$
C $h\rho g$ D $2h\rho g$



73 9702/12/O/N/09/Q11

The diagrams show two ways of hanging the same picture.



In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

- A $R_1 = R_2$ $T_1 = T_2$ C $R_1 > R_2$ $T_1 < T_2$
B $R_1 = R_2$ $T_1 > T_2$ D $R_1 < R_2$ $T_1 = T_2$

74 9702/12/M/J/11/Q21

Two solid substances P and Q have atoms of mass M_P and M_Q respectively. They have n_P and n_Q atoms per unit volume.

The density of P is greater than the density of Q.

What **must** be correct?

- A $M_P > M_Q$ B $n_P > n_Q$ C $M_P n_P > M_Q n_Q$ D $\frac{M_P}{n^P} > \frac{M_Q}{n^Q}$

75 9702/12/O/N/09/Q12

The diagram shows two pulley wheels connected by a belt.



Wheel Q is driven by a motor and rotates clockwise at a constant rate. Wheel Q puts tension in the top portion of the belt, which in turn drives the wheel P. The lower portion of the belt is slack and has no tension. The weight of the belt and frictional forces are negligible.

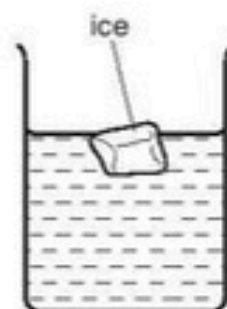
The diameter of P is 150 mm. The diameter of Q is 100 mm. The torque applied to Q is 3.0 N m.

What is the tension in the belt and the torque on wheel P?

	tension in top of belt / N	torque on wheel P / N m
A	20	2.0
B	30	4.5
C	40	2.0
D	60	4.5

76 9702/12/O/N/09/Q16

The diagram shows an ice cube floating in water.



Both the ice cube and the water are at 0°C.

Which statement correctly compares the molecular properties of the ice and those of the water?

- A The mean inter-molecular potential energies are the same for both the ice molecules and the water molecules.
- B The mean inter-molecular separations are the same for both the ice and the water.
- C The mean kinetic energies are the same for both the ice molecules and the water molecules.
- D The mean total energies are the same for both the ice molecules and the water molecules.

77 9702/12/O/N/09/Q17

In the kinetic model of gases, what is pressure equal to?

- A the number of atoms hitting and rebounding from a surface of the gas container
- B the number of atoms hitting and rebounding from a unit area of the gas container surface
- C the force exerted by the atoms hitting and rebounding from a surface of the gas container
- D the force exerted by the atoms hitting and rebounding from a unit area of the gas container surface

78 9702/12/O/N/09/Q18

A rectangular metal bar exerts a pressure of 15 200 Pa on the horizontal surface on which it rests. If the height of the metal bar is 80 cm, what is the density of the metal?

- A 190 kg m^{-3} B 1900 kg m^{-3} C $19\,000 \text{ kg m}^{-3}$ D $190\,000 \text{ kg m}^{-3}$

79 9702/12/M/J/10/Q12

Forces of 3 N, 4 N and 5 N act at one point on an object. The angles at which the forces act can vary.

What is the value of the **minimum** resultant force of these forces?

- A 0 B between 0 and 2 N C 2 N D between 2 N and 4 N

80 9702/12/M/J/10/Q13

An object, immersed in a liquid in a tank, experiences an upthrust.

What is the physical reason for this upthrust?

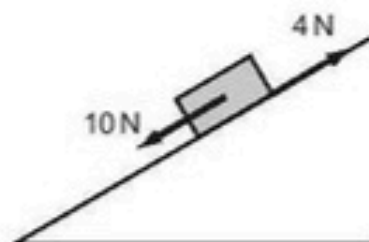
- A The density of the body differs from that of the liquid.
B The density of the liquid increases with depth.
C The pressure in the liquid increases with depth.
D The value of g in the liquid increases with depth.

81 9702/12/M/J/10/Q14

A brick weighing 20 N rests on an inclined plane. The weight of the brick has a component of 10 N parallel with the plane. The brick also experiences a frictional force of 4 N.

What is the acceleration of the brick down the plane? Assume that the acceleration of free fall g is equal to 10 m s^{-2} .

- A 0.3 m s^{-2} B 0.8 m s^{-2}
C 3.0 m s^{-2} D 8.0 m s^{-2}



82 9702/12/M/J/10/Q17

When ice melts, it contracts.

Which row is correct for ice turning into water?

	distance between atoms	density
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

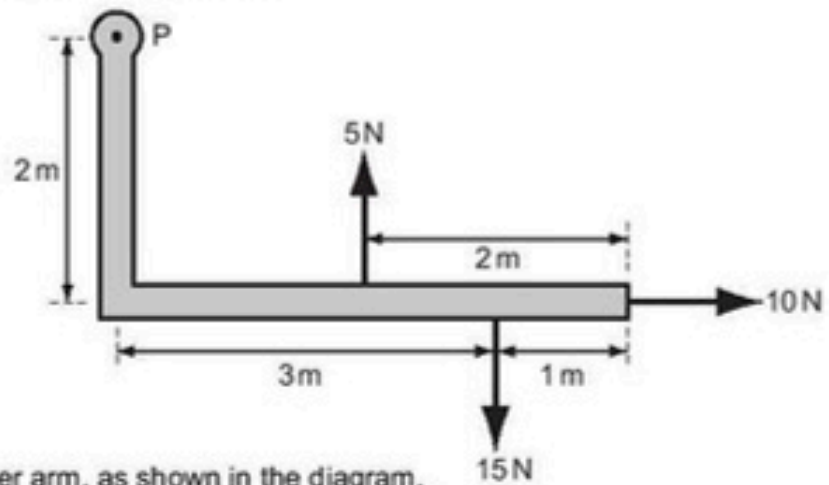
83 9702/12/M/J/10/Q18

Atmospheric pressure at sea level has a value of 100 kPa. The density of sea water is 1020 kg m^{-3} .

At what depth in the sea would the total pressure be 110 kPa?

- A 1.0 m B 9.8 m C 10 m D 11 m

- 84 **9702/11/O/N/10/Q13**
A rigid L-shaped lever arm is pivoted at point P.

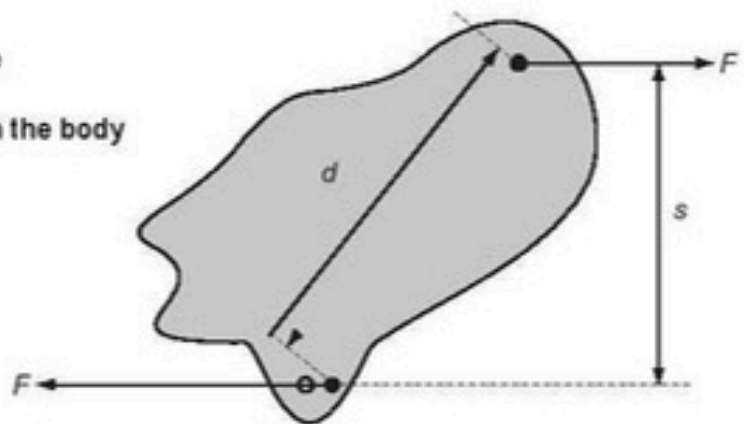


Three forces act on the lever arm, as shown in the diagram.
What is the magnitude of the resultant moment of these forces about point P?

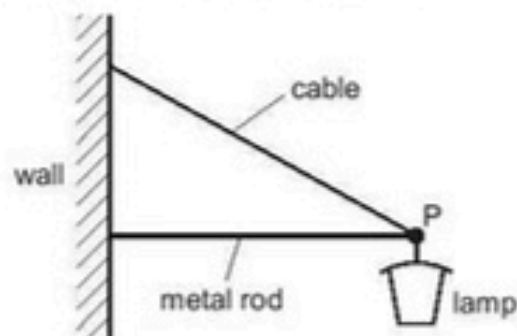
- A 15 Nm B 20 Nm C 35 Nm D 75 Nm

- 85 **9702/11/O/N/10/Q14**
Two parallel forces, each of magnitude F , act on a body as shown.
What is the magnitude of the torque on the body produced by these forces?

- A Fd B Fs
C $2Fd$ D $2Fs$



- 86 **9702/11/O/N/10/Q15***
A street lamp is fixed to a wall by a metal rod and a cable.



Which vector triangle represents the forces acting at point P?



87 9702/11/O/N/10/Q19*

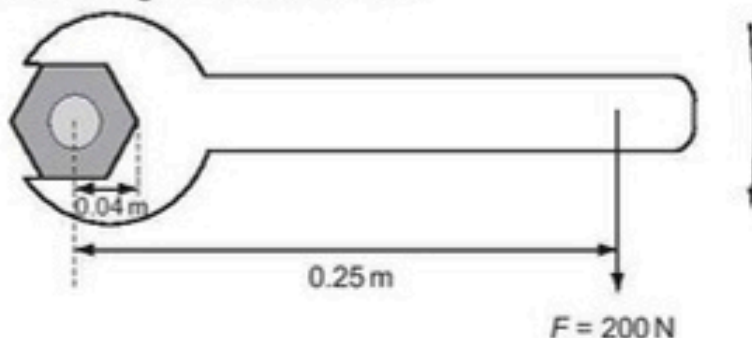
The Mariana Trench in the Pacific Ocean has a depth of about 10 km.

Assuming that sea water is incompressible and has a density of about 1020 kg m^{-3} , what would be the approximate pressure at that depth?

- A 10^5 Pa B 10^6 Pa C 10^7 Pa D 10^8 Pa

88 9702/12/O/N/10/Q12

A spanner is used to tighten a nut as shown.



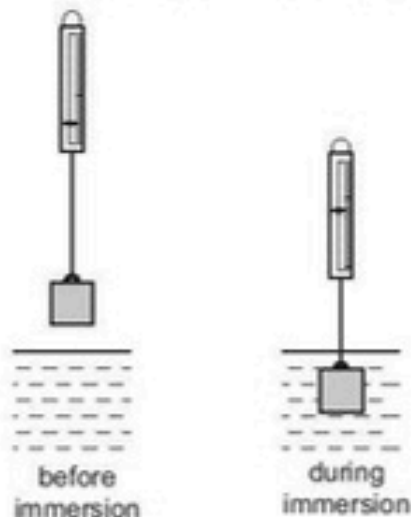
A force F is applied at right-angles to the spanner at a distance of 0.25 m from the centre of the nut. When the nut is fully tightened, the applied force is 200 N.

What is the resistive torque, in an anticlockwise direction, preventing further tightening?

- A 8 Nm B 42 Nm C 50 Nm D 1250 Nm

89 9702/12/O/N/10/Q13

The diagrams show a metal cube suspended from a spring balance before and during immersion in water.



A reduction in the balance reading occurs as a consequence of the immersion. Which statement is correct?

- A The balance reading will be further reduced if the cube is lowered further into the water.
 B The balance reading during immersion corresponds to the upthrust of the water on the cube.
 C The forces acting on the vertical sides of the cube contribute to the change in the balance reading.
 D The gravitational pull on the cube is unchanged by the immersion.

90 9702/12/O/N/10/Q18

Below are four short paragraphs describing the molecules in a beaker of water at 50°C .

Which paragraph correctly describes the molecules?

- A The molecules all travel at the same speed. This speed is not large enough for any of the molecules to leave the surface of the water. There are attractive forces between the molecules.
- B The molecules have a range of speeds. Some molecules travel sufficiently fast to leave the surface of the water. There are no forces between the molecules.
- C The molecules have a range of speeds. Some molecules travel sufficiently fast to leave the surface of the water. There are attractive forces between the molecules.
- D The molecules have a range of speeds. The fastest molecules are unable to leave the surface of the water. There are attractive forces between the molecules.

91 9702/12/O/N/10/Q20

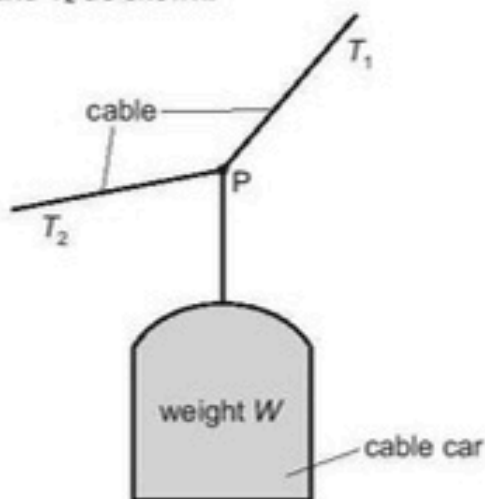
The pressure at sea level is approximately $100\,000\text{ Pa}$. The density of sea water is 1030 kg m^{-3} . What is the approximate pressure 80 m below the surface of the sea?

- A $100\,000\text{ Pa}$ B $180\,000\text{ Pa}$ C $800\,000\text{ Pa}$ D $900\,000\text{ Pa}$

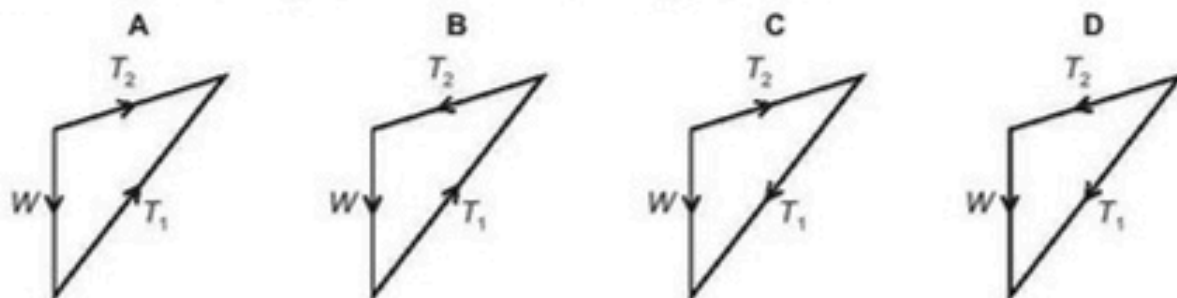
92 9702/11/M/J/11/Q11

A cable car of weight W hangs in equilibrium from its cable at point P.

The cable has tensions T_1 and T_2 as shown.



Which diagram correctly represents the forces acting at point P?

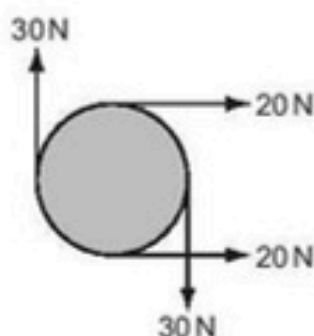


93 9702/11/M/J/11/Q12

The diagram shows four forces applied to a circular object.

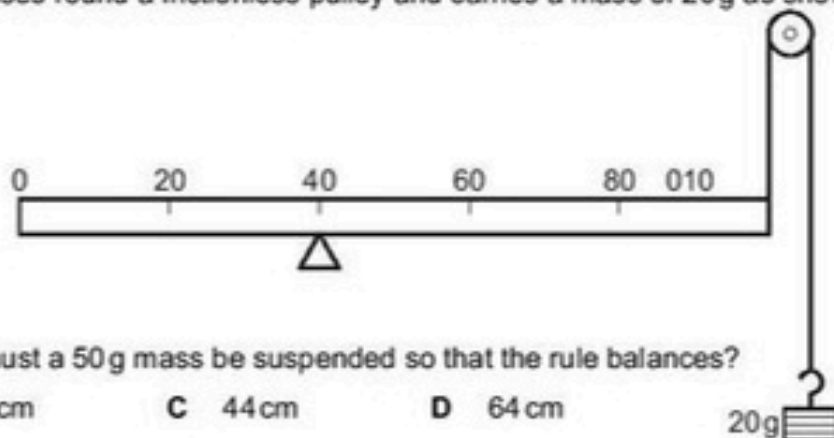
Which row describes the resultant force and resultant torque on the object?

	resultant force	resultant torque
A	zero	zero
B	zero	non-zero
C	non-zero	zero
D	non-zero	non-zero



94 9702/13/O/N/14/Q14

A uniform metre rule of mass 100 g is supported by a pivot at the 40 cm mark and a string at the 100 cm mark. The string passes round a frictionless pulley and carries a mass of 20 g as shown in the diagram.



At which mark on the rule must a 50 g mass be suspended so that the rule balances?

- A 4 cm B 36 cm C 44 cm D 64 cm

95 9702/11/M/J/11/Q19

1.5 m^3 of water is mixed with 0.50 m^3 of alcohol. The density of water is 1000 kg m^{-3} and the density of alcohol is 800 kg m^{-3} .

What is the density of the mixture with volume 2.0 m^3 ?

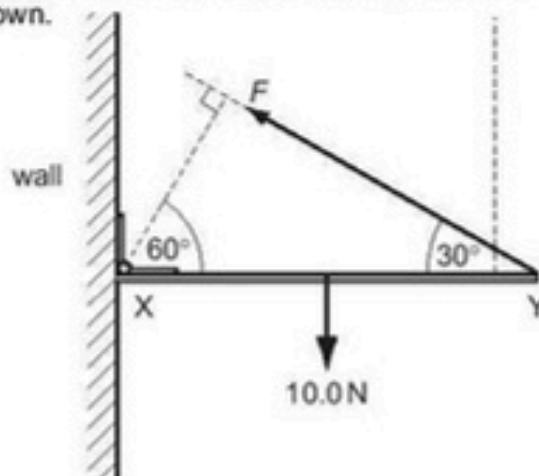
- A 850 kg m^{-3} B 900 kg m^{-3} C 940 kg m^{-3} D 950 kg m^{-3}

96 9702/12/M/J/11/Q16

A uniform rod XY of weight 10.0 N is freely hinged to a wall at X. It is held horizontal by a force F acting from Y at an angle of 30° to the horizontal, as shown.

What is the value of F ?

- A 5.0 N
B 8.7 N
C 10.0 N
D 20.0 N



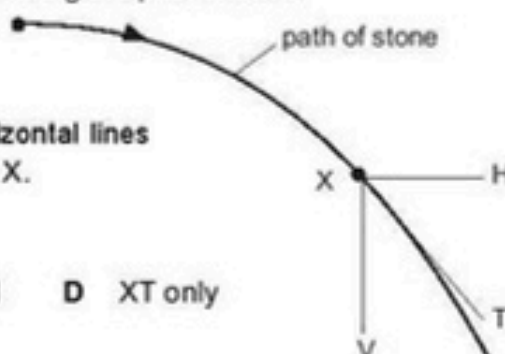
97 9702/12/M/J/11/Q14

A stone is projected horizontally in a vacuum and moves along the path shown.

X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.

Along which directions do forces act on the stone at X?

- A XV only B XH only C XV and XH D XT only



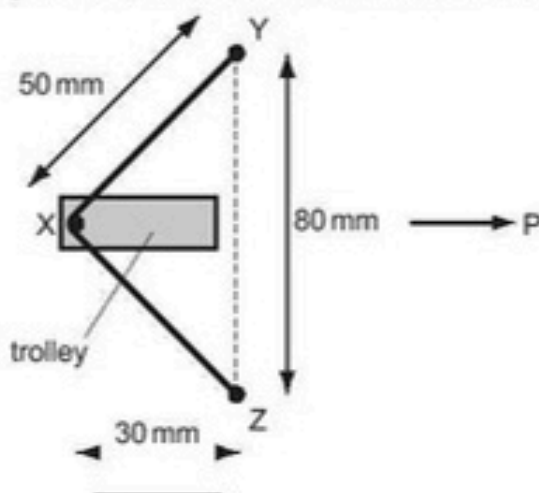
98 9702/12/M/J/11/Q17

The diagram shows two fixed pins, Y and Z. A length of elastic is stretched between Y and Z and around pin X, which is attached to a trolley.

X is at the centre of the elastic and the trolley is to be propelled in the direction P at right angles to YZ. The tension in the elastic is 4 N.

What is the force accelerating the trolley in the direction P when the trolley is released?

- A 2.4 N B 3.2 N
C 4.8 N D 6.4 N



99 9702/12/M/J/11/Q20

The diagram shows two vessels, P and Q, both with sides inclined at 45° .



Vessel P tapers outwards and vessel Q tapers inwards, as shown.

Both vessels contain a liquid. The depth of the liquid in the vessels is the same. The liquid in vessel P is twice as dense as the liquid in vessel Q.

What is the ratio $\frac{\text{pressure due to the liquid on the base of P}}{\text{pressure due to the liquid on the base of Q}}$?

- A $\frac{2}{1}$ B $\frac{\sqrt{2}}{1}$ C $\frac{1}{\sqrt{2}}$ D $\frac{1}{2}$

100 9702/13/O/N/13/Q12

A mass accelerates uniformly when the resultant force acting on it

- A is zero.
B is constant but not zero.
C increases uniformly with respect to time.
D is proportional to the displacement from a fixed point.

101 9702/12/O/N/15/Q19

The table summarises some descriptions of evaporation.
Which row of the table is correct?

	involves a change in state from liquid to vapour	occurs at a fixed temperature	involves a reduction in the average kinetic energy of the remaining atoms
A	true	true	true
B	true	false	true
C	true	false	false
D	false	true	false

102 9702/11/M/J/15/Q14

What is the **definition** of the force on a body?

- A the mass of the body multiplied by its acceleration
- B the power input to the body divided by its velocity
- C the rate of change of momentum of the body
- D the work done on the body divided by its displacement

103 9702/01/O/N/05/Q17

Why does the pressure increase when a sealed container of gas is heated?

- A The gas molecules collide more often with each other.
- B The gas molecules expand when they are heated.
- C The gas molecules travel faster and hit the walls of the container more often.
- D There are more gas molecules present to collide with the walls of the container.

104 9702/11/O/N/11/Q22

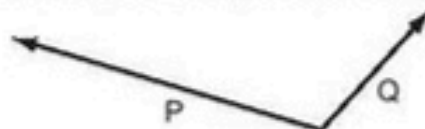
Pollen grains are suspended in a liquid and are illuminated strongly. When observed under a microscope they are seen to be in continuous random motion.

What is the reason for this?

- A convection currents in the liquid
- B evaporation of the liquid
- C molecules of the liquid colliding with the pollen grains
- D pollen grains colliding with each other

105 9702/12/O/N/11/Q12

Two possible displacements of an object are represented by the vectors P and Q.



Which vector best represents the resultant displacement ($P - Q$) of the object?



106 9702/12/O/N/11/Q19

In an experiment to demonstrate Brownian motion, smoke particles in a container are illuminated by a strong light source and observed through a microscope.

The particles are seen as small specks of light that are in motion.

What causes the Brownian motion?

- A collisions between the smoke particles and air molecules
- B collisions between the smoke particles and the walls of the container
- C convection currents within the air as it is warmed by the light source
- D kinetic energy gained by the smoke particles on absorption of light

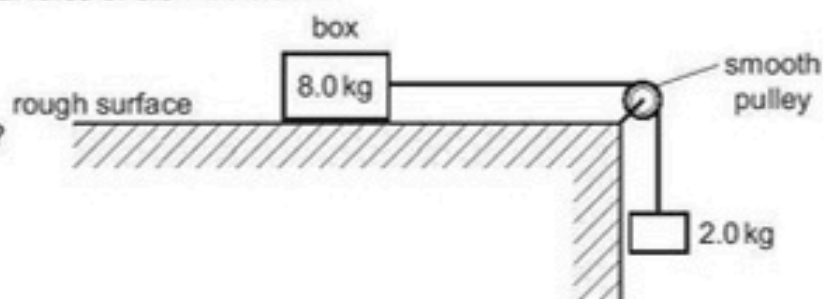
107 9702/01/O/N/08/Q11

A box of mass 8.0 kg rests on a horizontal, rough surface. A string attached to the box passes over a smooth pulley and supports a 2.0 kg mass at its other end.

When the box is released, a frictional force of 6.0 N acts on it.

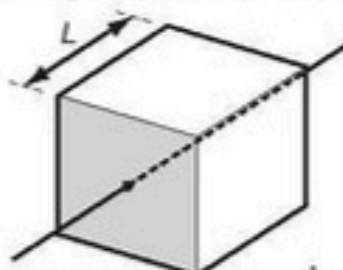
What is the acceleration of the box?

- A 1.4 ms^{-2}
- B 1.7 ms^{-2}
- C 2.0 ms^{-2}
- D 2.5 ms^{-2}

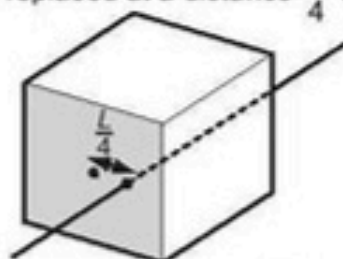


108 9702/12/M/J/12/Q15

The diagram shows a solid cube with weight W and sides of length L . It is supported by a frictionless spindle that passes through the centres of two opposite vertical faces. One of these faces is shaded.



The spindle is now removed and replaced at a distance $\frac{L}{4}$ to the right of its original position.



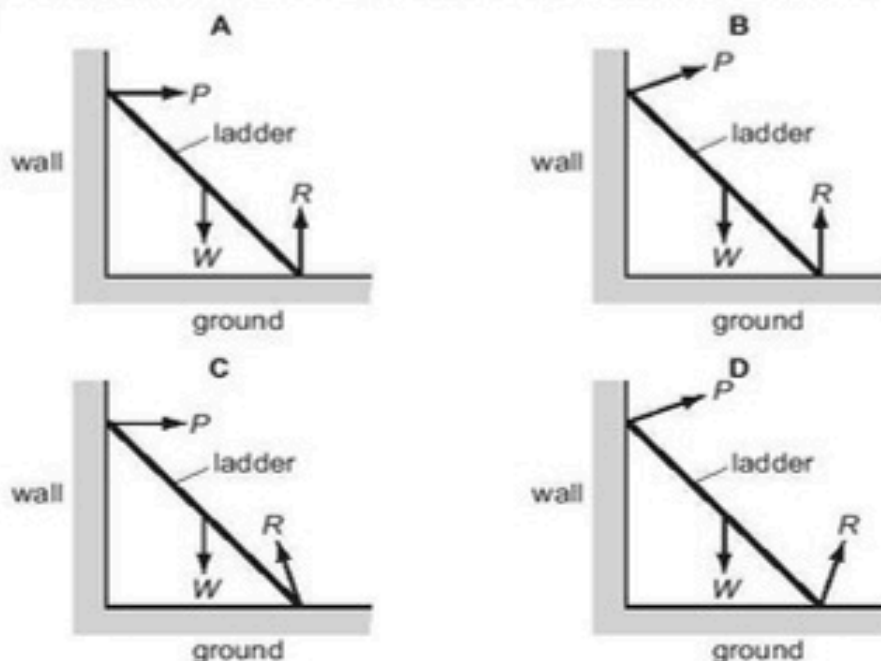
When viewing the shaded face, what is the torque of the couple that will now be needed to stop the cube from toppling?

- A $\frac{WL}{2}$ anticlockwise
- B $\frac{WL}{2}$ clockwise
- C $\frac{WL}{4}$ anticlockwise
- D $\frac{WL}{4}$ clockwise

109 9702/12/M/J/15/Q14

A ladder is positioned on icy (frictionless) ground and is leant against a rough wall. At the instant of release it begins to slide.

Which diagram correctly shows the directions of the forces P , W and R acting on the ladder as it begins to slide?



110 9702/12/O/N/11/Q20

A horizontal plate of area 0.036 m^2 is beneath the surface of a liquid of density 930 kg m^{-3} . The force on the plate due to the pressure of the liquid is 290 N .

What is the depth of the plate beneath the surface of the liquid?

- A 0.88 m B 1.13 m C 8.7 m D 9.1 m

111 9702/12/M/J/12/Q22

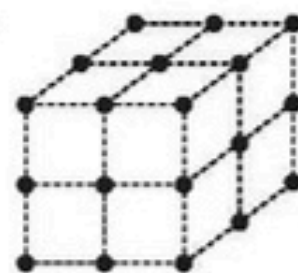
The diagram shows the arrangement of atoms in a particular crystal.

Each atom is at the corner of a cube.

The mass of each atom is $3.5 \times 10^{-25} \text{ kg}$

The density of the crystal is $9.2 \times 10^3 \text{ kg m}^{-3}$.

What is the shortest distance between the centres of two adjacent atoms?



- A $3.8 \times 10^{-29} \text{ m}$ B $6.2 \times 10^{-15} \text{ m}$ C $3.4 \times 10^{-10} \text{ m}$ D $3.0 \times 10^{-9} \text{ m}$

112 9702/12/M/J/12/Q20

A mass of gas enclosed in a cylinder by a piston is heated gently. At the same time, the piston is moved so that the pressure remains constant.

As a result of this, what will **not** occur?

- A The average velocity of the molecules will increase.
 B The mean separation of the molecules will increase.
 C The molecules will travel greater distances between collisions.
 D The number of collisions per second of the molecules on the piston will increase.

113 9702/12/M/J/12/Q21

In an experiment to demonstrate Brownian motion, a transparent container is filled with smoke particles suspended in air.

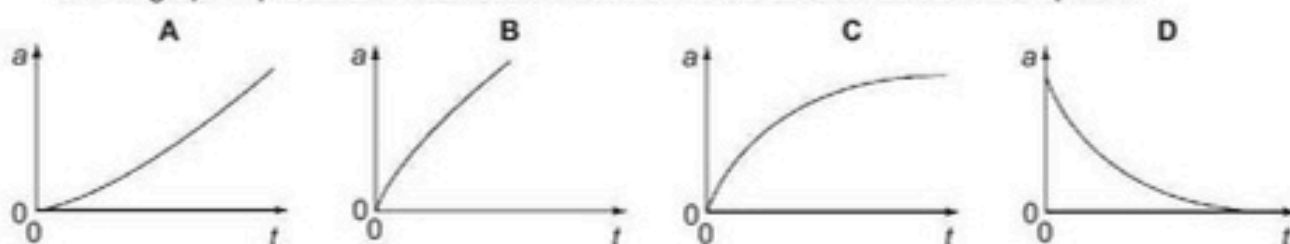
What can be seen when the contents of the container are strongly illuminated and viewed through a microscope?

- A air molecules that are colliding with smoke particles
- B air molecules that are moving in straight lines
- C smoke particles that are moving in random zigzag paths
- D smoke particles that are moving in straight lines

114 9702/12/M/J/12/Q16

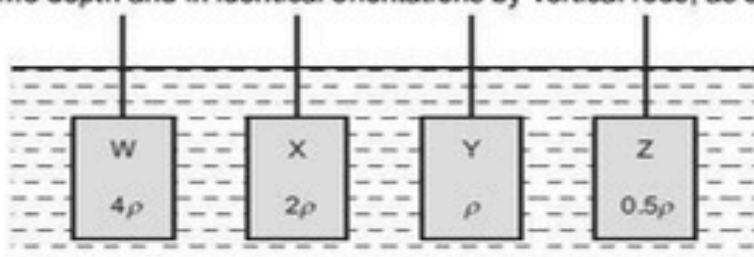
A sphere is released from rest in a viscous fluid.

Which graph represents the variation with time t of the acceleration a of the sphere?



115 9702/13/M/J/15/Q14

Four cuboids with identical length, breadth and height are immersed in water. The cuboids are held at the same depth and in identical orientations by vertical rods, as shown.



Water has density ρ .

Cuboid W is made of material of density 4ρ .

Cuboid X is made of material of density 2ρ .

Cuboid Y is made of material of density ρ .

Cuboid Z is made of material of density 0.5ρ .

Which statement is correct?

- A The upthrust of the water on each of the cuboids is the same.
- B The upthrust of the water on W is twice the upthrust of the water on X.
- C The upthrust of the water on X is twice the upthrust of the water on W.
- D The upthrust of the water on Y is zero.

116 9702/13/M/J/12/Q11

Each option gives a correct word equation involving force.

Which option gives the definition of force?

- A force = energy divided by displacement
- B force = mass $\times$ acceleration
- C force = pressure $\times$ area
- D force = rate of change of momentum

117 9702/12/M/J/12/Q23

A pipe is closed at one end and contains gas, trapped by a column of water.

The atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

The density of water is 1000 kg m^{-3} .

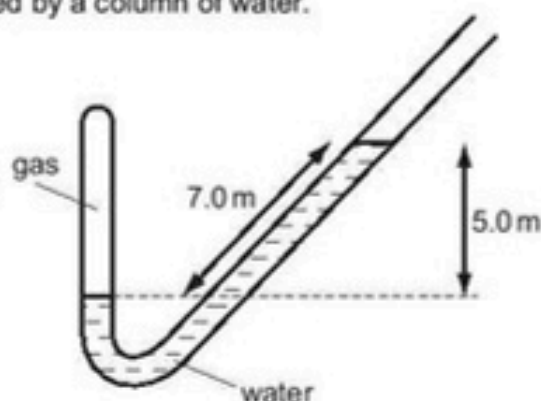
What is the pressure of the gas? (Use $g = 10 \text{ ms}^{-2}$.)

A $0.3 \times 10^5 \text{ Pa}$

C $1.5 \times 10^5 \text{ Pa}$

B $0.5 \times 10^5 \text{ Pa}$

D $1.7 \times 10^5 \text{ Pa}$



118 9702/13/M/J/12/Q14

A cylindrical block of wood has cross-sectional area A and weight W . It is totally immersed in water with its axis vertical. The block experiences pressures p_t and p_b at its top and bottom surfaces respectively.

Which expression is equal to the upthrust on the block?

A $(p_b - p_t)A + W$

B $(p_b - p_t)$

C $(p_b - p_t)A$

D $(p_b - p_t)A - W$

119 9702/13/M/J/12/Q15

A car of mass m travels at constant speed up a slope at an angle θ to the horizontal, as shown in the diagram. Air resistance and friction provide a resistive force F .

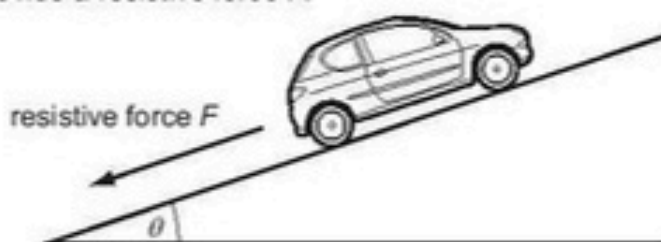
What force is needed to propel the car at this constant speed?

A $mg \cos \theta$

B $mg \sin \theta$

C $mg \cos \theta + F$

D $mg \sin \theta + F$



120 9702/13/M/J/12/Q24

At room temperature, the density of liquid mercury is five times greater than the density of solid aluminium.

What is the reason for this?

A Aluminium atoms are spaced widely apart.

B Aluminium atoms move more freely than mercury atoms.

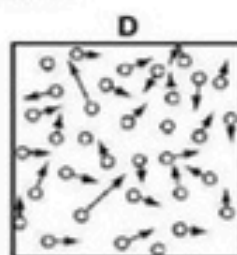
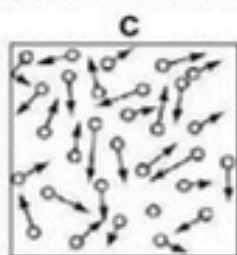
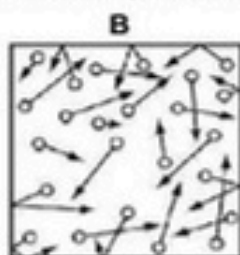
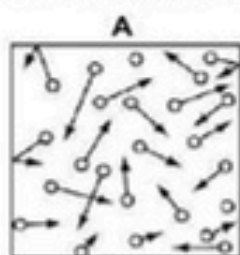
C Atoms in a liquid take up less space than atoms in a solid.

D Mercury atoms have greater mass than aluminium atoms.

121 9702/13/M/J/12/Q25

Each box shows identical molecules of a gas represented by circles with arrows to show the direction of travel and the speed of the molecule. A longer arrow represents a higher speed.

Which box contains a gas of the highest density and the lowest temperature?

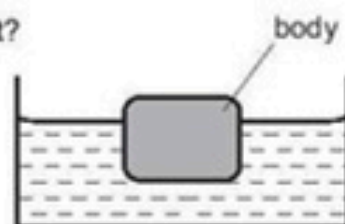


122 9702/12/O/N/12/Q16

A stationary body floats in water.

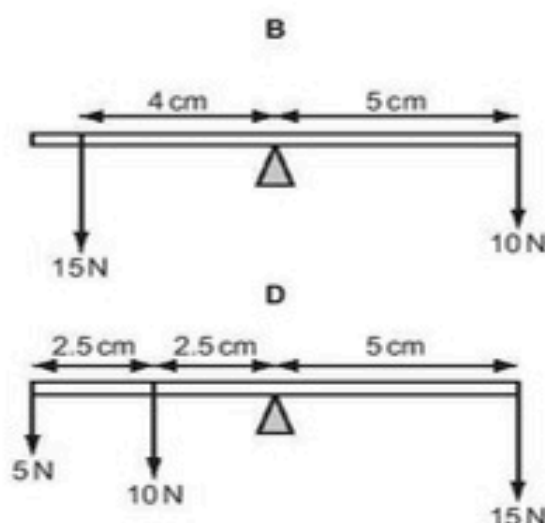
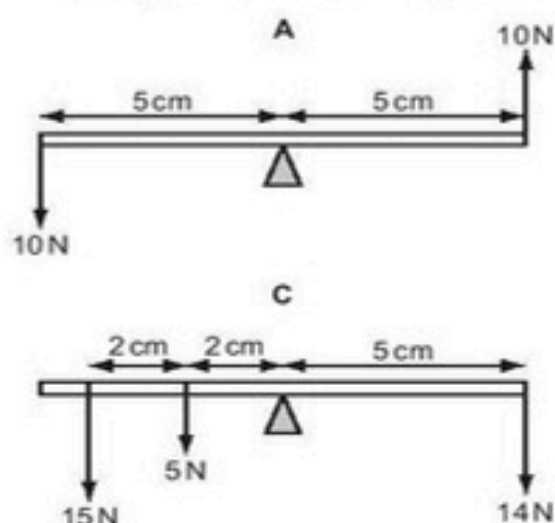
Which statement about the forces acting on the body is correct?

- A The gravitational force is equal to the viscous force.
- B The gravitational force is greater than the upthrust.
- C The upthrust is zero.
- D The viscous force is zero.



123 9702/12/O/N/12/Q17

A rigid uniform beam is pivoted horizontally at its midpoint. Different vertical forces are applied to different positions on the beam. In which diagram is the beam in equilibrium?

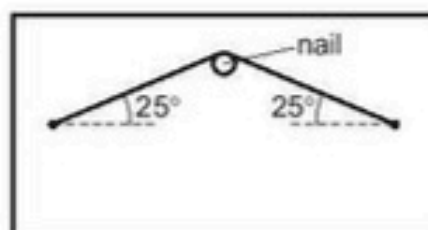


124 9702/12/O/N/12/Q18

A picture on a wall is supported by a wire looped over a nail.

The mass of the picture is 4.2 kg.
What is the tension in the supporting wire?

- | | |
|---------|--------|
| A 5.0 N | B 23 N |
| C 49 N | D 97 N |

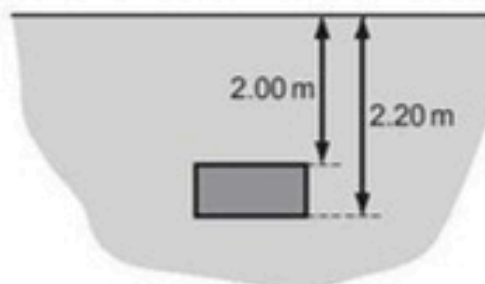


125 9702/13/O/N/12/Q23

The diagram shows a rectangular block of mass 8.2 kg immersed in sea water of density $1.10 \times 10^3 \text{ kg m}^{-3}$.

What is the difference in pressure between the top and bottom surfaces of the block?

- | | |
|--------------------------------|--------------------------------|
| A $2.2 \times 10^2 \text{ Pa}$ | C $1.8 \times 10^4 \text{ Pa}$ |
| B $2.2 \times 10^3 \text{ Pa}$ | D $2.3 \times 10^4 \text{ Pa}$ |



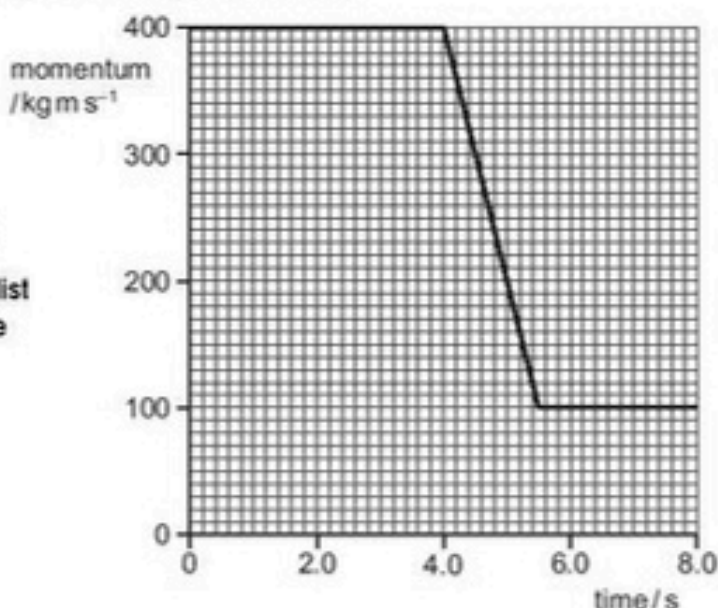
126 9702/12/O/N/12/Q23

Which row correctly describes the spacing and motion of the molecules in water and in ice when both are at a temperature of 0°C ?

	spacing	motion
A	molecules in ice are further apart than molecules in water	molecules in both ice and water have the same average speed
B	molecules in ice are further apart than molecules in water	molecules in ice travel more slowly than those in water
C	molecules in ice are closer than molecules in water	molecules in ice travel more slowly than those in water
D	molecules in ice are closer than molecules in water	molecules in both ice and water have the same average speed

127 9702/13/O/N/12/Q14

The graph shows the momentum of a cyclist over a period of 8.0 s.



At time 4.0 s, she applies the brakes.

What is the resultant force on the cyclist during the period when the brakes are applied?

- A 55 N B 200 N
C 270 N D 450 N

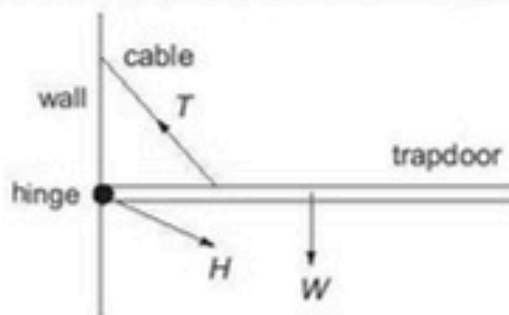
128 9702/11/M/J/13/Q16

A hinged trapdoor is held closed in the horizontal position by a cable.

Three forces act on the trapdoor: the weight W of the door, the tension T in the cable and the force H at the hinge.

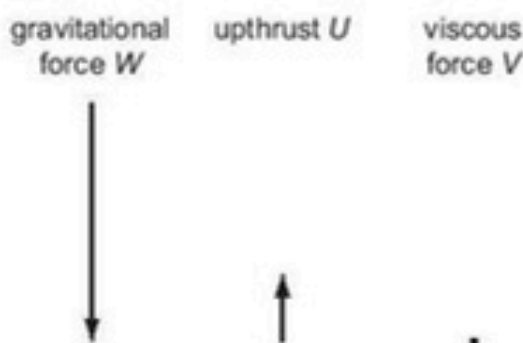
Which list gives the three forces in increasing order of magnitude?

- A H, T, W B T, H, W
C W, H, T D W, T, H

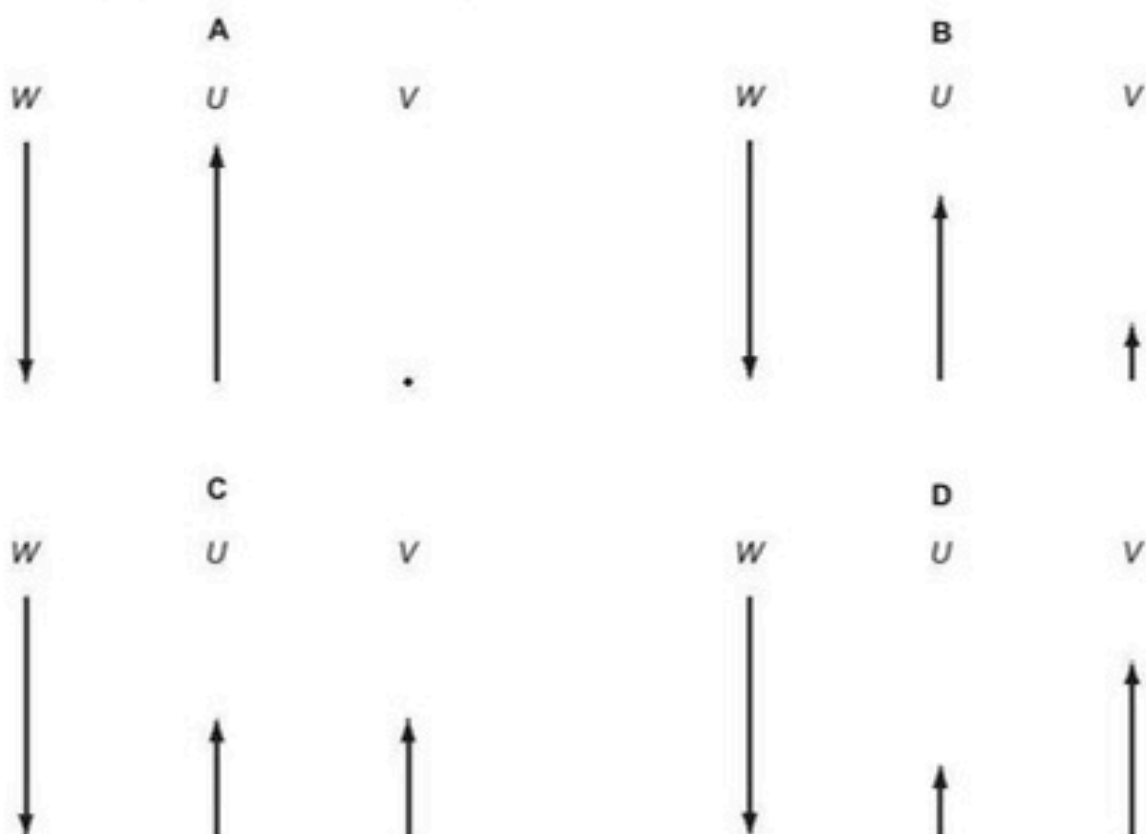


129 9702/13/O/N/12/Q15

A hailstone, initially stationary at the base of a cloud, falls vertically towards the Earth. The diagram shows the magnitudes and directions of the forces acting on the hailstone as it starts to drop.



Which diagram shows the magnitudes and directions of these forces when the hailstone attains a terminal (constant) speed in the air (of uniform density)?



130 9702/13/O/N/12/Q19

A piston in a gas supply pump has an area of 400 cm^2 and it moves a distance of 25 cm during one stroke.

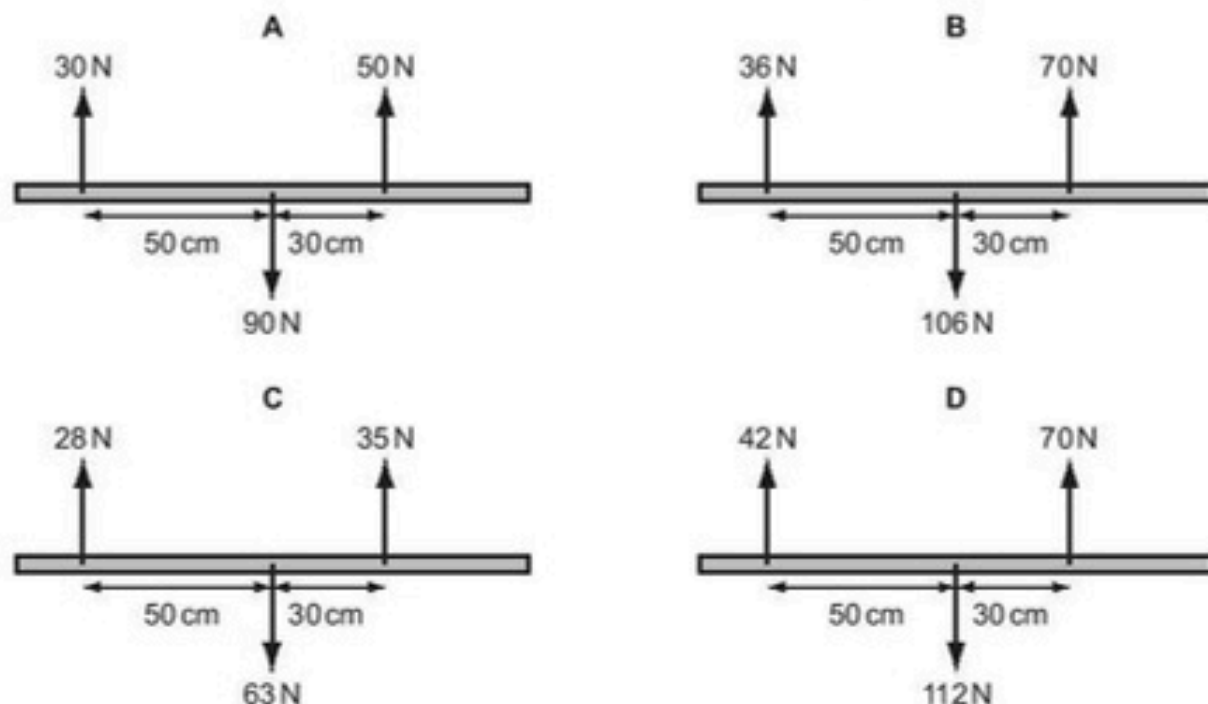
The pump moves the gas against a fixed pressure of 3000 Pa .

How much work is done by the piston during one stroke?

- A 30 J B $3.0 \times 10^3 \text{ J}$ C $3.0 \times 10^5 \text{ J}$ D $3.0 \times 10^7 \text{ J}$

131 9702/13/O/N/12/Q16

Four beams of the same length each have three forces acting on them. Which beam has both zero resultant force and zero resultant torque acting?



132 9702/13/O/N/12/Q22

Which row correctly describes the ordering and motion of the molecules in water and in ice when both are at a temperature of 0°C ?

	ordering	motion
A	a regular pattern of molecules in ice but not in water	molecules in both ice and water have the same average speed
B	a regular pattern of molecules in ice but not in water	molecules in ice travel more slowly than those in water
C	a regular pattern of molecules in both ice and water	molecules in ice travel more slowly than those in water
D	a regular pattern of molecules in both ice and water	molecules in both ice and water have the same average speed

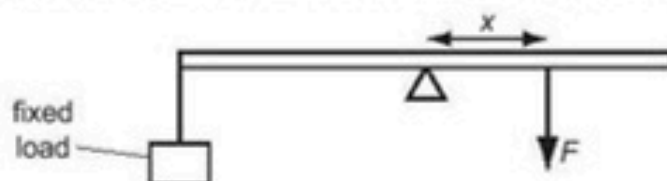
133 9702/11/M/J/13/Q20

Which row correctly states the characteristics of the process of evaporation?

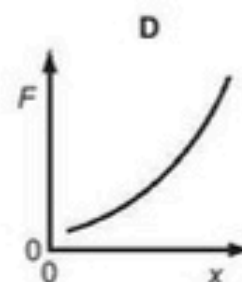
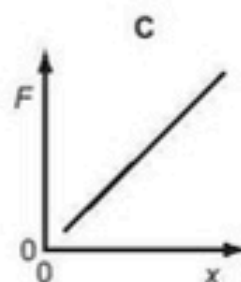
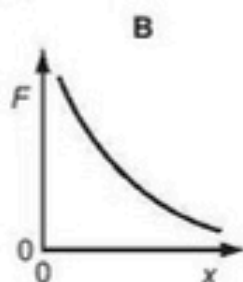
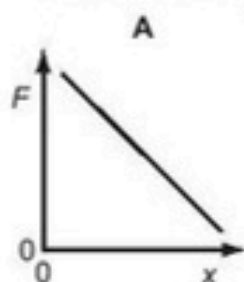
	requires heat energy	occurs only at a particular temperature	can cause a change of temperature
A	✓	✓	x
B	✓	x	✓
C	x	✓	x
D	x	x	✓

134 9702/11/M/J/13/Q14

A horizontal bar is supported on a pivot at its centre of gravity. A fixed load is attached to one end of the bar. To keep the bar in equilibrium, a force F is applied at a distance x from the pivot.

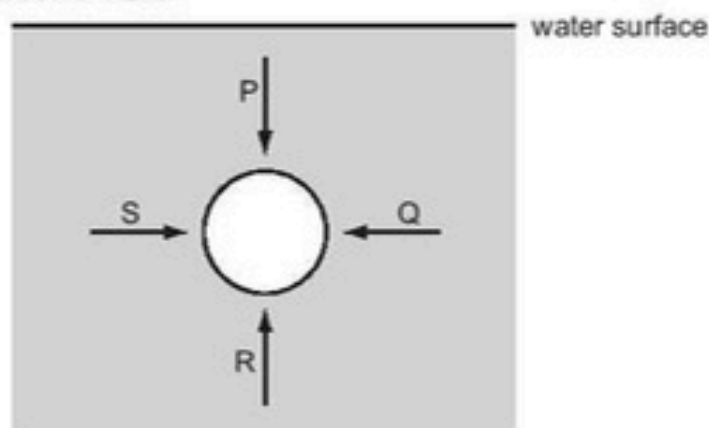


How does F vary with x ?



135 9702/11/M/J/13/Q15

The diagram represents a sphere under water. P , Q , R and S are forces acting on the sphere, due to the pressure of the water.



Each force acts perpendicularly to the sphere's surface. P and R act in opposite directions vertically. Q and S act in opposite directions horizontally.

Which information about the magnitudes of the forces is correct?

- | | |
|-----------------------|--------------------------------------|
| A $P < R$ and $S = Q$ | C $P = R$ and $S = Q$ and $P \neq S$ |
| B $P > R$ and $S = Q$ | D $P = R$ and $S = Q$ and $P = S$ |

136 9702/12/M/J/13/Q13

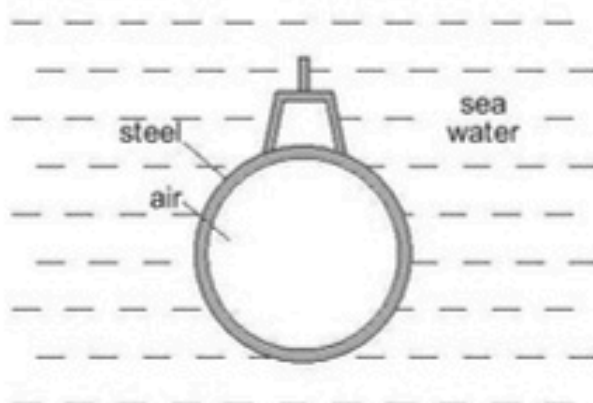
All external forces on a body cancel out.

Which statement **must** be correct?

- A The body does not move.
- B The momentum of the body remains unchanged.
- C The speed of the body remains unchanged.
- D The total energy (kinetic and potential) of the body remains unchanged.

137 9702/01/M/J/08/Q15

A submarine is in equilibrium in a fully submerged position.



What causes the upthrust on the submarine?

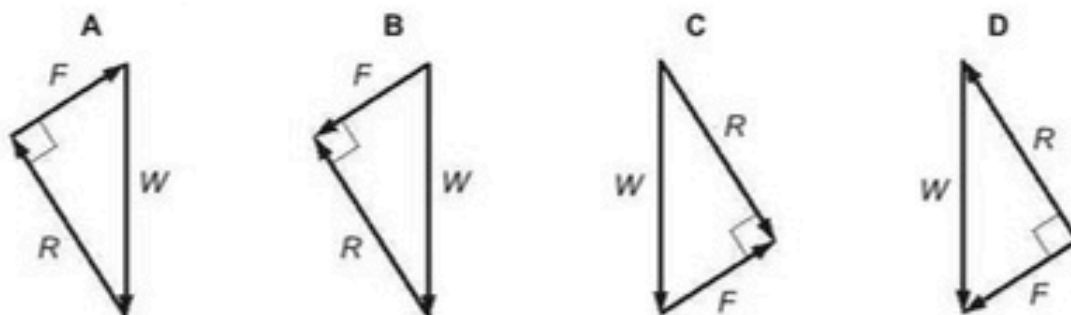
- A The air in the submarine is less dense than sea water.
- B The sea water exerts a greater upward force on the submarine than the weight of the steel.
- C The submarine displaces its own volume of sea water.
- D There is a difference in water pressure acting on the top and on the bottom of the submarine.

138 9702/12/M/J/13/Q12

A vehicle is at rest on a slope. It is considered to have three forces acting on it to keep it in equilibrium.

They are its weight W , a normal reaction force R and a frictional force F .

Which triangle of forces is correct?

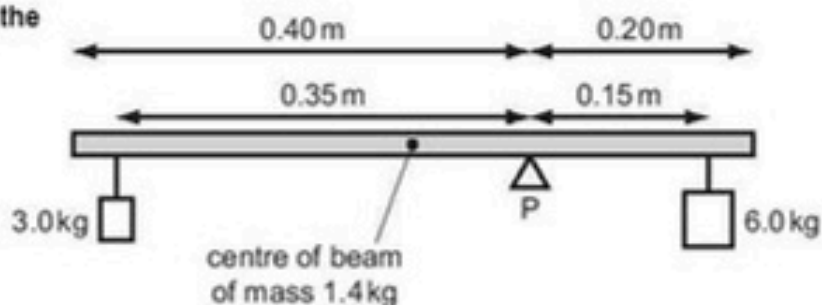


139 9702/12/M/J/13/Q14

A uniform beam of mass 1.4 kg is pivoted at P as shown. The beam has a length of 0.60 m and P is 0.20 m from one end. Loads of 3.0 kg and 6.0 kg are suspended 0.35 m and 0.15 m from the pivot as shown.

What torque must be applied to the beam in order to maintain it in equilibrium?

- A 0.010 Nm
- B 0.10 Nm
- C 0.29 Nm
- D 2.8 Nm



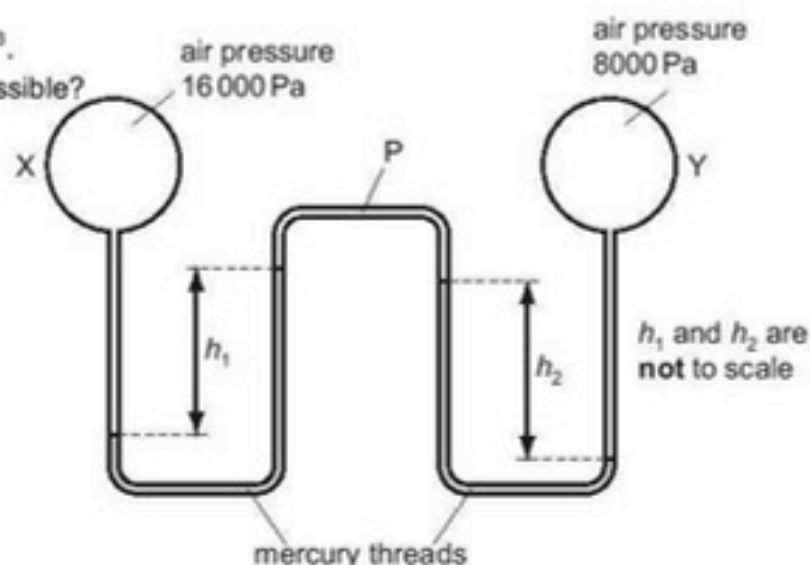
140 9702/12/M/J/13/Q20

Two bulbs X and Y containing air at different pressures are connected by a tube P which contains two mercury threads.

The density of mercury is 13600 kg m^{-3} .

Which pair of values of h_1 and h_2 is possible?

	h_1 / cm	h_2 / cm
A	4.0	2.0
B	6.0	6.0
C	12.0	18.0
D	18.0	12.0

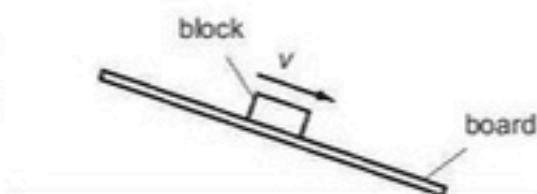


141 9702/13/M/J/13/Q13

A wooden block rests on a rough board. The end of the board is then raised until the block slides down the plane of the board at constant velocity v .

Which row describes the forces acting on the block when sliding with constant velocity?

	frictional force on block	resultant force on block
A	down the plane	down the plane
B	down the plane	zero
C	up the plane	down the plane
D	up the plane	zero



142 9702/12/O/N/13/Q11

A beam of α -particles collides with a lead sheet. Each α -particle in the beam has a mass of $6.6 \times 10^{-27} \text{ kg}$ and a speed of $1.5 \times 10^7 \text{ ms}^{-1}$.

5.0×10^4 α -particles per second collide with an area of 1.0 cm^2 of lead. Almost all of the α -particles are absorbed by the lead so that they have zero speed after collision.

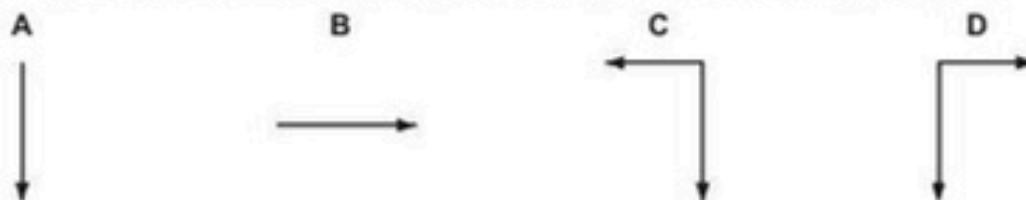
What is an estimate of the average pressure exerted on the lead by the α -particles?

- A $5.0 \times 10^{-15} \text{ Pa}$ B $5.0 \times 10^{-13} \text{ Pa}$ C $5.0 \times 10^{-11} \text{ Pa}$ D $5.0 \times 10^{-9} \text{ Pa}$

143 9702/12/O/N/13/Q12

An object in air is thrown upwards and towards the left.

Which diagram shows the force(s) acting on the body when it is at its highest point?



144 9702/13/M/J/13/Q17*

Ice at a temperature of 0°C is a rare example of a solid that floats on its liquid form, in this case water, when they are both at the same temperature.

What is the explanation for this?

- A The average speed of the molecules in the ice is greater than the average speed of the molecules in the water.
- B The average speed of the molecules in the water is greater than the average speed of the molecules in the ice.
- C The mean separation of the molecules in the ice is greater than the mean separation of the molecules in the water.
- D The mean separation of the molecules in the water is greater than the mean separation of the molecules in the ice.

145 9702/13/M/J/13/Q18

The formula for hydrostatic pressure is $p = \rho gh$.

Which equation, or principle of physics, is used in the derivation of this formula?

- | | |
|---|--|
| A density = $\frac{\text{mass}}{\text{volume}}$ | C atmospheric pressure decreases with height |
| B potential energy = mgh | D density increases with depth |

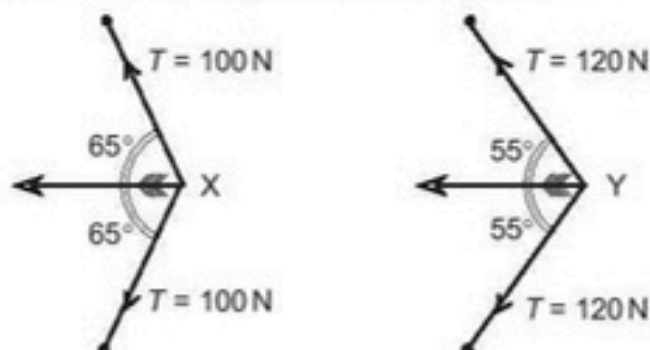
146 9702/12/O/N/13/Q14

An archer draws his bowstring back to position X. The bowstring and arrow are shown. The tension T in the string is also shown. Then he draws the bowstring back further to position Y.

The resultant force on the arrow is greater when the arrow is released from position Y.

What is the increase in force?

- | | |
|--------|--------|
| A 15 N | B 27 N |
| C 40 N | D 53 N |



147 9702/12/O/N/13/Q18

A car travelling on a level road at a steady 20 m s^{-1} against a constant resistive force develops a power of 40 kW.

What is the magnitude of the resistive force?

- | | | | |
|---------|---------|----------|----------|
| A 200 N | B 800 N | C 2000 N | D 4000 N |
|---------|---------|----------|----------|

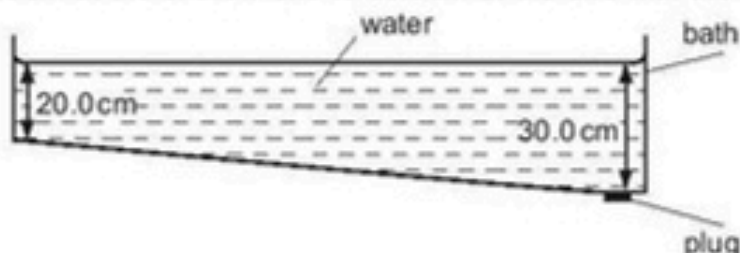
148 9702/12/O/N/13/Q22

Water in a bath varies in depth from 20.0 cm at the shallow end to 30.0 cm at the end with the plug.

The density of the water is 1000 kg m^{-3} .

What is the pressure of the water acting on the plug?

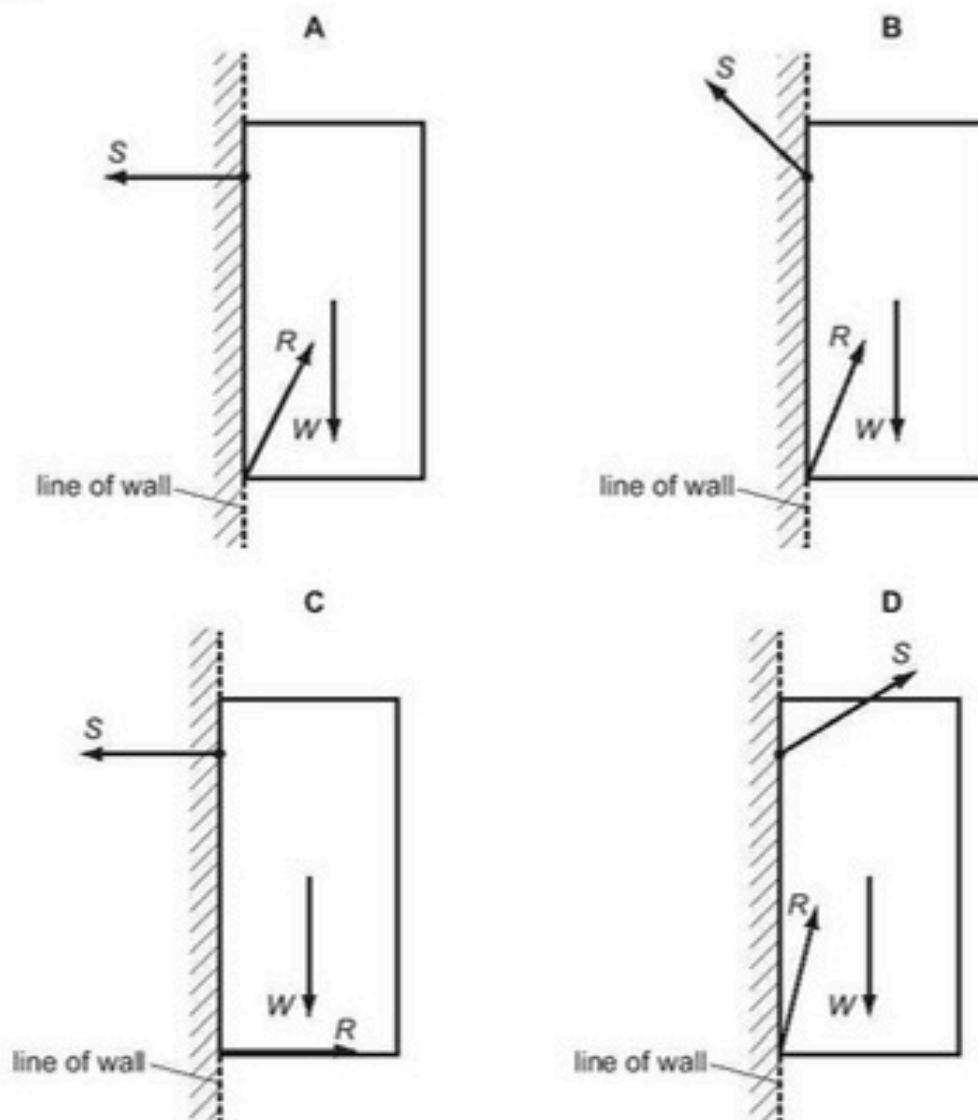
- | | |
|-----------|-----------|
| A 1960 Pa | B 2450 Pa |
| C 2940 Pa | D 4900 Pa |



149 9702/13/O/N/13/Q14

A cupboard is attached to a wall by a screw.

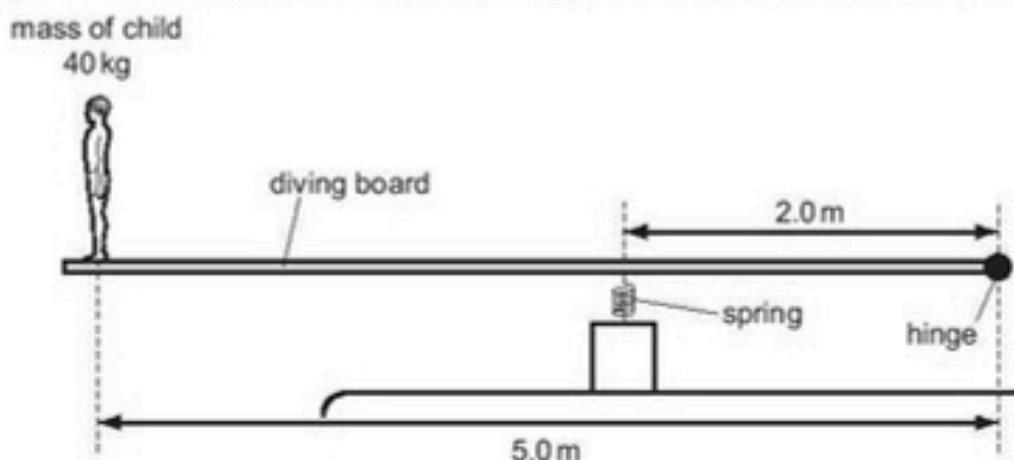
Which force diagram shows the cupboard in equilibrium, with the weight W of the cupboard, the force S that the screw exerts on the cupboard and the force R that the wall exerts on the cupboard?



Space for working

150 9702/13/O/N/13/Q15

A diving board of length 5.0 m is hinged at one end and supported 2.0 m from this end by a spring of spring constant 10 kN m^{-1} . A child of mass 40 kg stands at the far end of the board.



What is the extra compression of the spring caused by the child standing on the end of the board?

- A 1.0 cm B 1.6 cm C 9.8 cm D 16 cm

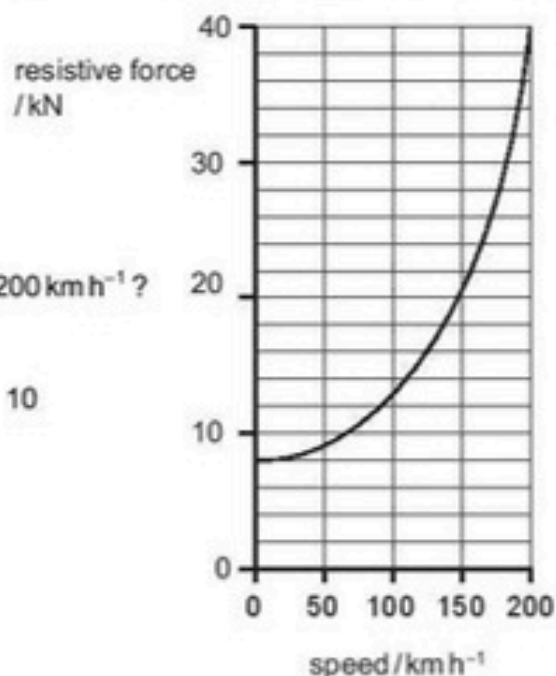
151 9702/13/O/N/13/Q16

The graph shows how the total resistive force acting on a train varies with its speed.

Part of this force is due to wheel friction, which is constant. The rest is due to wind resistance.

What is the ratio $\frac{\text{wind resistance}}{\text{wheel friction}}$ at a speed of 200 km h^{-1} ?

- A 4 B 5 C 8 D 10



152 9702/13/O/N/13/Q20

Gold has a density of 19.3 g cm^{-3} .

The volume occupied by a single atom of gold may be considered to be a cube with sides of length $2.6 \times 10^{-8} \text{ cm}$.

What is the mass of a gold atom?

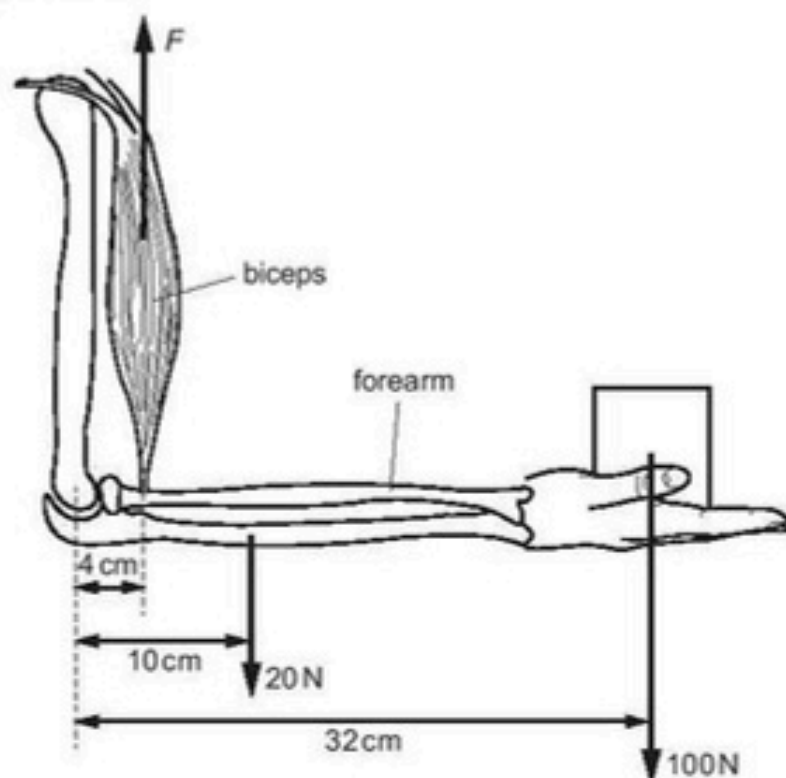
- A $3.4 \times 10^{-25} \text{ g}$ B $3.4 \times 10^{-22} \text{ g}$ C $1.3 \times 10^{-17} \text{ g}$ D $1.3 \times 10^{-14} \text{ g}$

153 9702/11/M/J/14/Q12

A man holds a 100 N load stationary in his hand. The combined weight of the forearm and hand is 20 N. The forearm is held horizontal, as shown.

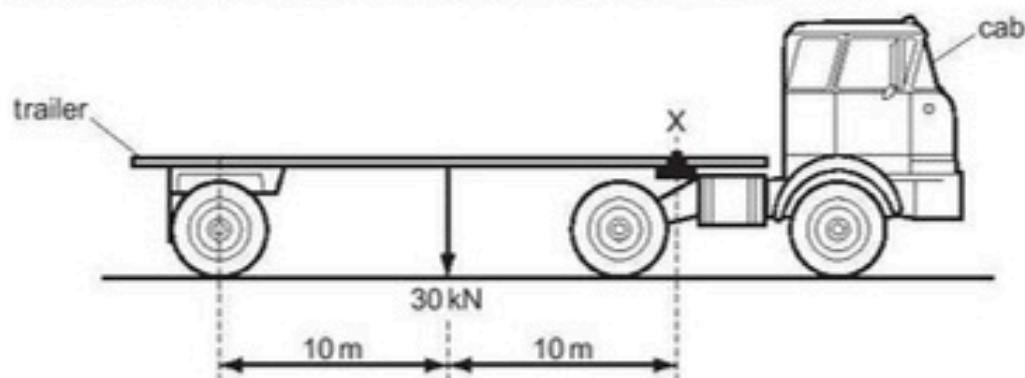
What is the vertical force F needed in the biceps?

- A 750 N B 800 N
C 850 N D 900 N



154 9702/12/O/N/11/Q14

A trailer of weight 30 kN is hitched to a cab at X, as shown in the diagram.



What is the upward force exerted by the cab on the trailer at X?

- A 3 kN B 15 kN C 30 kN D 60 kN

155 9702/13/O/N/13/Q21

In an experiment to demonstrate Brownian motion, a transparent container is filled with smoke particles suspended in air.

What can be seen when the contents of the container are strongly illuminated and viewed through a microscope?

- A molecules in random motion C smoke particles in random motion
B molecules vibrating regularly D smoke particles vibrating regularly

156 9702/11/M/J/14/Q17

Which statement about boiling and evaporation is correct?

- A Boiling can only occur at the surface of a liquid.
- B Evaporation can only occur at a fixed temperature.
- C Only boiling involves a change of phase.
- D When some of a liquid evaporates, the rest of the liquid becomes cooler.

157 9702/11/M/J/14/Q18

There is one temperature, about 0.01°C , at which water, water vapour and ice can co-exist in equilibrium.

Which statement about the properties of the molecules at this temperature is correct?

- A Ice molecules are closer to one another than water molecules.
- B The mean kinetic energy of water molecules is greater than the mean kinetic energy of ice molecules.
- C Water vapour molecules are less massive than water molecules.
- D Water vapour molecules have the same mean speed as both ice and water molecules.

158 9702/12/M/J/14/Q10

A tractor of mass 1000 kg is connected by a tow-bar to a trailer of mass 1000 kg . The total resistance to motion has a constant value of 4000 N . One quarter of this resistance acts on the trailer.

When the tractor and trailer are moving along horizontal ground at a constant speed of 6 m s^{-1} , what is the force exerted on the tractor by the tow-bar?

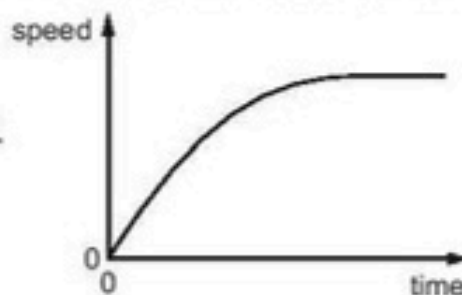
- A 0 N B 1000 N C 3000 N D 4000 N

159 9702/12/M/J/14/Q11

The graph shows the variation with time of the speed of a raindrop falling vertically through air.

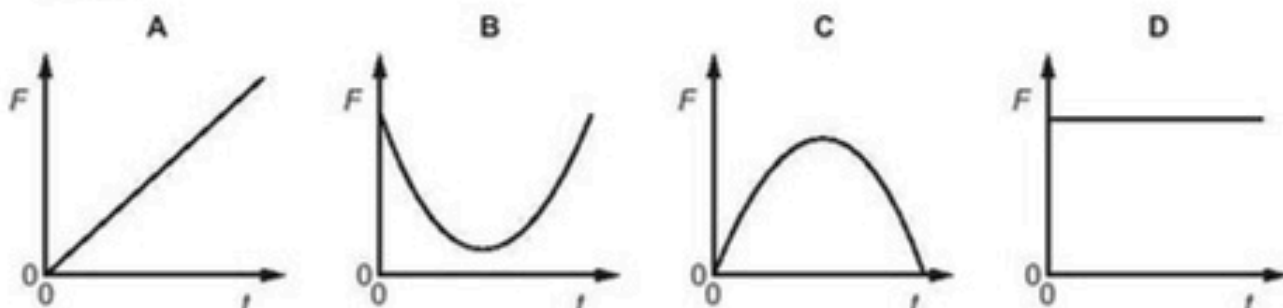
Which statement is correct?

- A The acceleration decreases to produce a steady speed.
- B The acceleration increases as the speed increases.
- C The air resistance decreases as the speed increases.
- D The resultant force increases as the speed increases.



160 9702/13/M/J/14/Q10

A tennis ball is dropped onto a table and bounces back up. The table exerts a force F on the ball. Which graph best shows the variation with time t of the force F while the ball is in contact with the table?



161 9702/12/M/J/14/Q17

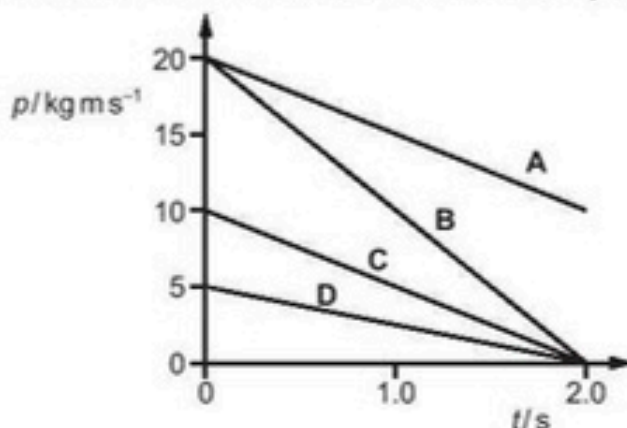
If the Universe was such that the speed of the molecules in a substance increased with temperature but at any particular temperature the speed of all the molecules in a substance was the same, which process would **not** occur?

- A boiling B condensation C evaporation D melting

162 9702/13/M/J/14/Q11

A resultant force of 10N acts on a body for a time of 2.0 s.

Which graph could show the variation with time t of the momentum p of the body?



163 9702/12/O/N/13/Q20

Below are four short paragraphs describing the molecules in a beaker of water at 50 °C.

Which paragraph correctly describes the molecules?

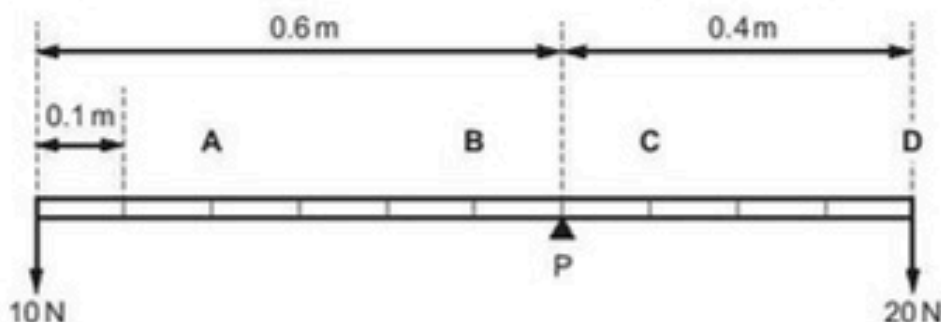
- A The molecules all travel at the same speed. This speed is not large enough for any of the molecules to leave the surface of the water. There are attractive forces between the molecules.
- B The molecules have a range of speeds. Some molecules travel sufficiently fast to leave the surface of the water. There are no forces between the molecules.
- C The molecules have a range of speeds. The fastest molecules are unable to leave the surface of the water. There are attractive forces between the molecules.
- D The molecules have a range of speeds. Some molecules travel sufficiently fast to leave the surface of the water. There are attractive forces between the molecules.

164 9702/13/M/J/14/Q15

A uniform beam is pivoted at P as shown. Weights of 10 N and 20 N are attached to its ends.

The length of the beam is marked at 0.1 m intervals. The weight of the beam is 100 N.

At which point should a further weight of 20 N be attached to achieve equilibrium?



165 9702/13/M/J/14/Q13

What is the condition for an object to be in equilibrium?

- A The object's velocity and the resultant torque on it must both be zero.
- B The object's velocity must be zero.
- C The resultant force and the resultant torque on the object must both be zero.
- D The resultant force on the object must be zero.

166 9702/13/M/J/14/Q17

A shot-put champion accelerates a 7.0 kg metal ball in a straight line. The ball moves from rest to a speed of 12 m s^{-1} in a distance of 1.2 m.

What is the average resultant force on the metal ball?

- A 70 N B 210 N C 420 N D 840 N

167 9702/13/M/J/14/Q22

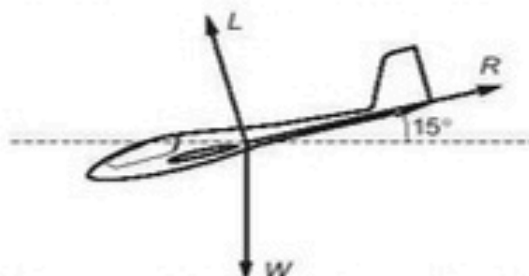
The Mariana Trench in the Pacific Ocean has a depth of about 10 km.

Assuming that sea water is incompressible and has a density of about 1020 kg m^{-3} , what would be the approximate pressure at that depth?

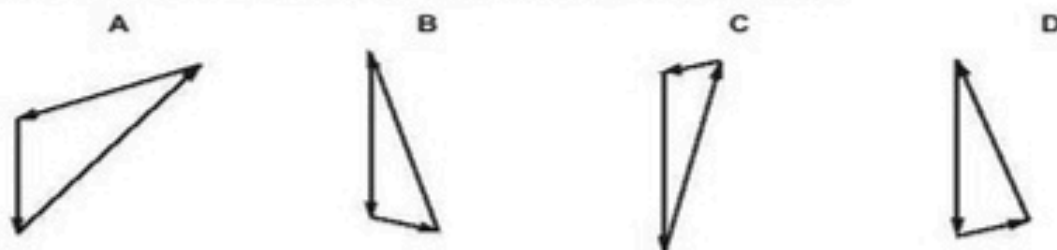
- A 10^5 Pa B 10^6 Pa C 10^7 Pa D 10^8 Pa

168 9702/12/O/N/14/Q10

A glider is descending at constant speed at an angle of 15° to the horizontal. The diagram shows the directions of the lift L , air resistance R and weight W acting on the glider.



Which vector triangle could represent the forces acting on the glider?



169 9702/12/O/N/14/Q11

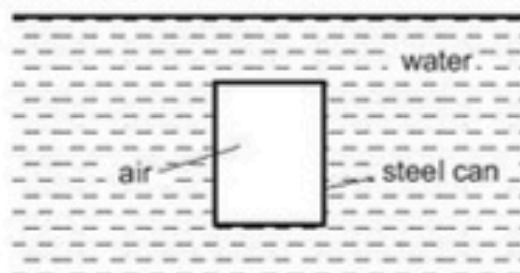
A ball is falling at terminal speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

What is the order of increasing magnitude of these three forces?

- A upthrust $\rightarrow$ viscous drag $\rightarrow$ weight
- B viscous drag $\rightarrow$ upthrust $\rightarrow$ weight
- C viscous drag $\rightarrow$ weight $\rightarrow$ upthrust
- D weight $\rightarrow$ upthrust $\rightarrow$ viscous drag

170 9702/13/O/N/14/Q12

A sealed cylindrical steel can is situated below the surface of water.



What is the origin of the upthrust that acts on the can?

- A The air pressure in the can is less than the water pressure outside the can.
- B The average density of the air and steel is less than the density of water.
- C The water pressure on the bottom of the can is greater than the water pressure on the top.
- D The weight of displaced water acts upwards on the can.

171 9702/12/M/J/10/Q18

Atmospheric pressure at sea level has a value of 100kPa.

The density of sea water is 1020 kg m^{-3} .

At which depth in the sea would the total pressure be 110kPa?

- A 1.0 m
- B 9.8 m
- C 10 m
- D 11 m

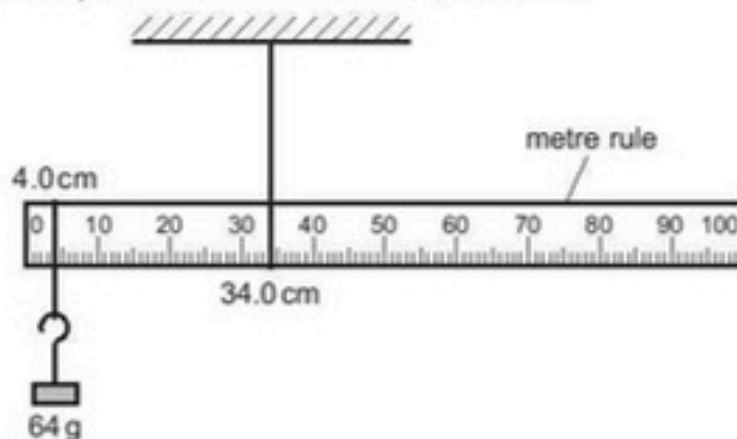
172 9702/11/M/J/15/Q14

What is the **definition** of the force on a body?

- A the mass of the body multiplied by its acceleration
- B the power input to the body divided by its velocity
- C the rate of change of momentum of the body
- D the work done on the body divided by its displacement

173 9702/11/M/J/15/Q15

A uniform metre rule is pivoted at the 34.0cm mark, as shown.



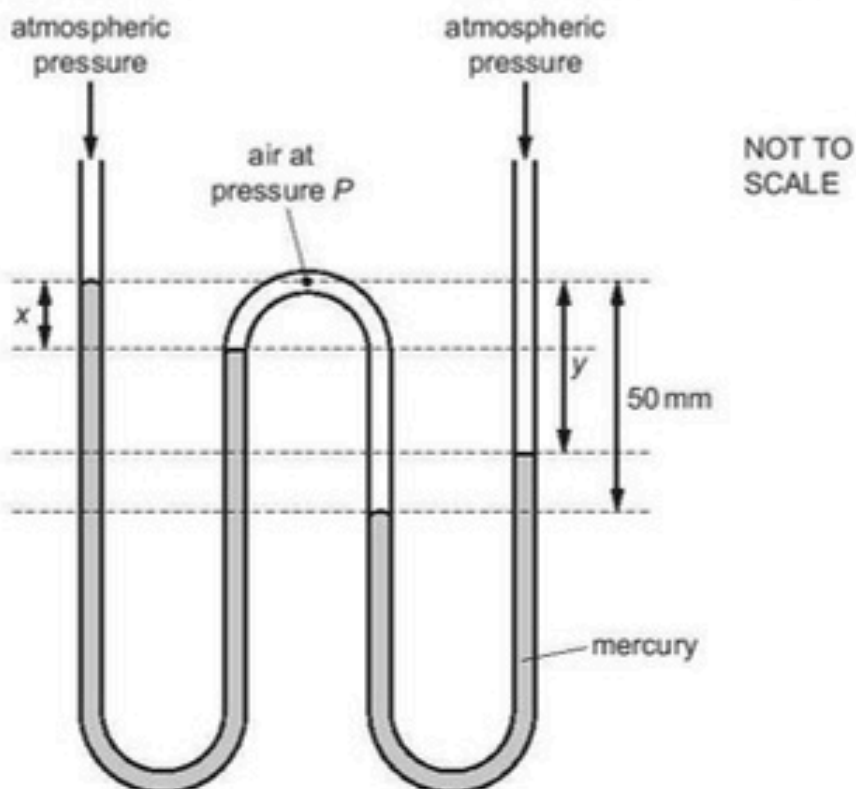
The rule balances when a 64 g mass is hung from the 4.0 cm mark.

What is the mass of the metre rule?

- A 38 g
- B 44 g
- C 120 g
- D 136 g

174 9702/13/M/J/15/Q21

A W-shaped tube contains two amounts of mercury, each open to the atmosphere. Air at pressure P is trapped in between them. The diagram shows two vertical distances x and y .



Atmospheric pressure is equal to the pressure that would be exerted by a column of mercury of height 760 mm. The pressure P is expressed in this way.

Which values of x , y and P are possible?

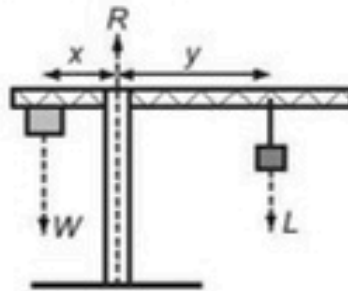
	x/mm	y/mm	$P/\text{mm of mercury}$
A	20	20	780
B	20	30	780
C	30	20	810
D	30	30	790

175 9702/12/M/J/17/Q13

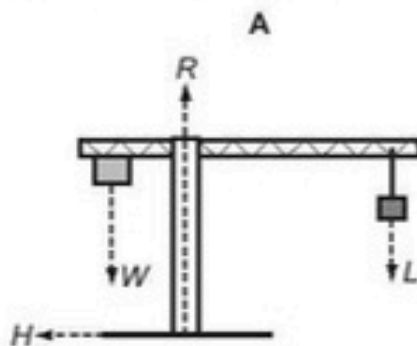
What are the SI base units of the quantity $\frac{\text{pressure}}{\text{density}}$?

- A s^{-2} B $\text{kg}^2 \text{s}^{-2}$ C $\text{kg}^2 \text{m}^2 \text{s}^{-2}$ D $\text{m}^2 \text{s}^{-2}$

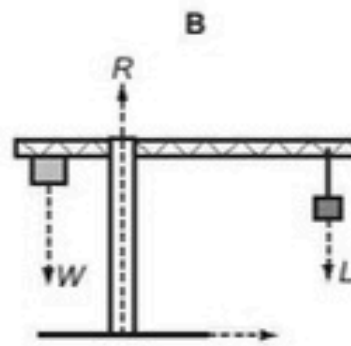
The diagram shows a crane supporting a load L .



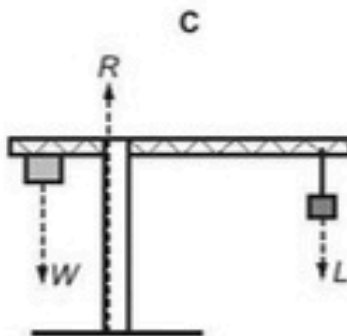
A mass provides a balancing load W . The position of the load is such that the system is perfectly balanced with $Wx = Ly$. The ground provides a reaction force R . The distance x does not change. If the load is moved further out so that the distance y increases and the crane does not topple, which statement is correct?



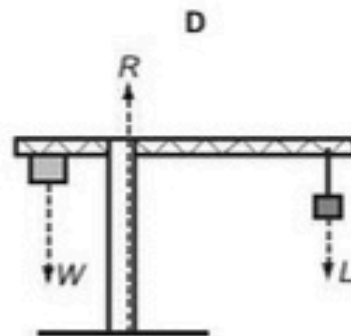
A horizontal force H acts on the base of the support column towards the left.



A horizontal force H acts on the base of the support column towards the right.



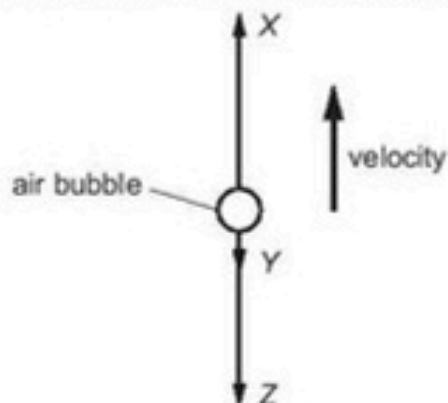
The reaction force R moves to the left.



The reaction force R moves to the right.

177 9702/13/M/J/15/Q15

An air bubble in a tank of water is rising with constant velocity. The forces acting on the bubble are X, Y and Z as shown.

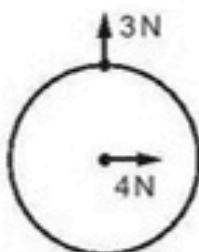


What describes the three forces?

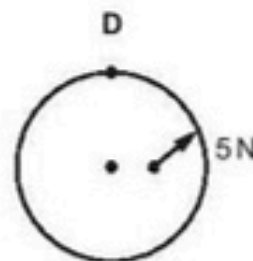
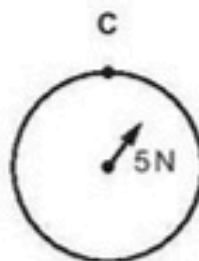
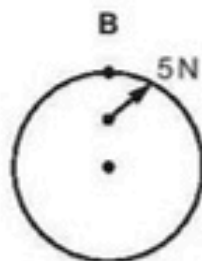
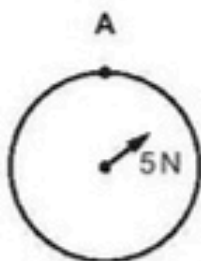
- A Z is the viscous drag on the bubble, Y is the weight of the bubble, X is the upthrust on the bubble and $X = Y + Z$.
- B Z is the viscous drag on the bubble, Y is the weight of the bubble, X is the upthrust on the bubble and $X > Y + Z$.
- C Z is the weight of the bubble, Y is the viscous drag on the bubble, X is the upthrust on the bubble and $X = Y + Z$.
- D Z is the weight of the bubble, Y is the viscous drag on the bubble, X is the upthrust on the bubble and $X > Y + Z$.

178 9702/12/M/J/13/Q3

Two forces act on a circular disc as shown.

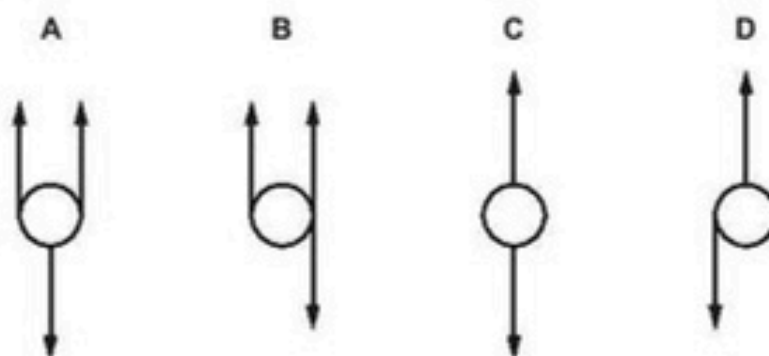


Which diagram shows the line of action of the resultant force?

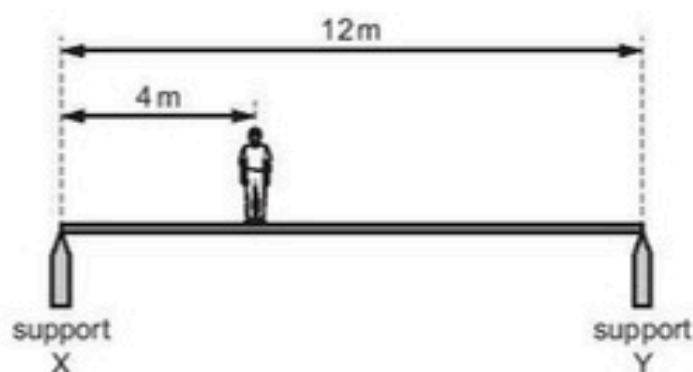


- 179 9702/11/M/J/17/Q12
A sphere is acted upon by various forces, all of the same magnitude.

Which system of forces provides a resultant torque but zero resultant force on the sphere?



- 180 9702/11/M/J/17/Q13
A uniform horizontal footbridge is 12m long and weighs 4000 N. It rests on two supports X and Y as shown.



A man of weight 600 N is a distance of 4 m from support X.

What is the upward force on the footbridge from support X?

- A 2200 N B 2300 N C 2400 N D 2600 N

- 181 9702/11/M/J/17/Q14
A metal block has a mass of 750 g. 60% of the mass is magnesium and the remainder is copper.

The density of magnesium is 1.7 g cm^{-3} .

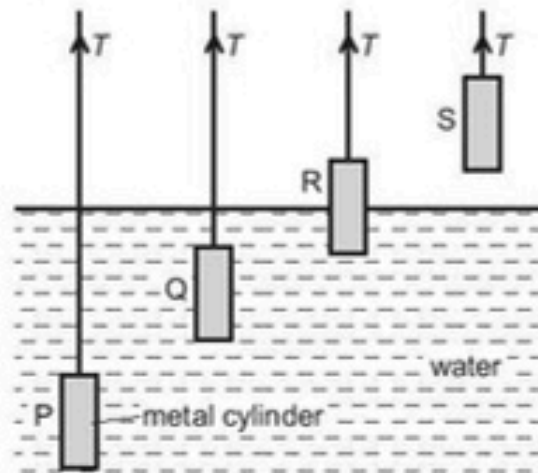
The density of copper is 9.0 g cm^{-3} .

What is the density of the block?

- A 2.5 g cm^{-3} B 4.6 g cm^{-3} C 5.4 g cm^{-3} D 10.7 g cm^{-3}

182 9702/12/M/J/17/Q10

A metal cylinder is suspended vertically in equilibrium by a cord. The diagram shows the cylinder in four different positions P, Q, R and S.

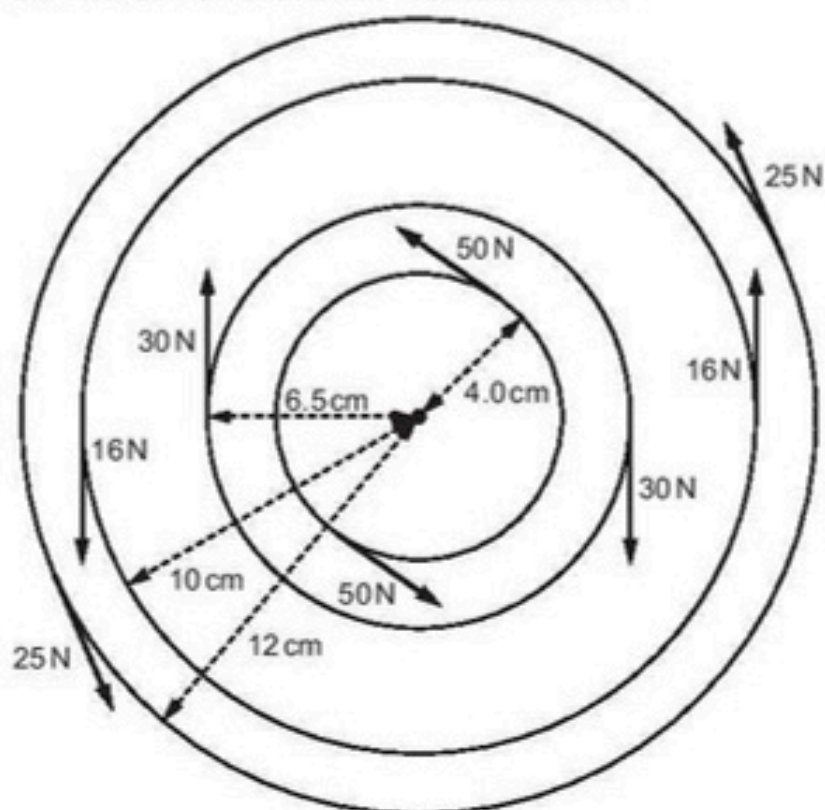


Which statement explains the variation of the tension T in the cord?

- A At P and at Q, the tension T in the cord is the same because the difference in pressure between the top and bottom of the cylinder is the same.
- B At Q, the tension T in the cord is less than at P because, at smaller depth, liquid pressure is smaller.
- C At R, the tension T in the cord is less than at P because atmospheric pressure is less than water pressure.
- D At S, the tension T in the cord is greater than at P because atmospheric pressure at S exerts no force on the top or bottom of the cylinder.

183 9702/12/M/J/17/Q11

In a machine, many couples act on a rotating object as shown.



What is the resultant torque acting on the rotating object?

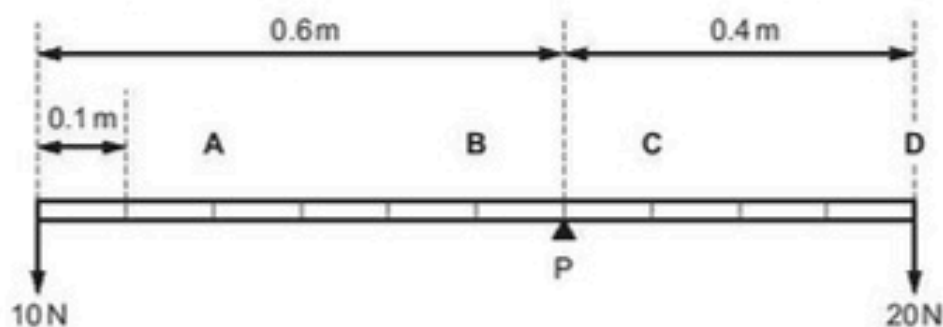
- A 4.7 Nm B 8.6 Nm C 9.3 Nm D 17.1 Nm

184 9702/12/M/J/17/Q12

A uniform beam is pivoted at P as shown. Weights of 10 N and 20 N are attached to its ends.

The length of the beam is marked at 0.1 m intervals. The weight of the beam is 100 N.

At which point should a further weight of 20 N be attached to achieve equilibrium?



185 9702/12/M/J/17/Q13

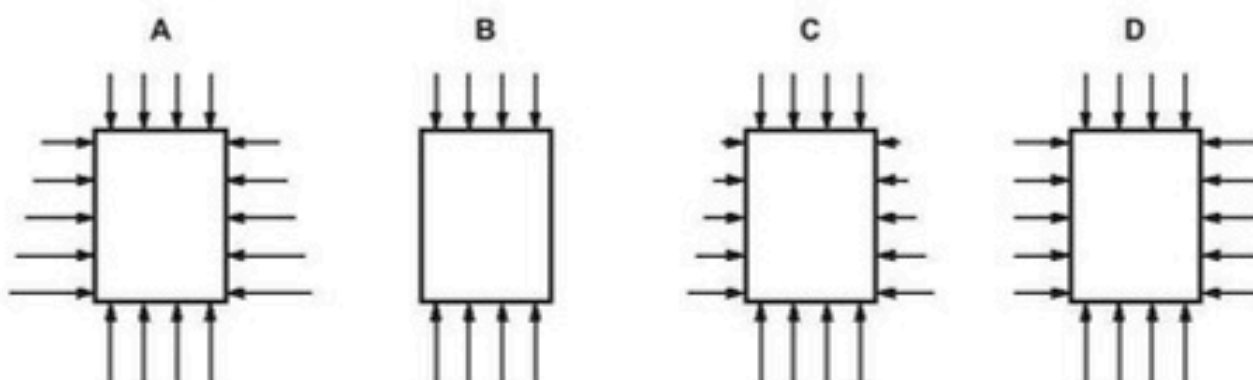
What are the SI base units of the quantity $\frac{\text{pressure}}{\text{density}}$?

- A s^{-2} B $\text{kg}^2 \text{s}^{-2}$ C $\text{kg}^2 \text{m}^2 \text{s}^{-2}$ D $\text{m}^2 \text{s}^{-2}$

186 9702/11/O/N/17/Q12

A block is submerged vertically in a liquid. The four diagrams show, to scale, the forces exerted by the liquid on the block.

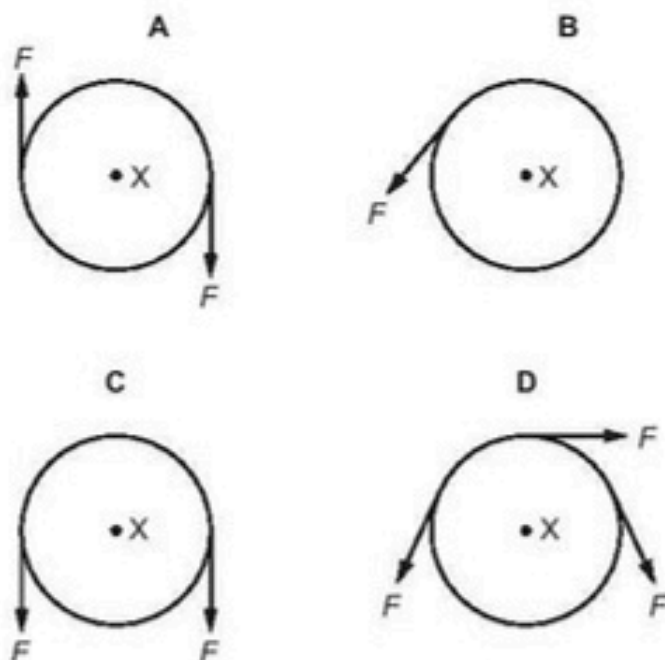
Which diagram correctly shows a possible situation as viewed from the side?



187 9702/11/O/N/17/Q13

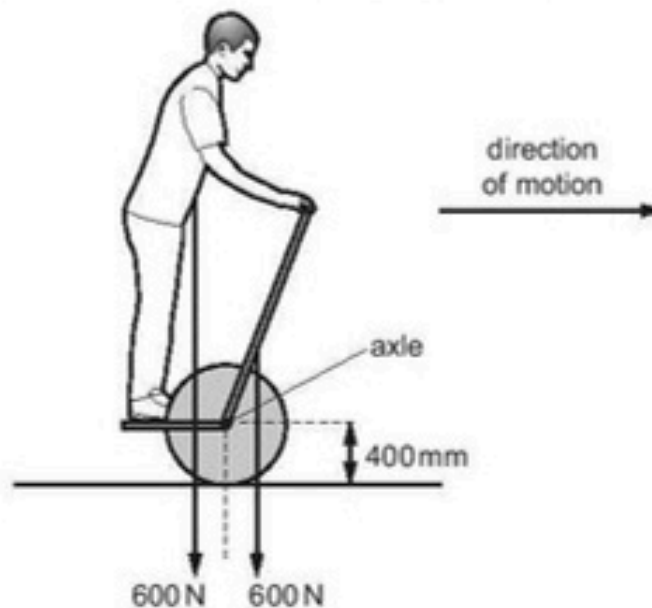
A rigid circular disc of radius r has its centre at X . A number of forces of equal magnitude F act at the edge of the disc. All the forces are in the plane of the disc.

Which arrangement of forces provides a moment of magnitude $2Fr$ about X ?



188 9702/11/O/N/17/Q14

The diagram shows a motorised vehicle for carrying one person.



The vehicle has two wheels on one axle. The passenger stands on a platform between the wheels.

The weight of the machine is 600 N. Its centre of mass is 200 mm in front of the axle. The wheel radius is 400 mm.

When stationary, a passenger of weight 600 N stands with his centre of mass 200 mm behind the axle to balance the machine.

The motor is now switched on to provide a horizontal force of 90 N at the ground to move the vehicle forwards.

How far and in which direction must the passenger move his centre of mass to maintain balance?

- | | |
|--------------------------|---------------------------|
| A 60 mm backwards | C 140 mm backwards |
| B 60 mm forwards | D 140 mm forwards |

189 9702/11/O/N/17/Q15

The derivation of the pressure equation $\Delta p = \rho g \Delta h$ uses a number of relationships between quantities.

Which relationship is **not** used in the derivation of this equation?

- | | |
|--|---|
| A density = $\frac{\text{mass}}{\text{volume}}$ | B potential energy = mass $\times$ acceleration of free fall $\times$ height |
| C pressure = $\frac{\text{force}}{\text{area}}$ | D weight = mass $\times$ acceleration of free fall |

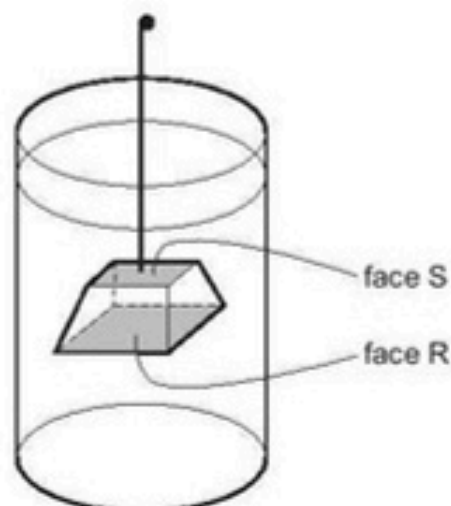
190 9702/12/O/N/17/Q11

The diagram shows a block of copper suspended in water.

The block experiences an upthrust from the water.

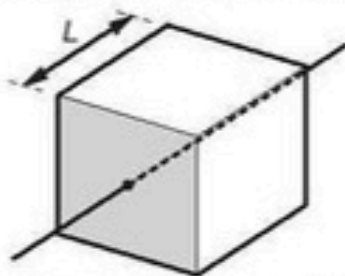
Which statement is the basis of an explanation for this upthrust?

- A Copper is more dense than water.
- B The area of face R is greater than the area of face S.
- C The density of water increases with depth.
- D The pressure of water increases with depth.

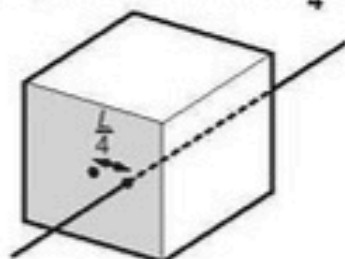


191 9702/12/O/N/17/Q12

The diagram shows a solid cube with weight W and sides of length L . It is supported at rest by a frictionless spindle that passes through the centres of two opposite vertical faces. One of these faces is shaded.



The spindle is now removed and replaced at a distance $\frac{L}{4}$ to the right of its original position.

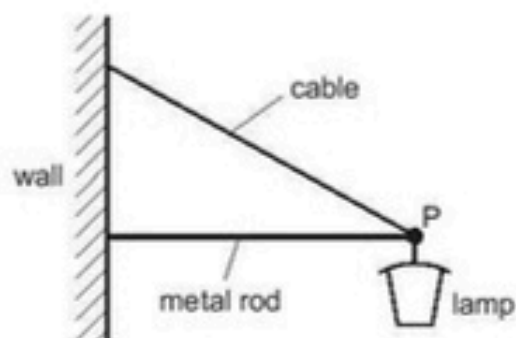


When viewing the shaded face, what is the torque of the couple that will now be needed to keep the cube at rest?

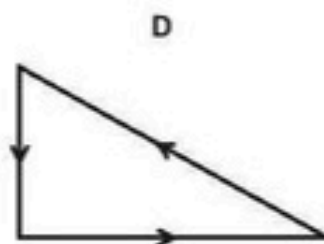
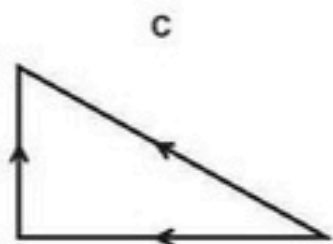
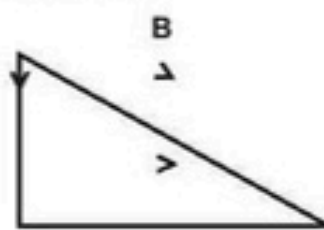
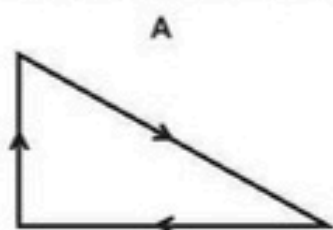
- | | |
|--------------------------------|----------------------------|
| A $\frac{WL}{4}$ anticlockwise | B $\frac{WL}{4}$ clockwise |
| C $\frac{WL}{2}$ anticlockwise | D $\frac{WL}{2}$ clockwise |

192 9702/12/O/N/17/Q13

A street lamp is fixed to a wall by a metal rod and a cable.

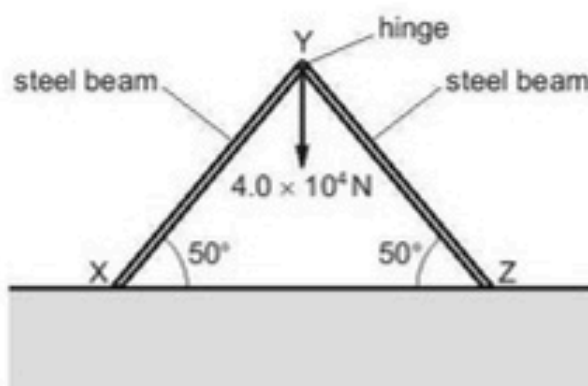


Which vector triangle could represent the forces acting at point P?



193 9702/12/O/N/17/Q14

Two rigid steel beams XY and YZ are fixed at their lower ends and are hinged at Y. Each beam is inclined at 50° to the horizontal, as shown. A weight of $4.0 \times 10^4 \text{ N}$ hangs from Y. The structure is in equilibrium.

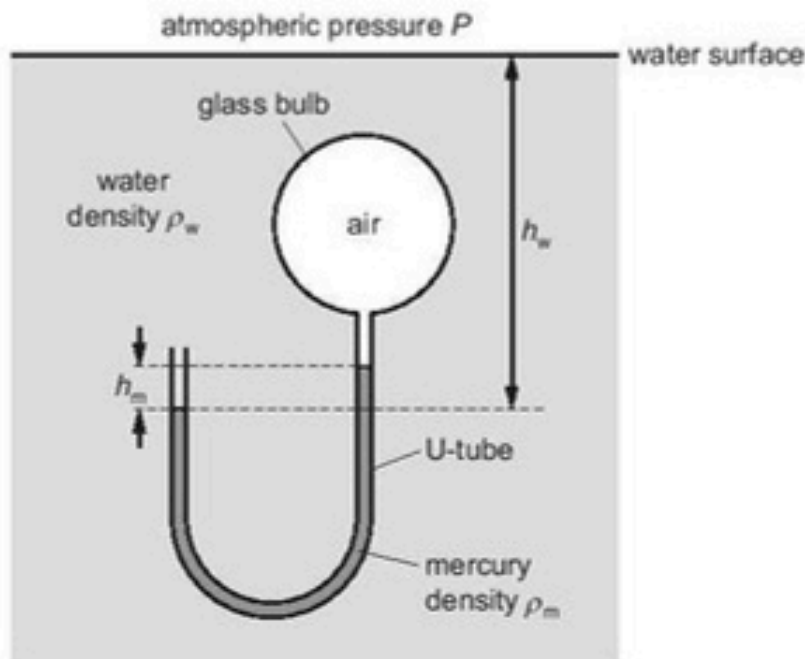


What is the force exerted by each beam on the hinge at Y?

- A $2.6 \times 10^4 \text{ N}$ B $3.1 \times 10^4 \text{ N}$ C $5.2 \times 10^4 \text{ N}$ D $6.2 \times 10^4 \text{ N}$

194 9702/12/O/N/17/Q15

Air is trapped inside a glass bulb which is immersed in water and attached to a U-tube containing mercury. The densities of water and mercury are ρ_w and ρ_m respectively. The surface of the water is open to the atmosphere where atmospheric pressure is P .



The acceleration of free fall is g .

What is the pressure of the air in the glass bulb?

- A $P + g\rho_w h_w + g\rho_m h_m$
- B $P + g\rho_w h_w - g\rho_m h_m$
- C $g\rho_w h_w + g\rho_m h_m$
- D $g\rho_w h_w - g\rho_m h_m$

195 9702/13/O/N/17/Q14

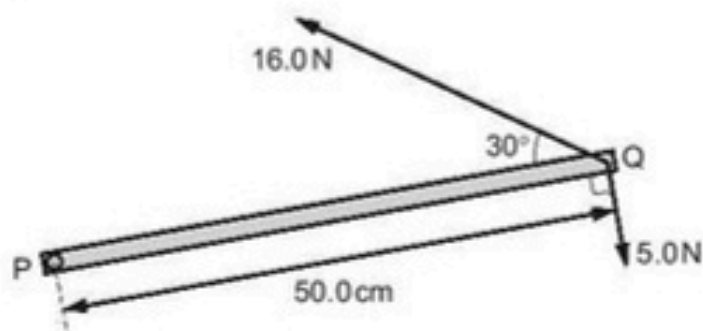
A thin horizontal plate of area 0.036 m^2 is beneath the surface of a liquid of density 930 kg m^{-3} . The force on one side of the plate due to the pressure of the liquid is 290 N .

What is the depth of the plate beneath the surface of the liquid?

- A 0.88 m
- B 1.1 m
- C 1.8 m
- D 8.7 m

196 9702/11/M/J/18/Q10

A horizontal metal bar PQ of length 50.0 cm is hinged at end P. The diagram shows the metal bar viewed from above.



Two forces of 16.0 N and 5.0 N are in the horizontal plane and act on end Q as shown in the diagram.

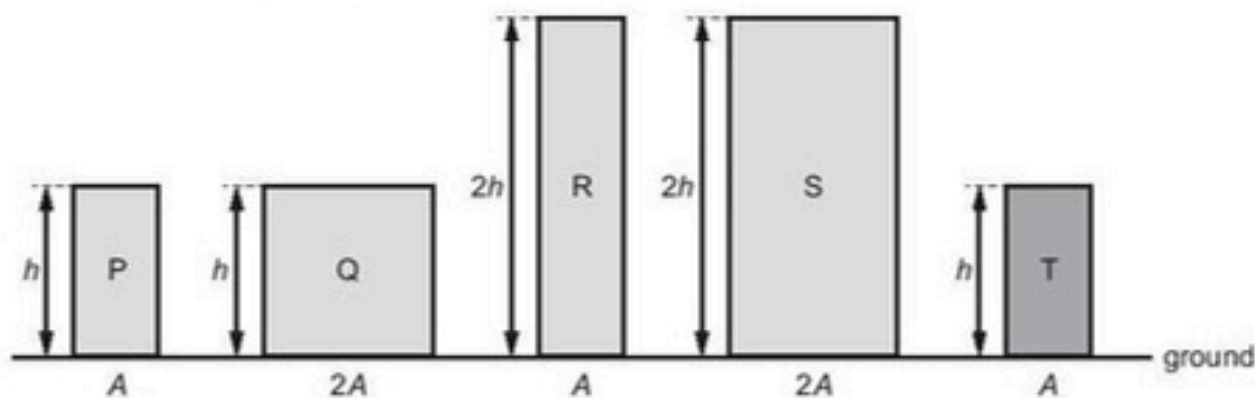
What is the total moment about P due to the two forces?

- A 1.5 Nm B 4.4 Nm C 6.5 Nm D 9.4 Nm

197 9702/11/M/J/18/Q11

Blocks P, Q, R and S are made from material of the same density. Block T is made from a material of twice the density of the material of the other blocks.

The cross-sectional area and height of each of the blocks are shown.



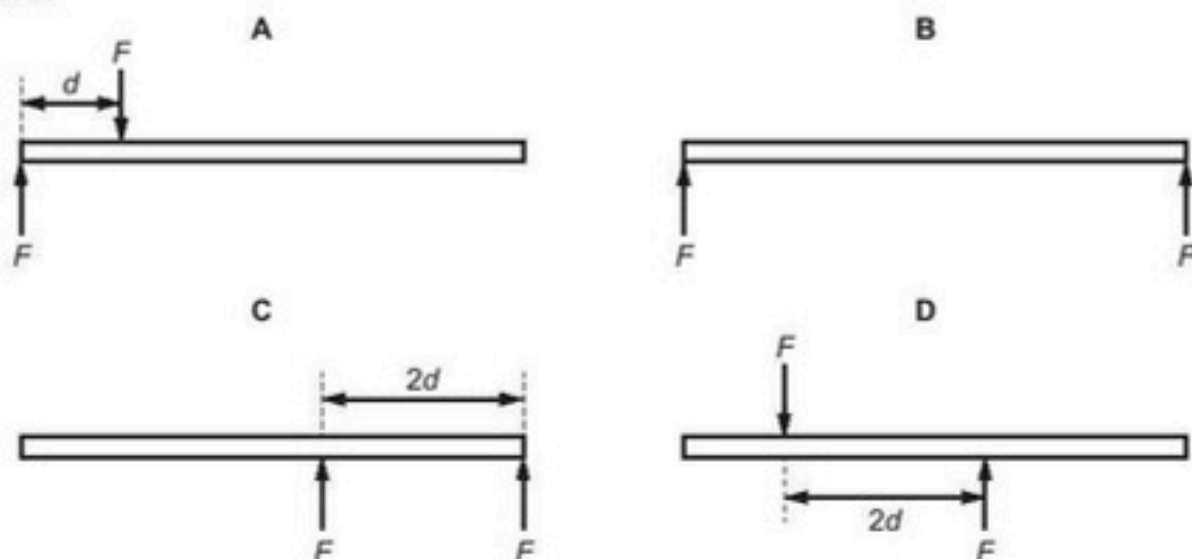
Which two blocks exert the same pressure on the ground?

- A P and T B Q and R C Q and S D S and T

198 9702/11/M/J/18/Q12

Two parallel forces, each of magnitude F , act on a rod of length $5d$.

Which diagram shows the positions of the two forces that will produce the largest torque on the rod?



199 9702/11/M/J/18/Q13

Liquids X and Y are stored in large open tanks. Liquid X has a density of 800 kg m^{-3} and liquid Y has a density of 1200 kg m^{-3} .

At which depths are the pressures equal?

	depth in liquid X / m	depth in liquid Y / m
A	8	20
B	10	15
C	15	10
D	20	8

200 9702/12/M/J/18/Q10

The density of water is 1.0 g cm^{-3} and the density of glycerine is 1.3 g cm^{-3} .

Water is added to a measuring cylinder containing 40 cm^3 of glycerine so that the density of the mixture is 1.1 g cm^{-3} . Assume that the mixing process does not change the total volume of the liquid.

What is the volume of water added?

- A 40 cm^3 B 44 cm^3 C 52 cm^3 D 80 cm^3

201 9702/12/M/J/18/Q11

An astronaut throws a stone horizontally near to the surface of the Moon, where there is no atmosphere.

Which row describes the horizontal and vertical forces acting on the stone after release?

	horizontal force	vertical force
A	non-zero and constant	constant
B	non-zero and constant	decreasing
C	zero	constant
D	zero	decreasing

202 9702/12/M/J/18/Q12

A cylindrical block of wood has cross-sectional area A and weight W . It is totally immersed in water with its axis vertical. The block experiences pressures p_1 and p_2 at its top and bottom surfaces respectively.

Which expression is equal to the upthrust on the block?

- A $(p_2 - p_1)$ C $(p_2 - p_1)A - W$
 B $(p_2 - p_1)A$ D $(p_2 - p_1)A + W$

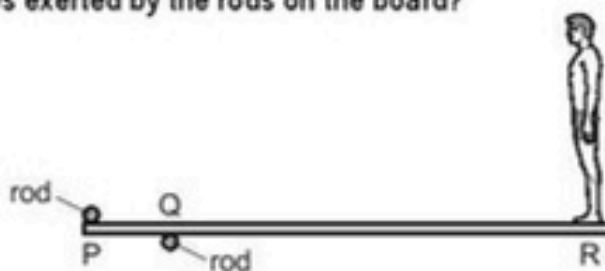
203 9702/12/M/J/18/Q13

A uniform diving-board is held by two fixed rods at points P and Q. A person stands at end R of the diving-board, as shown.

The forces exerted by the rods on the board are vertical. The board remains in equilibrium as the person slowly moves towards point Q from end R.

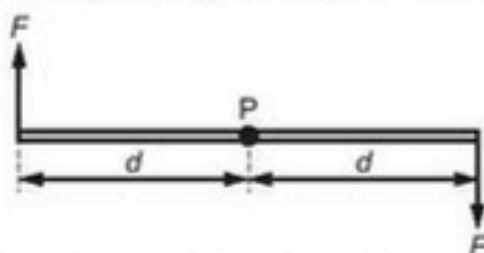
Which row describes the changes to the forces exerted by the rods on the board?

	force at P	force at Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases



204 9702/12/M/J/18/Q14

Two forces, each of magnitude F , act in opposite directions on a rod.



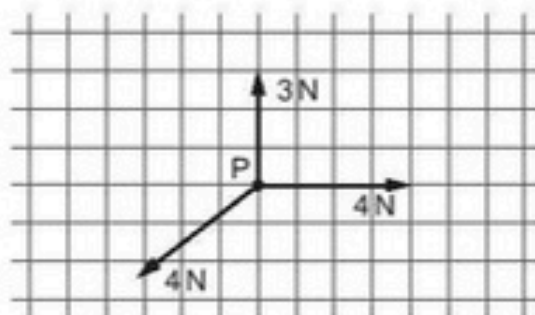
Each force acts on the rod at a distance d from the pivot P .

What is the torque of this couple about P ?

- A 0 B $F \times d$ C $2F \times d$ D $2F \times 2d$

205 9702/12/M/J/18/Q15

The vector diagram shows three coplanar forces acting on an object at P .



The magnitude of the resultant of these three forces is 1 N.

What is the direction of this resultant force?

- A B C D

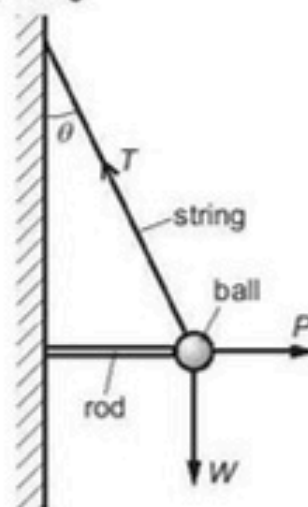
206 9702/12/M/J/18/Q17

The diagram shows a ball of weight W hanging in equilibrium from a string.

The string is at an angle θ to the vertical. The tension in the string is T . The ball is held away from the wall by a horizontal force P from a metal rod.

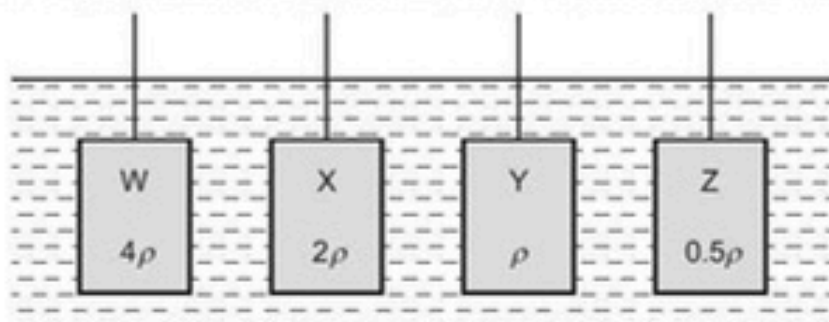
What is the relationship between the magnitudes of T , P and W ?

- A $P = T \cos \theta$ and $W = T \sin \theta$
 B $T = P + W$
 C $T^2 = P^2 + W^2$
 D $W = P \tan \theta$ and $W = T \cos \theta$



207 9702/13/M/J/18/Q10

Four cuboids with identical lengths, breadths and heights are immersed in water. The cuboids are held at the same depth and in identical orientations by vertical rods, as shown.



Water has density ρ .

Cuboid W is made of material of density 4ρ .

Cuboid Y is made of material of density ρ .

Cuboid X is made of material of density 2ρ .

Cuboid Z is made of material of density 0.5ρ .

Which statement is correct?

- A The upthrust of the water on each of the cuboids is the same.
- B The upthrust of the water on W is twice the upthrust of the water on X.
- C The upthrust of the water on X is twice the upthrust of the water on W.
- D The upthrust of the water on Y is zero.

208 9702/13/M/J/18/Q11

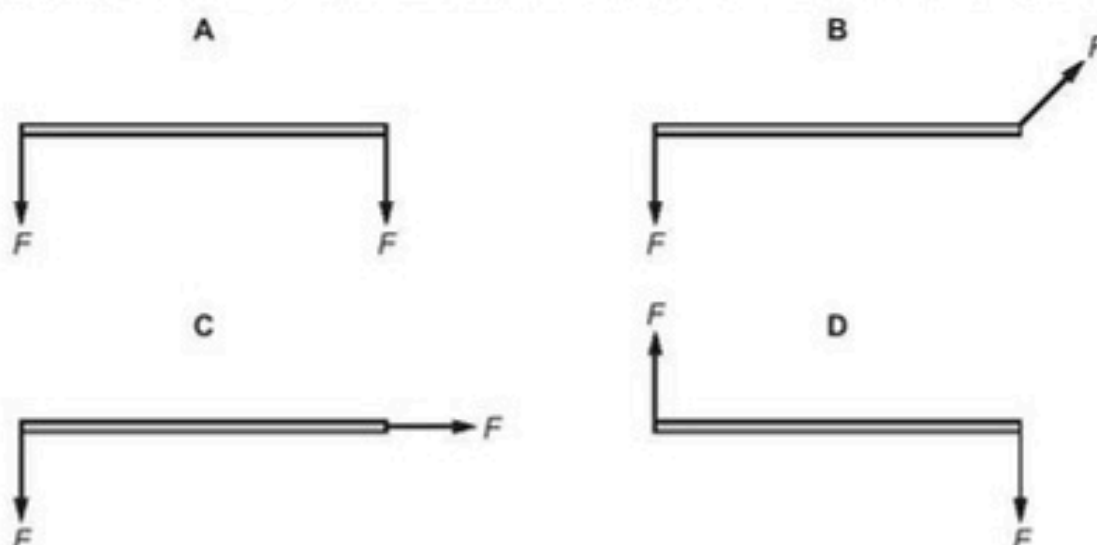
A rectangular block of lead of density $1.13 \times 10^3 \text{ kg m}^{-3}$ has sides of length 12.0 cm, 15.0 cm and 10.0 cm.

What is the maximum pressure the block can exert when resting on a table?

- A 1.13 kPa B 1.70 kPa C 11.1 kPa D 16.6 kPa

209 9702/13/M/J/18/Q12

Which diagram shows a couple formed by two forces, each of magnitude F , acting on a rod?



210 9702/13/M/J/18/Q13

Full-fat milk is made up of fat-free milk mixed with fat.

A volume of $1.000 \times 10^{-3} \text{ m}^3$ of full-fat milk has a mass of 1.035 kg. It contains 4.00% fat by volume.

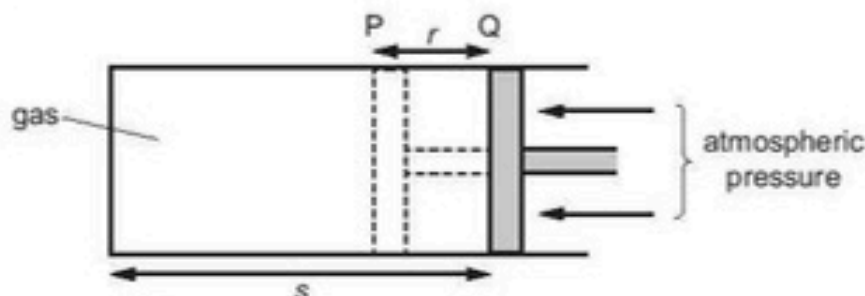
The density of fat-free milk is $1.040 \times 10^3 \text{ kg m}^{-3}$.

What is the density of fat?

- A $1.25 \times 10^2 \text{ kg m}^{-3}$ C $9.28 \times 10^2 \text{ kg m}^{-3}$
B $9.15 \times 10^2 \text{ kg m}^{-3}$ D $1.16 \times 10^3 \text{ kg m}^{-3}$

211 9702/13/M/J/18/Q14

Gas is trapped inside a cylinder by a piston of cross-sectional area A . The piston is **not** frictionless.



The gas is heated and this causes it to expand, pushing back the piston through distance r from position P to position Q. The length of the gas column is then s .

Which expression represents the amount of work done by the gas against the atmosphere during this expansion?

- A (atmospheric pressure) $\times Ar$
B (atmospheric pressure) $\times As$
C (pressure inside the gas) $\times Ar$
D (pressure inside the gas) $\times As$

212 9702/11/O/N/15/Q13

In which example is it **not** possible for the underlined body to be in equilibrium?

- A An aeroplane climbs at a steady rate.
B An aeroplane tows a glider at a constant altitude.
C A speedboat changes direction at a constant speed.
D Two boats tow a ship into harbour.

213 9702/11/O/N/18/Q8

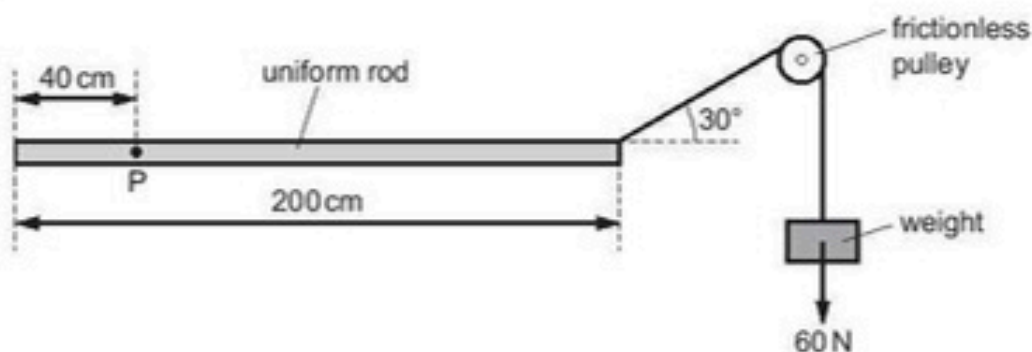
A ball of mass m is thrown vertically into the air. When the ball has speed v , the air resistance acting on the ball is F .

What is the magnitude of the acceleration of the ball when its speed is v as it rises and as it falls?

	acceleration when ball is rising	acceleration when ball is falling
A	$g - \frac{F}{m}$	$g - \frac{F}{m}$
B	$g - \frac{F}{m}$	$g + \frac{F}{m}$
C	$g + \frac{F}{m}$	$g - \frac{F}{m}$
D	$g + \frac{F}{m}$	$g + \frac{F}{m}$

214 9702/11/O/N/18/Q11

A uniform rod of length 200 cm is freely pivoted at point P. The rod is held horizontally in equilibrium by a 60 N weight that is attached to the rod by a string passing over a frictionless pulley.

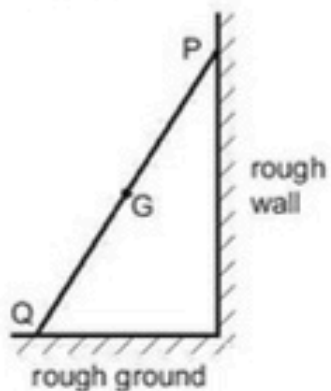


What is the weight of the rod?

- A 30 N B 60 N C 80 N D 140 N

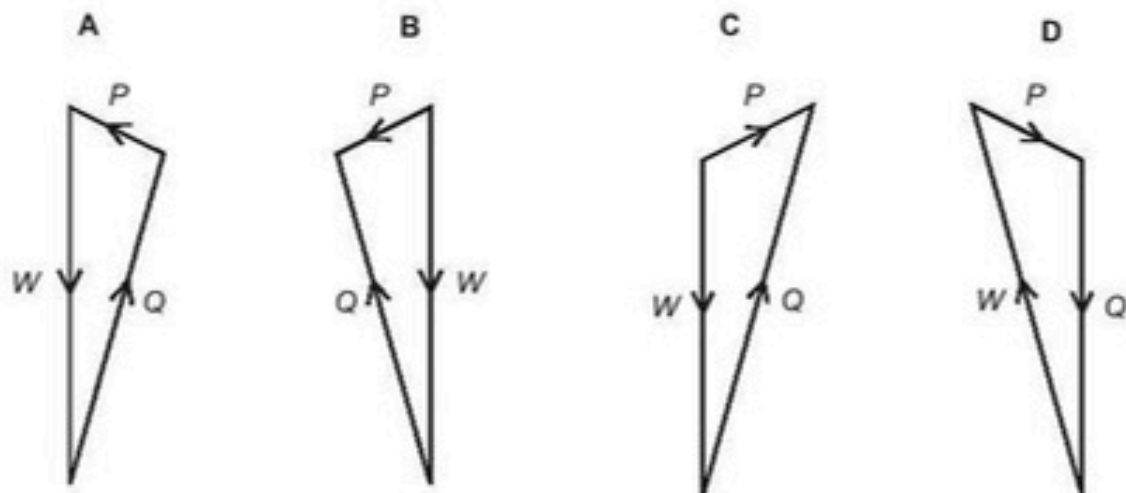
215 9702/11/O/N/18/Q12

A ladder rests in equilibrium on rough ground against a rough wall.



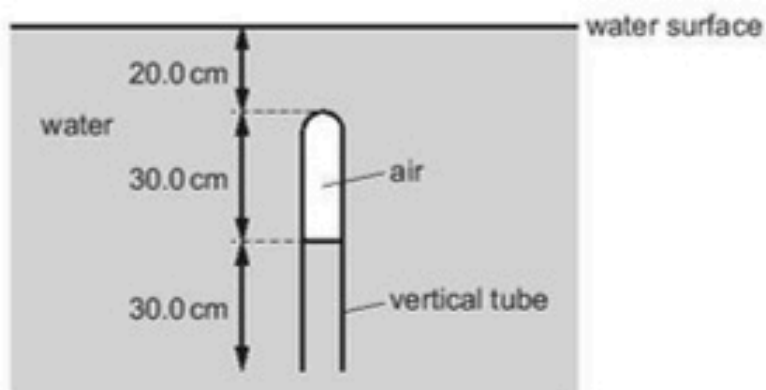
The weight W of the ladder acts through the centre of gravity G . Forces also act on the ladder at P and at Q . These forces are P and Q respectively.

Which vector triangle represents the forces on the ladder?



216 9702/11/O/N/18/Q13

A vertical tube, closed at one end, is immersed in water. A column of air is trapped inside the tube.



The density of water is 1000 kg m^{-3} .

What is the difference between the pressure of the air in the tube and the atmospheric pressure?

- A** 1960 Pa **B** 2940 Pa **C** 4910 Pa **D** 7850 Pa

217 9702/12/O/N/18/Q8

The acceleration of free fall on the surface of planet P is one tenth of that on the surface of planet Q.

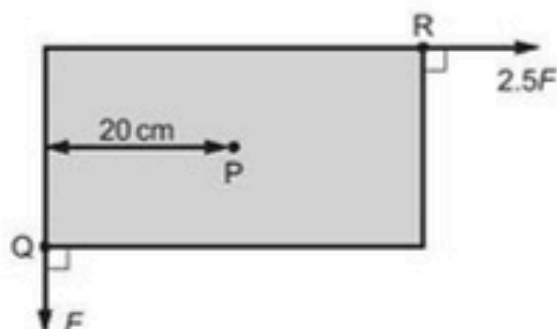
On the surface of P, a body has a mass of 1.0 kg and a weight of 1.0 N.

What are the mass and the weight of the same body on the surface of planet Q?

	mass on Q / kg	weight on Q / N
A	1.0	0.1
B	1.0	10
C	10	10
D	10	100

218 9702/12/O/N/18/Q12

A uniform rectangular board is supported by a frictionless pivot at its centre point P.



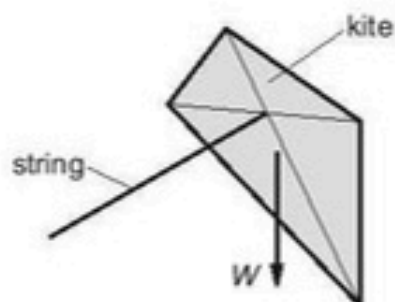
Two forces act in the plane of the board. Force F acts at corner Q and force $2.5F$ acts at corner R. The perpendicular distance between the line of action of force F and point P is 20 cm. The board is in equilibrium.

What is the area of the board?

- A 160 cm^2 B 320 cm^2 C 640 cm^2 D 1600 cm^2

219 9702/12/O/N/18/Q13

A kite is in equilibrium at the end of a string, as shown.



The kite has three forces acting on it: the weight W , the tension T in the string, and the force F from the wind.

Which vector diagram represents the forces acting on the kite?



220 9702/12/O/N/18/Q14

The density of the air in the atmosphere decreases as the height h above the surface of the Earth increases.

Which graph best shows the variation with height h of the pressure p of the air?



221 9702/13/O/N/18/Q10

A ship of mass $8.4 \times 10^7 \text{ kg}$ is approaching a harbour with speed 16.4 ms^{-1} . By using reverse thrust it can maintain a constant total stopping force of $920\,000 \text{ N}$.

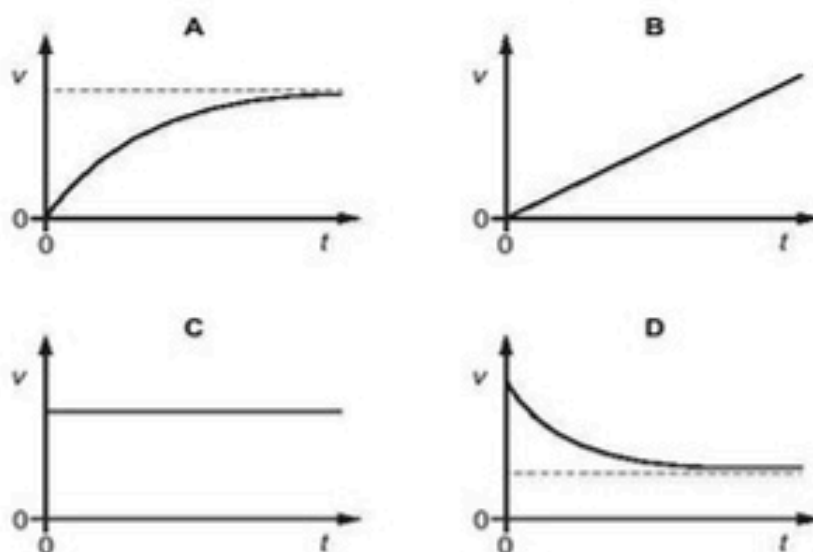
How long will it take to stop?

- A 15 seconds
- B 150 seconds
- C 25 minutes
- D 250 minutes

222 9702/13/O/N/18/Q11

A rigid, hollow sphere is immersed deep in water and released from rest. It experiences an upthrust which propels it towards the surface of the water.

Which graph best shows the variation with time t of its upward velocity v ?

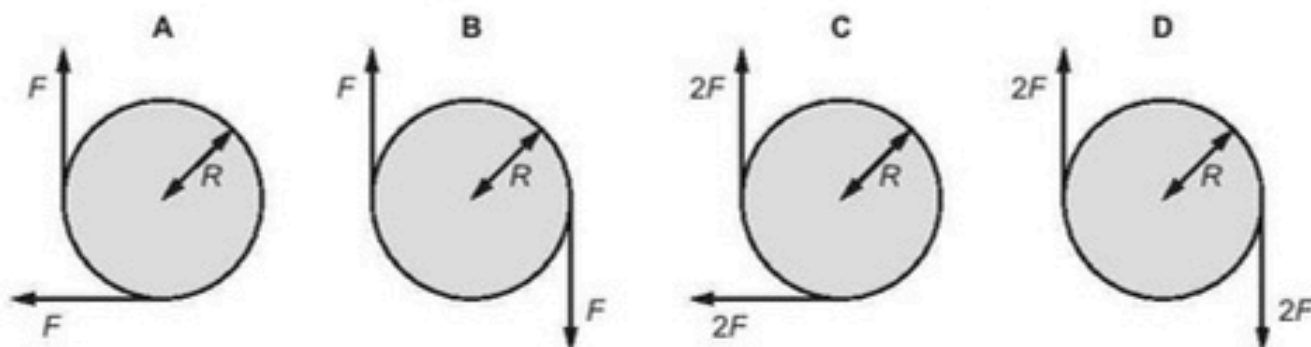


223 9702/13/O/N/18/Q12

A flat metal disc has radius R .

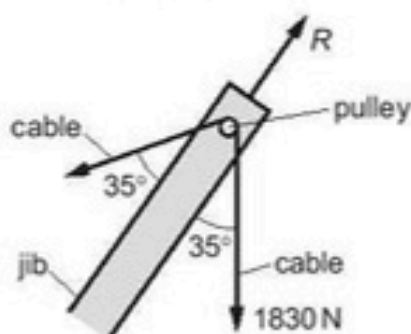
Forces of magnitude F are applied tangentially at the edge of the disc. The forces are in the plane of the disc.

Which arrangement of forces produces only a torque of magnitude $2FR$?



224 9702/13/O/N/18/Q13

The diagram shows the jib of a crane at an angle of 35° to the vertical. A cable passes over a frictionless pulley and carries a load of 1830 N.



The force R that the pulley exerts on the cable is in line with the jib. The cable and the pulley are in equilibrium.

What is the value of R ?

- A 1000 N B 1500 N C 2100 N D 3000 N

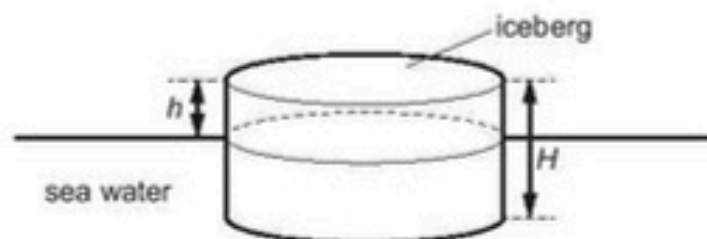
225 9702/13/O/N/18/Q14

What is a unit for density?

- A Nm^{-3} B gmm^{-1} C kgcm^{-2} D μgmm^{-3}

226 9702/11/M/J/19/Q12

A cylindrical iceberg of height H floats in sea water. The top of the iceberg is at height h above the surface of the water.



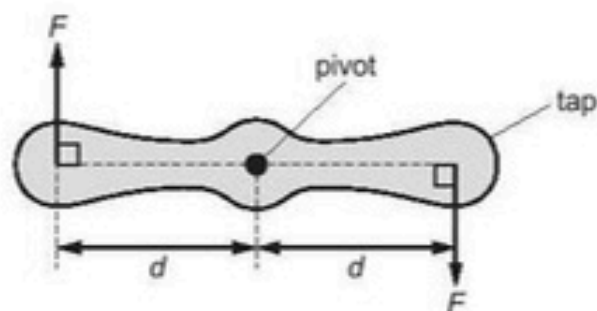
The density of ice is ρ_i and the density of sea water is ρ_w .

What is the height h of the iceberg above the sea water?

- A $\left(1 - \frac{\rho_i}{\rho_w}\right)H$ B $\left(\frac{\rho_i}{\rho_w} - 1\right)H$ C $\frac{\rho_w}{\rho_i}H$ D $\frac{\rho_i}{\rho_w}H$

227 9702/11/M/J/19/Q13

A couple is applied to a tap as shown.

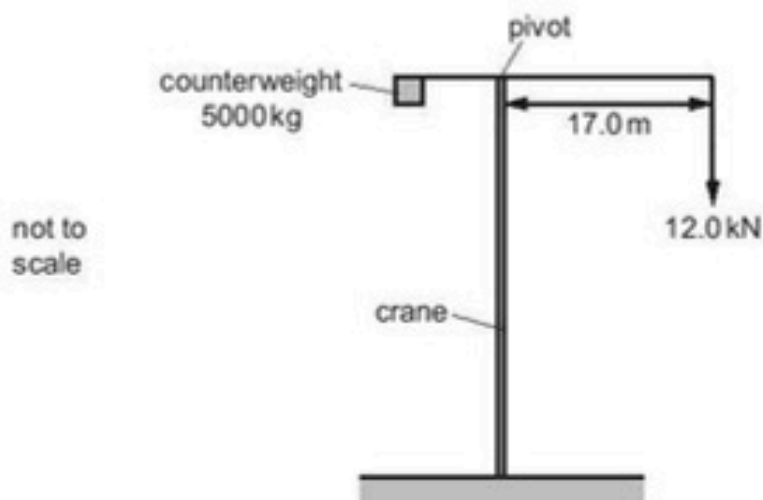


What is the torque of the couple?

- A $\frac{Fd}{2}$ B Fd C $2Fd$ D $4Fd$

228 9702/11/M/J/19/Q14

A crane uses a counterweight to stop it from toppling over when lifting a load, as shown.



The counterweight has a mass of 5000 kg. The crane is required to lift a load of 12.0 kN and the horizontal distance from the pivot to the load is 17.0 m.

How far from the pivot should the centre of gravity of the counterweight be positioned in order to keep the crane in equilibrium?

- A 0.0408 m B 0.240 m C 4.16 m D 40.8 m

229 9702/11/M/J/19/Q15

Three parallel forces act on an object. As a result of these forces, the object is in equilibrium.

What must be correct for these forces?

- A They all act along the same line.
 B They all have the same magnitude.
 C They do not all act along the same line.
 D They do not all have the same magnitude.

230 9702/11/M/J/19/Q16

An empty glass beaker has a mass of 103 g. When filled with water, it has a total mass of 361 g. When filled with cooking oil, it has a total mass of 351 g.

The density of water is 1.00 g cm^{-3} .

What is the density of the cooking oil?

- A 0.961 g cm^{-3} B 0.972 g cm^{-3} C 1.03 g cm^{-3} D 1.04 g cm^{-3}

231 9702/12/M/J/19/Q12

A submarine descends vertically at constant velocity. The three forces acting on the submarine are viscous drag, upthrust and weight.

Which relationship between their magnitudes is correct?

A weight < drag

C weight < upthrust

B weight = drag

D weight > upthrust

232 9702/12/M/J/19/Q15

A volume of 1.5 m^3 of water is mixed with 0.50 m^3 of alcohol. The density of water is 1000 kg m^{-3} and the density of alcohol is 800 kg m^{-3} .

The volume of the mixture is 2.0 m^3 .

What is the density of the mixture?

A 850 kg m^{-3}

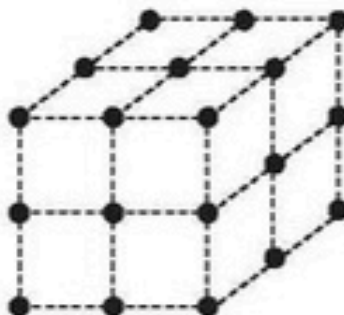
B 900 kg m^{-3}

C 940 kg m^{-3}

D 950 kg m^{-3}

233 9702/12/M/J/19/Q21

The diagram shows the arrangement of atoms in a particular crystal.



Each atom is at the corner of a cube.

The mass of each atom is $3.5 \times 10^{-25} \text{ kg}$. The density of the crystal is $9.2 \times 10^3 \text{ kg m}^{-3}$.

What is the shortest distance between the centres of two adjacent atoms?

A $3.8 \times 10^{-29} \text{ m}$

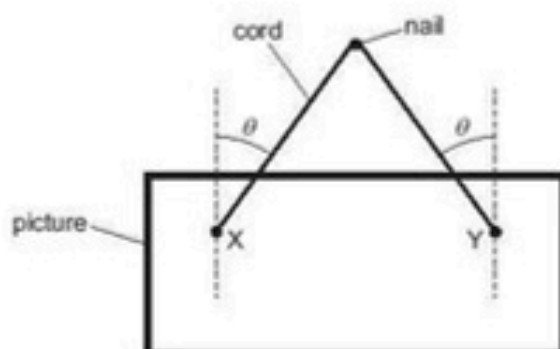
C $3.4 \times 10^{-10} \text{ m}$

B $6.2 \times 10^{-15} \text{ m}$

D $3.0 \times 10^{-9} \text{ m}$

234 9702/13/M/J/19/Q11

A picture is suspended from a nail by a single cord connected to two points X and Y on the picture. There is negligible friction between the cord and the nail so that the tension in both sides of the cord is the same. The picture hangs symmetrically, as shown.



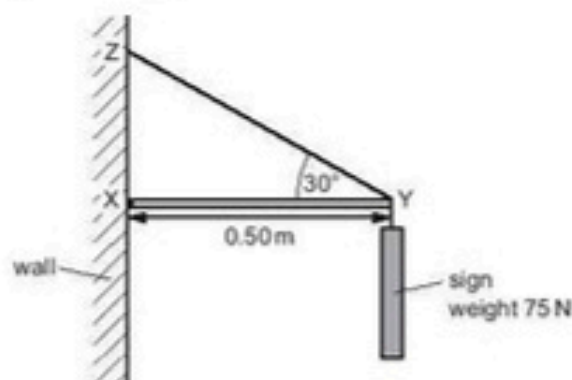
The tension in the cord is T . The angle between the cord and the vertical is θ on both sides.

Which statement is correct?

- A Increasing the length of the cord, with points X and Y in the same place on the picture, would reduce the tension in the cord.
- B Moving points X and Y further apart on the picture while keeping the length of the cord constant would reduce the tension in the cord.
- C Moving points X and Y to the top edge of the picture while keeping their distance apart constant and the length of the cord constant would reduce the tension in the cord.
- D The weight of the picture is equal to $T \cos \theta$.

235 9702/13/M/J/19/Q12

A shop sign weighing 75 N hangs from a frame attached to a vertical wall.



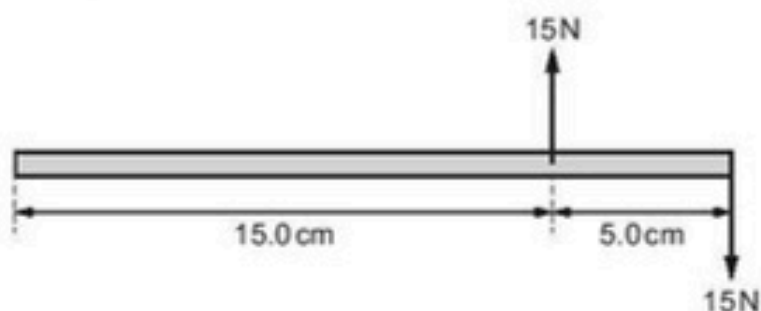
The frame consists of a horizontal rod XY and a rod YZ that is at an angle of 30° to the horizontal. Rod XY is attached to the wall by a hinge at X and has length 0.50 m. Assume that the weights of the rods are negligible.

What is the horizontal force exerted by the wall on rod XY?

- A 0 N
- B 43 N
- C 130 N
- D 150 N

236 9702/13/M/J/19/Q13

What is the torque of the couple shown?



- A 0.75 Nm B 1.50 Nm C 3.00 Nm D 5.25 Nm

237 9702/13/M/J/19/Q14

Water has a density of 1.0 g cm^{-3} .

Glycerine has a density of 1.3 g cm^{-3} .

A student measures out a volume of 40 cm^3 of glycerine into a container.

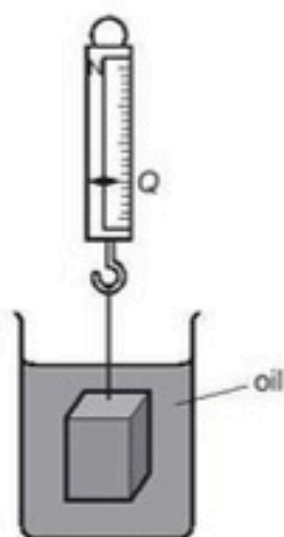
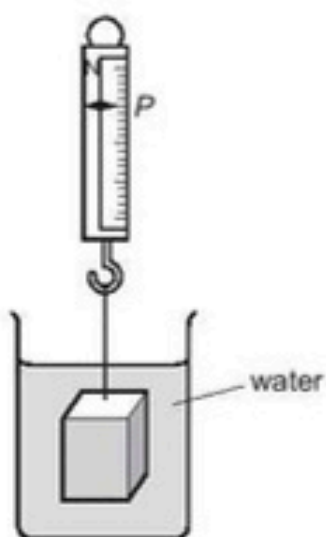
The student adds water to the container to make a mixture of water and glycerine. Assume that the total volume of water and glycerine does not change when the two liquids are mixed.

Which volume of water needs to be added to make a mixture of density 1.1 g cm^{-3} ?

- A 4.0 cm^3 B 8.0 cm^3 C 34 cm^3 D 80 cm^3

237 9702/11/O/N/19/Q11

An object of weight W is suspended from a newton meter. When the object is completely immersed in water, the newton meter reads P . When the object is completely immersed in oil, the newton meter reads Q .



What is the ratio $\frac{\text{density of oil}}{\text{density of water}}$?

A $\frac{W-P}{Q-P}$

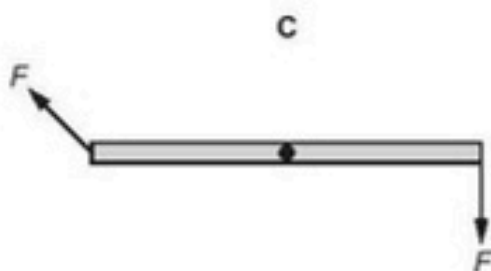
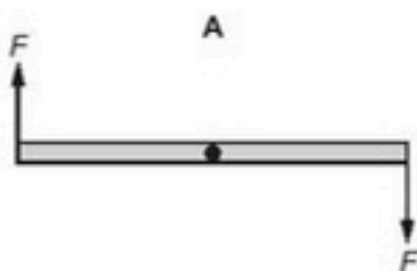
B $\frac{Q-P}{W-P}$

C $\frac{W-P}{W-Q}$

D $\frac{W-Q}{W-P}$

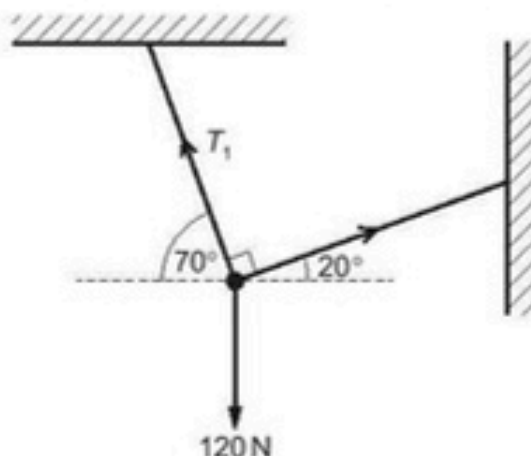
238 9702/11/O/N/19/Q12

Which diagram shows a couple?



239 9702/11/O/N/19/Q13

An object of weight 120 N is supported in equilibrium by two strings as shown.



What is the tension T_1 in the left-hand string?

- A 41 N B 77 N C 113 N D 128 N

240 9702/11/O/N/19/Q14

In a large container in an oil refinery, three oils of different densities are mixed. No chemical activity occurs.

The mixture consists of:

1200 kg of oil of density 1100 kg m^{-3}

1500 kg of oil of density 860 kg m^{-3}

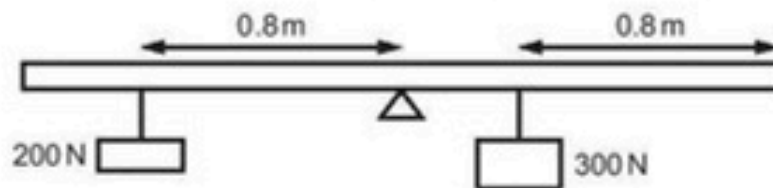
4000 kg of oil of density 910 kg m^{-3} .

What is the density of the mixture?

- A 927 kg m^{-3} B 933 kg m^{-3} C 957 kg m^{-3} D 1045 kg m^{-3}

241 9702/12/O/N/19/Q11

A rigid uniform bar of length 2.4 m is pivoted horizontally at its midpoint.



Weights are hung from two points on the bar as shown in the diagram. To maintain equilibrium, a couple is applied to the bar.

What is the torque and direction of this couple?

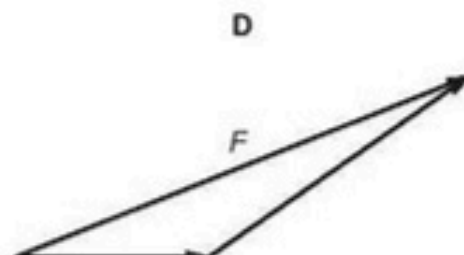
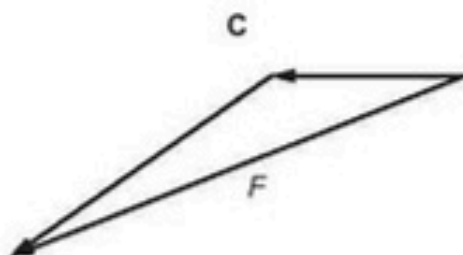
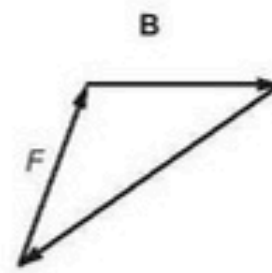
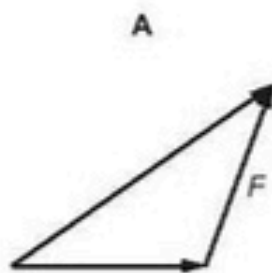
- | | |
|-----------------------|-----------------------|
| A 40 Nm clockwise | C 80 Nm clockwise |
| B 40 Nm anticlockwise | D 80 Nm anticlockwise |

242 9702/12/O/N/19/Q12

Two coplanar forces act on an object as shown.



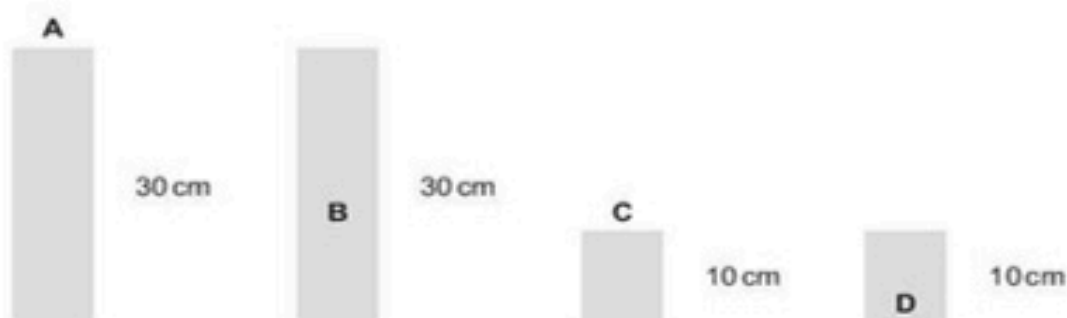
Which diagram shows the resultant F of these two forces?



243 9702/12/O/N/19/Q13

Four measuring cylinders are filled with the same liquid to the heights shown.

At which position is the pressure the greatest?



244 9702/13/O/N/19/Q10

A solid sphere, which is less dense than water, is held completely immersed in water a few metres below the surface. The density of the water is uniform.

The sphere is released. Immediately after release, the sphere rises.

Which row correctly describes the changes in the upthrust on the sphere and the resultant force on the sphere?

	upthrust on the sphere	resultant force on the sphere
A	constant	increasing
B	constant	decreasing
C	decreasing	increasing
D	decreasing	decreasing

245 9702/13/O/N/19/Q11

The diagram shows two pairs X and Y of parallel forces.

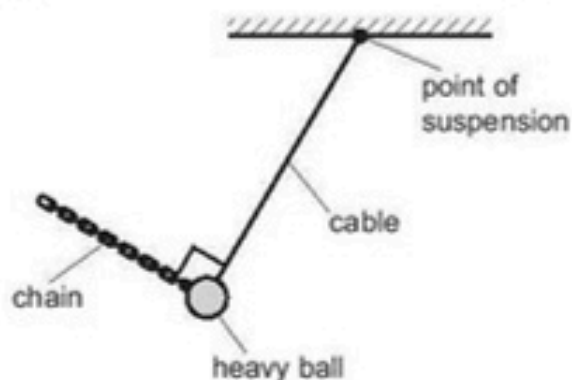


Which statement is correct?

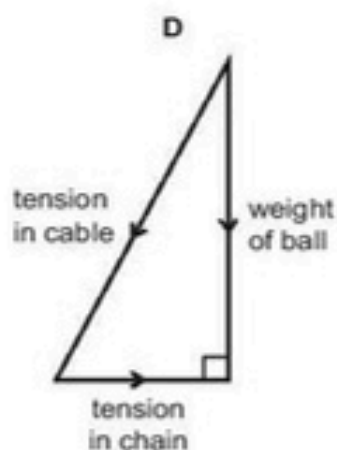
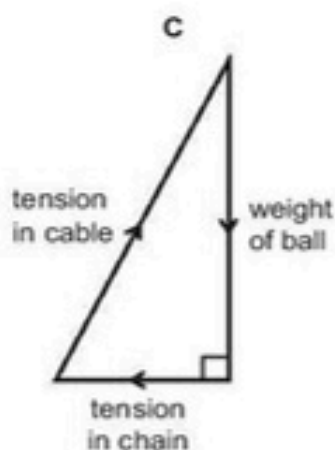
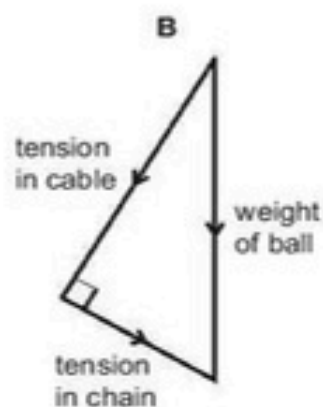
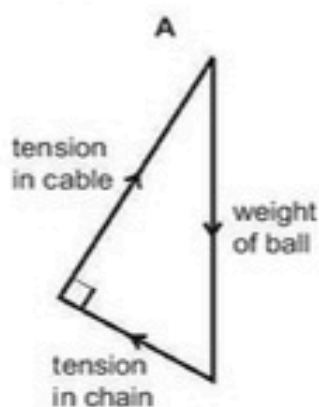
- A X is equivalent to a clockwise torque of 10 Nm and a force of 20 N to the left.
- B X is equivalent to a clockwise torque of 20 Nm only.
- C Y is equivalent to an anticlockwise torque of 30 Nm and a force of 20 N to the left.
- D Y is equivalent to an anticlockwise torque of 30 Nm only.

246 9702/13/O/N/19/Q12

A heavy ball hanging from a cable is held in equilibrium by a chain, as shown.



Which vector diagram shows the three forces acting on the ball?



247 9702/13/O/N/19/Q13

At sea level, atmospheric pressure is 100 kPa and the density of air is 1.3 kg m^{-3} .

Assume that the density of air decreases linearly with height above sea level.

What is an estimate of the total height of the atmosphere based on this information?

A 7.8 km

B 16 km

C 77 km

D 150 km

248. 9702/12/F/M/20/Q12

A uniform rod of weight 20 N and length 2.0 m is acted upon by two vertical forces, as shown.



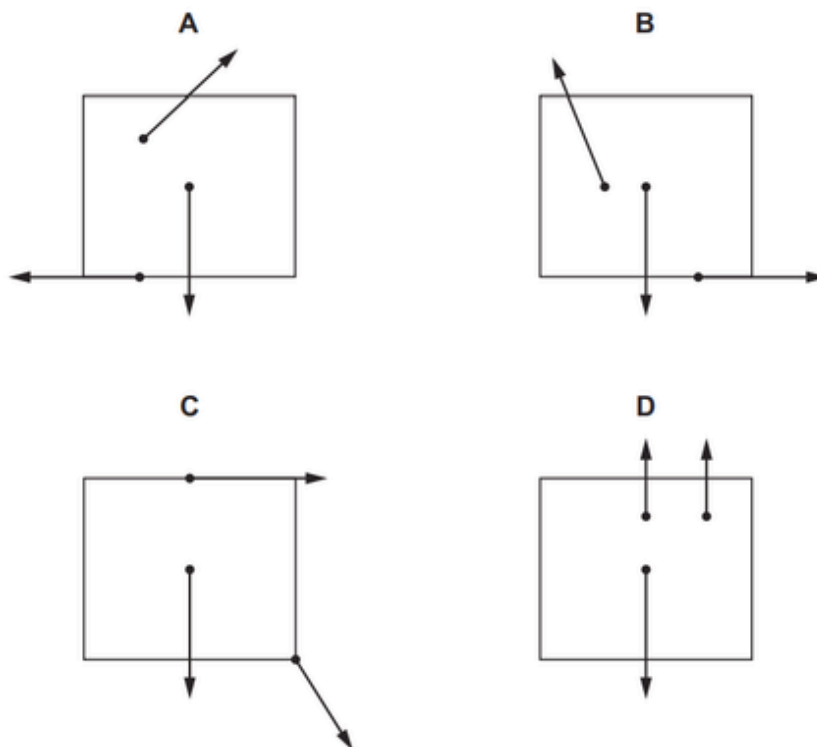
What are the resultant force acting on the rod and the resultant moment about the centre of gravity of the rod?

	resultant force /N	resultant moment /Nm
A	0	10
B	0	20
C	20	10
D	20	20

249. 9702/12/F/M/20/Q13

Three coplanar forces act on a block.

Which diagram shows the directions of the forces such that the block could be in equilibrium?



250. 9702/12/F/M/20/Q14

A cylinder contains a volume of 0.012 m^3 of gas at a pressure of $1.0 \times 10^5 \text{ Pa}$.

400 J of work is done **on** this gas, with its pressure remaining constant throughout.

What is the final volume of the gas?

- A** 0.0040 m^3 **B** 0.0080 m^3 **C** 0.016 m^3 **D** 0.020 m^3

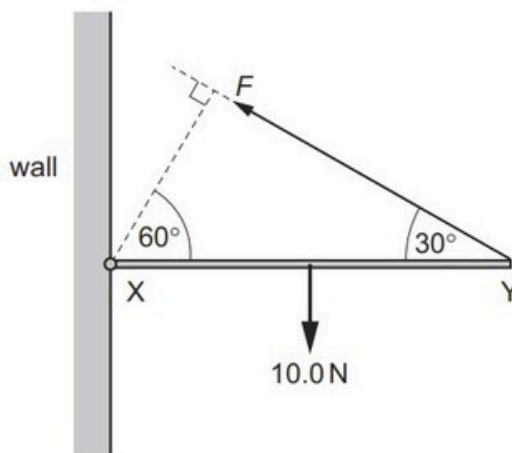
251. 9702/11/M/J/20/Q12

Which statement best describes a couple?

- A** a pair of forces of equal magnitude acting in opposite directions which produce rotational motion but not translational motion
- B** a pair of forces of equal magnitude acting in opposite directions which produce translational motion but not rotational motion
- C** a pair of forces of equal magnitude acting in the same direction which produce rotational motion but not translational motion
- D** a pair of forces of equal magnitude acting in the same direction which produce translational motion but not rotational motion

252. 9702/11/M/J/20/Q13

A uniform rod XY of weight 10.0 N is freely hinged to a wall at X. It is held horizontal by a force F acting from Y at an angle of 30° to the horizontal, as shown.



What is the value of F ?

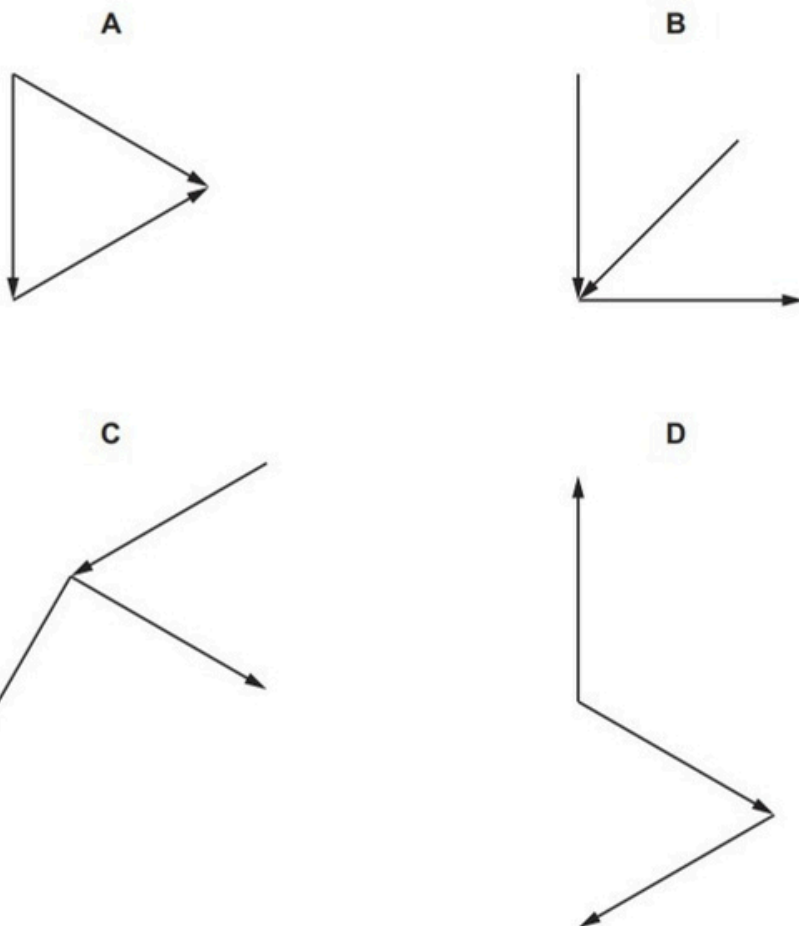
- A** 5.0 N **B** 8.7 N **C** 10.0 N **D** 20.0 N

253. 9702/11/M/J/20/Q14

Four combinations of vectors are shown, each representing all the forces acting on an object. The forces all act in the same plane.

The object is in equilibrium.

Which combination of vectors could represent the forces acting on the object?



254. 9702/11/M/J/20/Q15

A rectangular metal bar exerts a pressure of 15 200 Pa on the horizontal surface on which it rests.

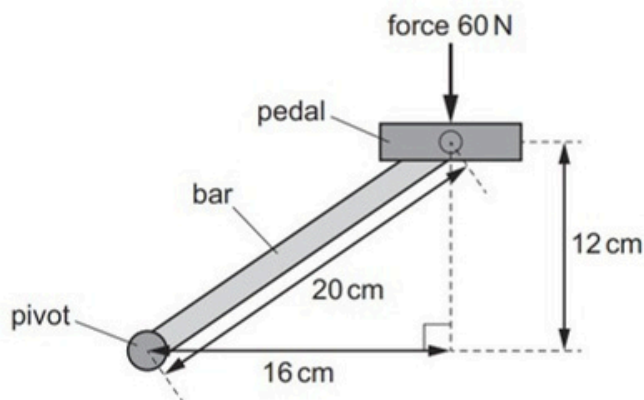
The height of the metal bar is 80 cm.

What is the density of the metal?

- A 190 kg m^{-3}
- B 1900 kg m^{-3}
- C $19\,000 \text{ kg m}^{-3}$
- D $190\,000 \text{ kg m}^{-3}$

255. 9702/12/M/J/20/Q11

A bicycle pedal is connected to a pivot by a metal bar, as shown.



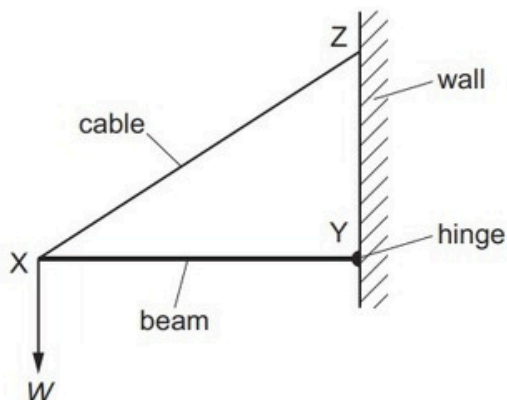
The force on the pedal is 60 N downwards.

What is the moment of this force about the pivot?

- A** 7.2 Nm **B** 9.6 Nm **C** 12 Nm **D** 1200 Nm

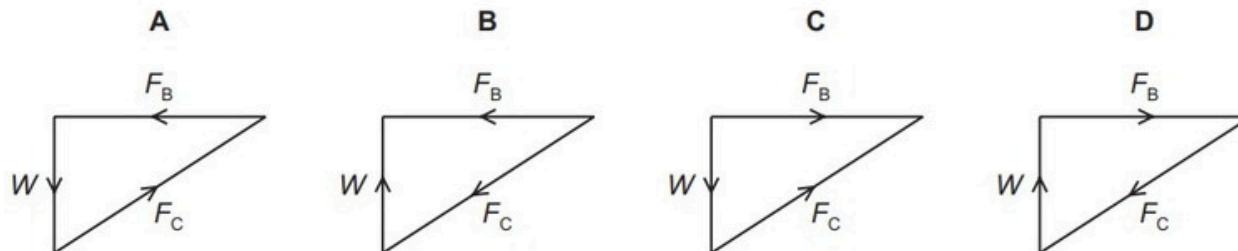
256. 9702/12/M/J/20/Q12

A thin horizontal beam XY is freely hinged at point Y to a vertical wall. The beam is held stationary by a cable XZ which is attached to the wall at point Z.



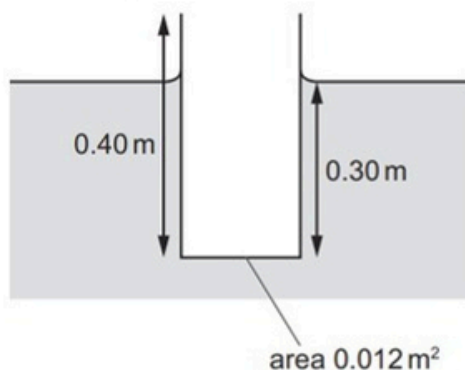
The beam supports a weight W at point X. The forces in the cable and the beam are F_C and F_B respectively.

Which vector triangle represents the forces acting on point X?



257. 9702/12/M/J/20/Q13

A pipe, open at one end, floats in a liquid as shown.



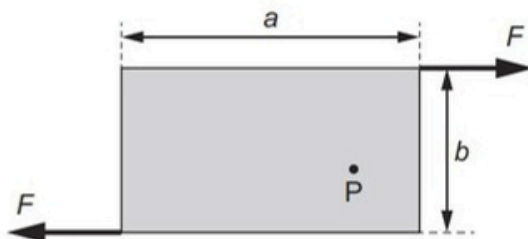
The cross-sectional area of the pipe is 0.012 m^2 . The weight of the pipe is 32 N.

What is the density of the liquid?

- A** 680 kg m^{-3} **B** 910 kg m^{-3} **C** 6700 kg m^{-3} **D** 8900 kg m^{-3}

258. 9702/13/M/J/20/Q12

Two forces, each of magnitude F , act along the edges of a rectangular metal plate, as shown.



The plate has length a and width b .

What is the torque about point P ?

- A** Fa **B** Fb **C** $2Fa$ **D** $2Fb$

259. 9702/13/M/J/20/Q15

Liquid Q has twice the density of liquid R.

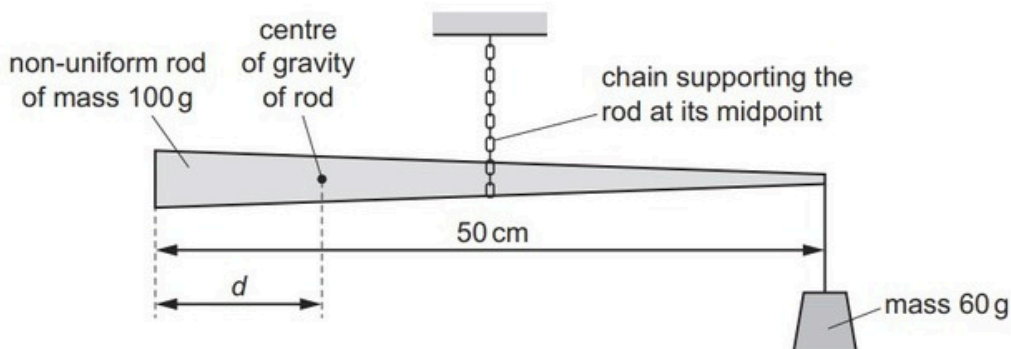
At depth x in liquid R, the pressure due to the liquid is 4 kPa.

At which depth in liquid Q is the pressure due to the liquid 7 kPa?

- A** $\frac{2x}{7}$ **B** $\frac{7x}{8}$ **C** $\frac{8x}{7}$ **D** $\frac{7x}{2}$

260. 9702/13/M/J/20/Q13

A non-uniform rod has a mass of 100 g and a length of 50 cm. It is supported by a chain at its midpoint. The rod is held in equilibrium by having a mass of 60 g suspended from its right-hand end, as shown.



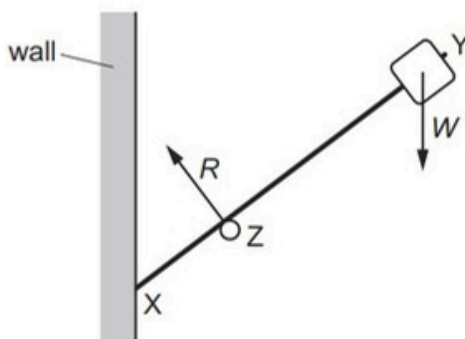
The centre of gravity of the rod is a distance d from its left-hand end.

What is the value of d ?

- A** 10 cm **B** 15 cm **C** 25 cm **D** 40 cm

261. 9702/13/M/J/20/Q14

A light rigid rod XY has an object of weight W fixed at one end. The rod is in equilibrium, resting on a support at Z and a vertical wall at X. The support exerts a force R on the rod as shown. The diagram shows the directions, but not the magnitudes, of the forces R and W .

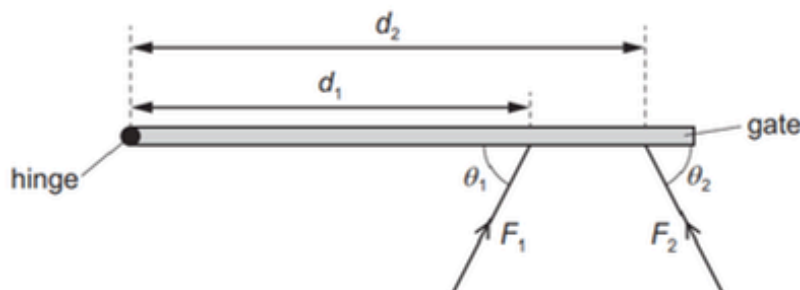


What is the direction of the force on the rod at X?

- A**  **B**  **C**  **D** 

262. 9702/11/O/N/20/Q12

Two people push a vertical gate to open it. The forces exerted by the people on the gate are shown.



One person is distance d_1 from the gate's hinge and pushes with horizontal force F_1 at angle θ_1 to the gate.

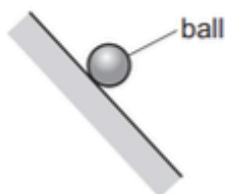
The other person is at distance d_2 from the hinge and pushes with horizontal force F_2 at an angle θ_2 to the gate.

What is the total moment about the hinge due to forces F_1 and F_2 ?

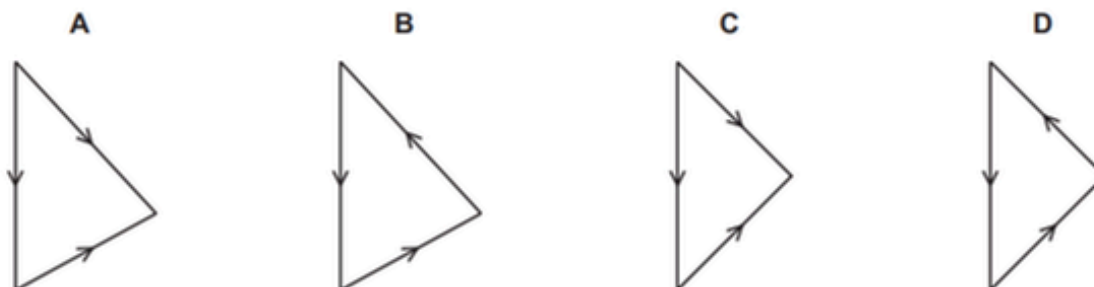
- A $(d_1 \times F_1 \cos \theta_1) + (d_2 \times F_2 \cos \theta_2)$
- B $(d_1 \times F_1 \sin \theta_1) + (d_2 \times F_2 \sin \theta_2)$
- C $(d_1 \times F_1 \cos \theta_1) - (d_2 \times F_2 \cos \theta_2)$
- D $(d_1 \times F_1 \sin \theta_1) - (d_2 \times F_2 \sin \theta_2)$

263. 9702/11/O/N/20/Q13

A ball is rolling down a slope at a constant speed. The three forces acting on the ball are its weight, the contact force normal to the slope and friction.

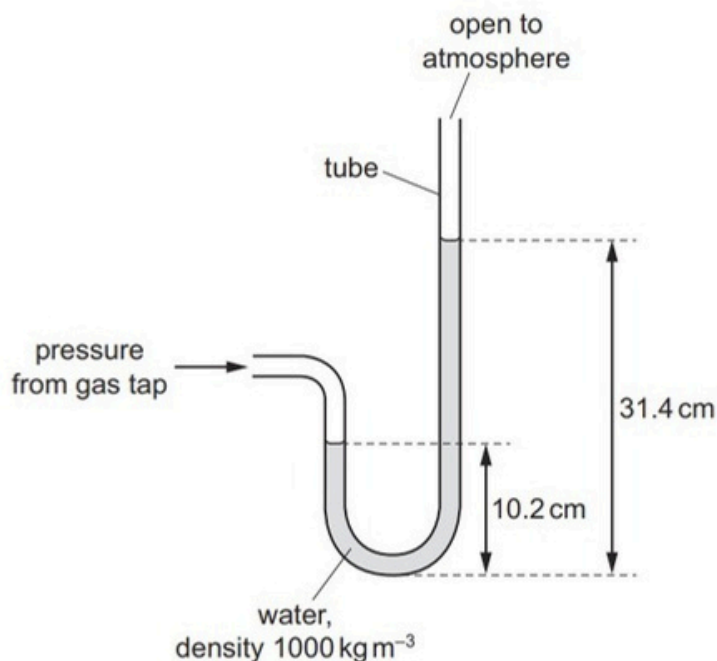


Which diagram could represent these three forces?



264. 9702/11/O/N/20/Q14

One end of a U-shaped tube is attached to a gas tap, with its other end open to the atmosphere. It contains water of density 1000 kg m^{-3} and the heights of both sides of the water column are shown.



The atmospheric pressure is 101 kPa.

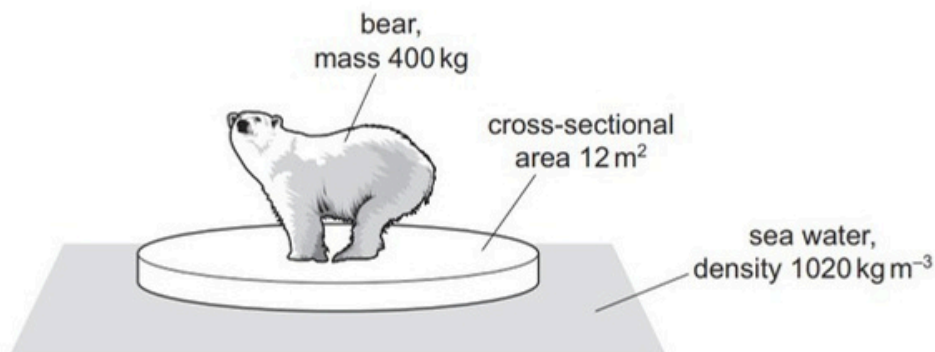
What is the pressure of the gas from the gas tap?

- A** 99 kPa **B** 100 kPa **C** 102 kPa **D** 103 kPa

265. 9702/12/O/N/20/Q15

A cylindrical block of ice of cross-sectional area 12 m^2 is floating, partially submerged, in the sea.

The density of the sea water is 1020 kg m^{-3} . A polar bear of mass 400 kg steps onto the block of ice.



The block of ice sinks a vertical distance d .

What is the value of d ?

- A** 3.3 mm **B** 3.3 cm **C** 0.32 m **D** 3.1 m

266. 9702/12/O/N/20/Q14

In a high-wire circus act, a man of mass 85 kg is standing at rest at the midpoint of the wire.



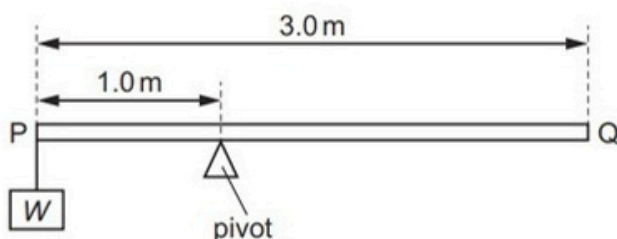
The wire on either side of the man is at an angle of 20° to the horizontal.

What is the tension T in the wire?

- A** 0.44 kN **B** 0.89 kN **C** 1.2 kN **D** 2.4 kN

267. 9702/12/O/N/20/Q13

The diagram shows a uniform beam PQ. The length of the beam is 3.0 m and its weight is 50 N. The beam is supported on a pivot 1.0 m from end P. A load of weight W is hung from end P. The beam is in equilibrium.



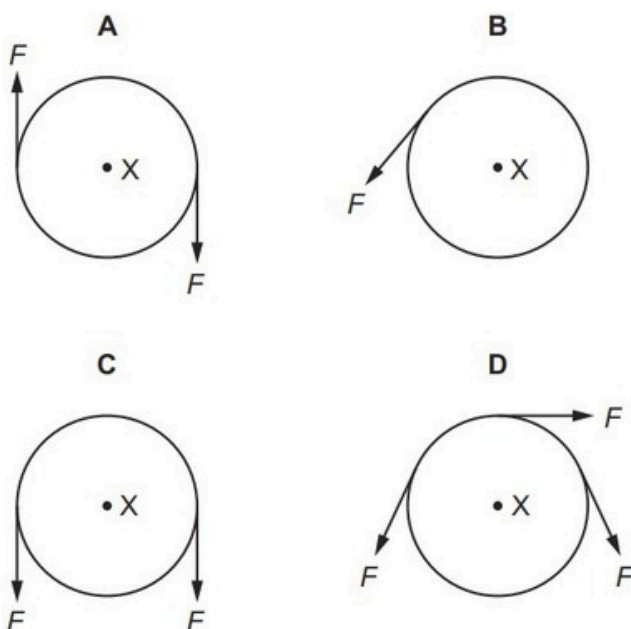
What is the value of W ?

- A** 25 N **B** 50 N **C** 75 N **D** 100 N

268. 9702/12/O/N/20/Q12

A rigid circular disc of radius r has its centre at X. A number of forces of equal magnitude F act at the edge of the disc. All the forces are in the plane of the disc.

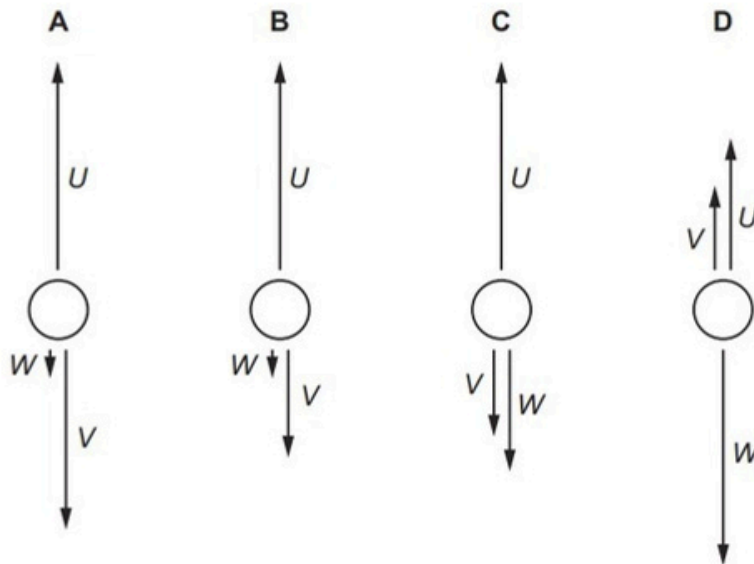
Which arrangement of forces provides a total moment of magnitude $2Fr$ about X?



269. 9702/13/O/N/20/Q11

An air bubble is rising through a liquid at a constant speed. The forces on it are the upthrust U , the viscous drag V and its weight W .

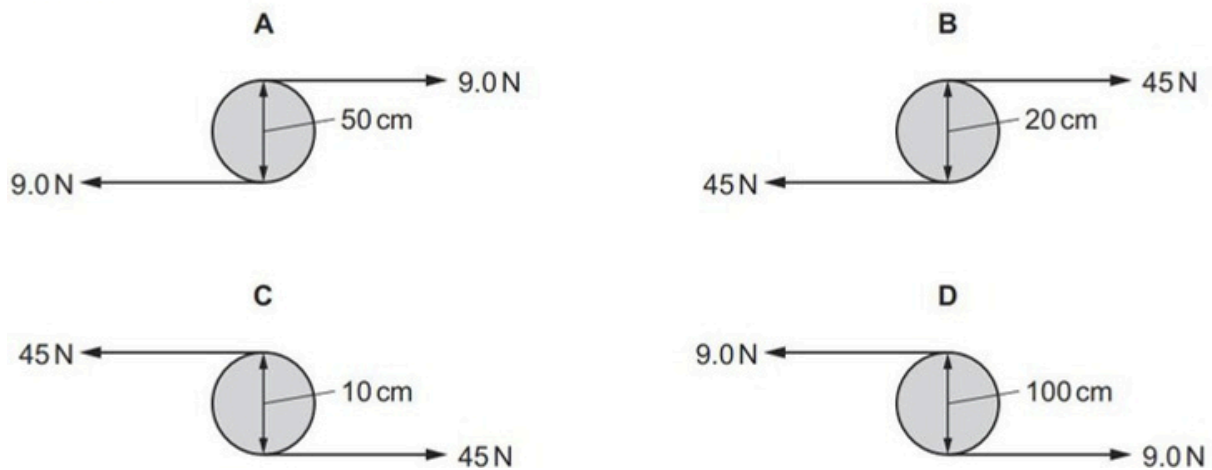
Which diagram shows the directions and relative sizes of the forces?



270. 9702/13/O/N/20/Q12

A couple applies a clockwise torque of 9.0 N m to a circular disc.

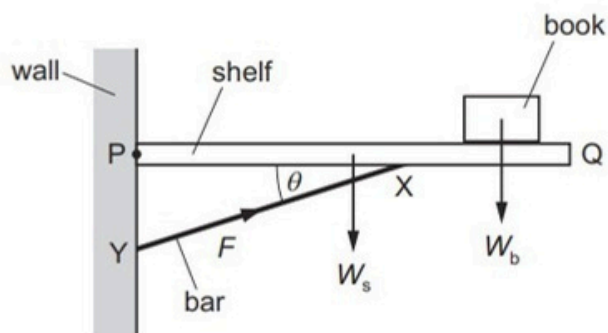
Which diagram shows this couple?



271. 9702/13/O/N/20/Q13

A shelf PQ is attached to a vertical wall at P and supports a book.

The shelf is held horizontal by a rigid bar XY, as shown.



The weight of the shelf is W_s and the weight of the book is W_b .

The bar is at an angle θ to the shelf and exerts a force F on the shelf.

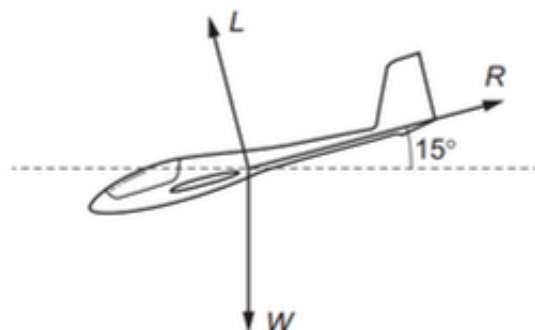
The shelf is in equilibrium.

What are the magnitudes of the horizontal and the vertical components of the force of the wall on the shelf at P?

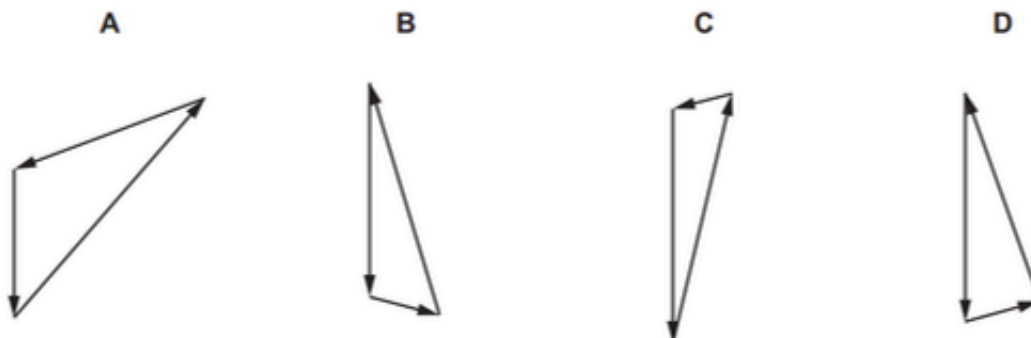
	horizontal component	vertical component
A	$F \cos \theta$	$(W_s + W_b - F \cos \theta)$
B	$F \cos \theta$	$(W_s + W_b - F \sin \theta)$
C	$F \sin \theta$	$(W_s + W_b - F \cos \theta)$
D	$F \sin \theta$	$(W_s + W_b - F \sin \theta)$

272. 9702/13/O/N/20/Q14

A glider is descending at constant speed at an angle of 15° to the horizontal. The diagram shows the directions of the lift L , air resistance R and weight W acting on the glider.

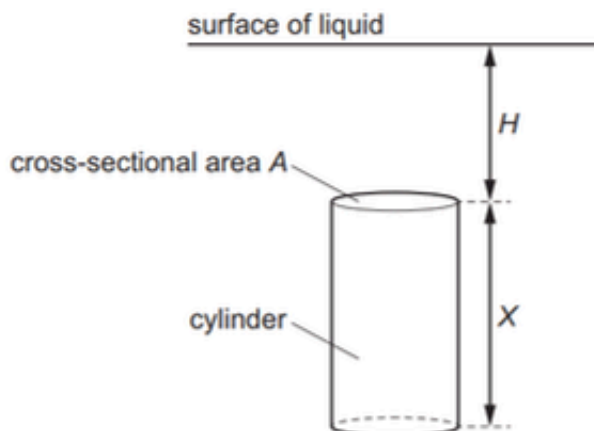


Which vector triangle could represent the forces acting on the glider?



273. 9702/13/O/N/20/Q15

A solid cylinder of density ρ_c , cross-sectional area A and length X is submerged in a liquid of density ρ_L . The upper face of the cylinder is at a depth H below the surface of the liquid, as shown.



The acceleration of free fall is g .

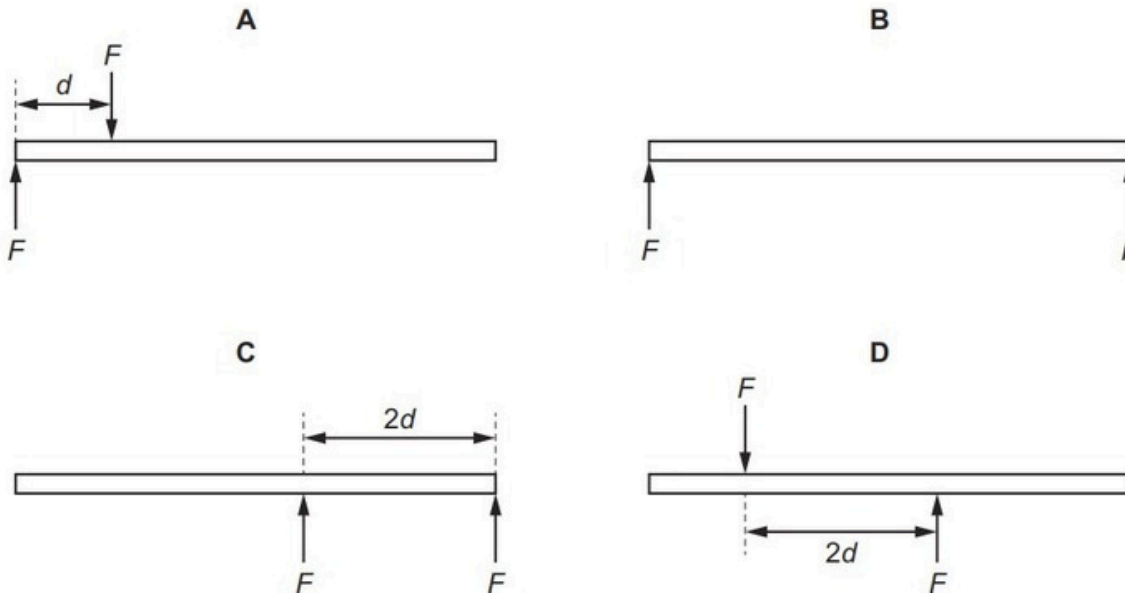
Which expression gives the magnitude of the upthrust force acting on the cylinder?

- A $\rho_c AHg$ B $\rho_c AXg$ C $\rho_L AHg$ D $\rho_L AXg$

274. 9702/12/F/M/21/Q12

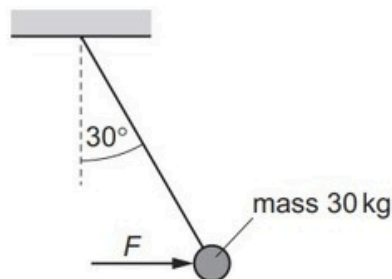
Two parallel forces, each of magnitude F , act on a rod of length $5d$.

Which diagram shows the positions of the two forces that will produce the largest torque on the rod?



275. 9702/12/F/M/21/Q13

A mass of 30 kg is suspended from the end of a wire. A horizontal force F acts on the mass so that it is in equilibrium, with the wire at an angle of 30° to the vertical, as shown.



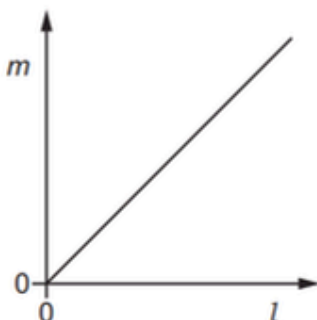
What is the magnitude of F ?

- A** 17 N **B** 150 N **C** 170 N **D** 510 N

276. 9702/12/F/M/21/Q14

A balance is used to measure the mass m of a number of cylindrical metal rods of length l . All the metal rods have the same radius r .

The graph shows the variation with l of m .



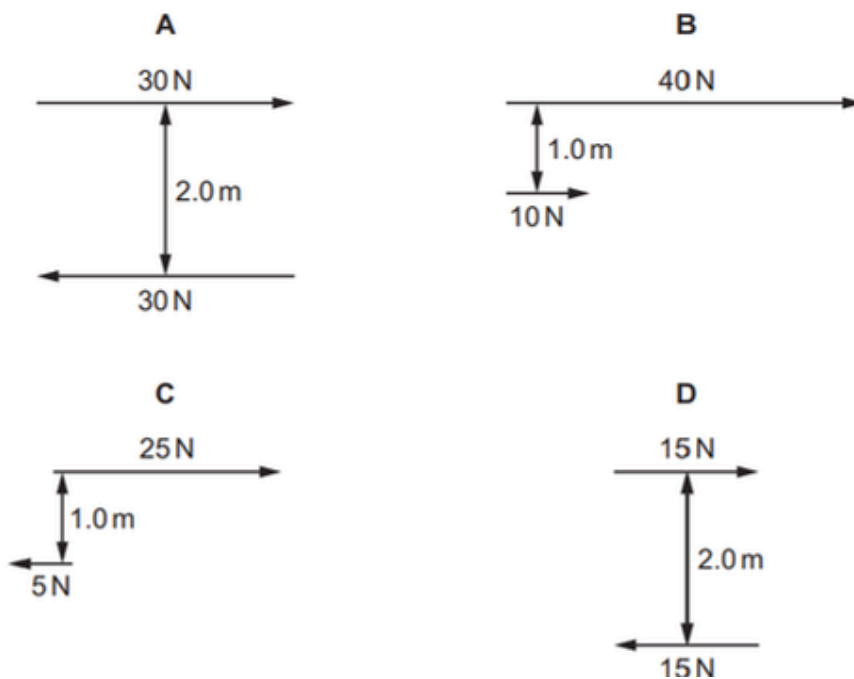
The gradient of the graph is G .

Which expression gives the density of the metal?

- A** $\frac{G}{2\pi r}$
 B $G2\pi r$
 C $\frac{G}{\pi r^2}$
 D $G\pi r^2$

277. 9702/11/M/J/21/Q12

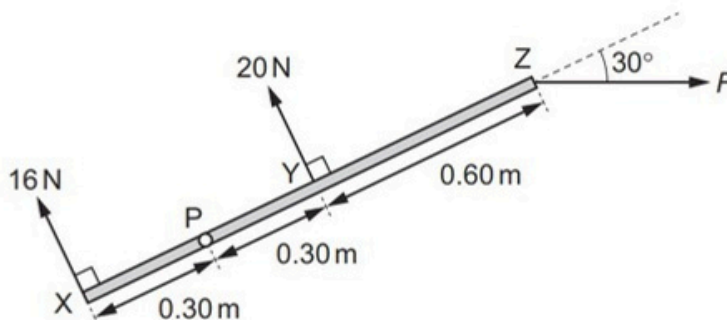
Which pair of forces forms a couple with a torque of 30 N m ?



278. 9702/11/M/J/21/Q13

A uniform rigid bar XZ with negligible mass is 1.20 m long. The bar is pivoted at point P. Three coplanar forces act on the bar as shown. Forces of 16 N and 20 N act perpendicularly to the bar at points X and Y respectively. Force F acts at point Z at an angle of 30° to the axis of the bar.

The distances along the bar of the pivot and of the forces are shown.



The bar experiences a resultant moment about P of 6.0 N m in a clockwise direction.

What is the magnitude of F ?

- A 9.2 N B 11 N C 16 N D 24 N

279. 9702/11/M/J/21/Q14

Water of depth 9.0 cm is covered by oil of depth 5.0 cm in a measuring cylinder.

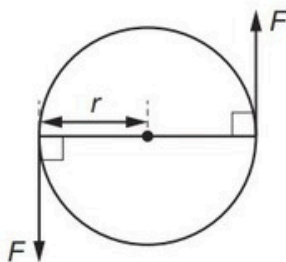
The density of the water is 1000 kg m^{-3} and the density of the oil is 800 kg m^{-3} .

What is the total pressure exerted on the base of the measuring cylinder due to the oil and water?

- A 390 Pa B 880 Pa C 1200 Pa D 1300 Pa

280. 9702/12/M/J/21/Q12

A disc of radius r is acted upon by two opposite forces, each of magnitude F . The forces form a couple, as shown.

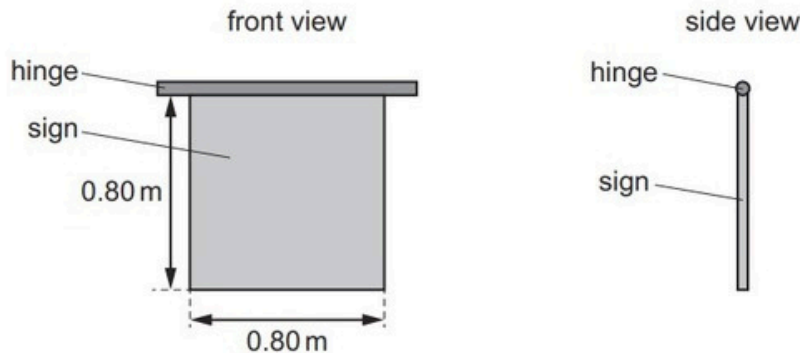


What is the torque of this couple?

- A $\frac{1}{2} Fr$ B Fr C $2Fr$ D $4Fr$

281. 9702/12/M/J/21/Q13

A uniform square sign of weight 40 N is suspended vertically from its top edge by a horizontal hinge, as shown.



The hinge is not frictionless. When the sign is displaced from the vertical by an external force and then released, it does not return to the vertical position.

The maximum torque exerted by the hinge on the sign is 6.0 N m. The sign is displaced by 90° so that it is horizontal and then gradually released.

At which angle to the vertical does the sign hang after release?

- A** 11° **B** 22° **C** 68° **D** 79°

282. 9702/12/M/J/21/Q14

Each foot of an elephant has a circular cross-section with a circumference of 1.4 m. The elephant has a mass of 5400 kg.

The elephant is standing still with all four feet on the ground. Assume the pressure under each foot is the same.

What is the approximate pressure exerted on the ground by each of the elephant's feet?

- A** 8.7 kPa **B** 35 kPa **C** 85 kPa **D** 340 kPa

283. 9702/13/M/J/21/Q12

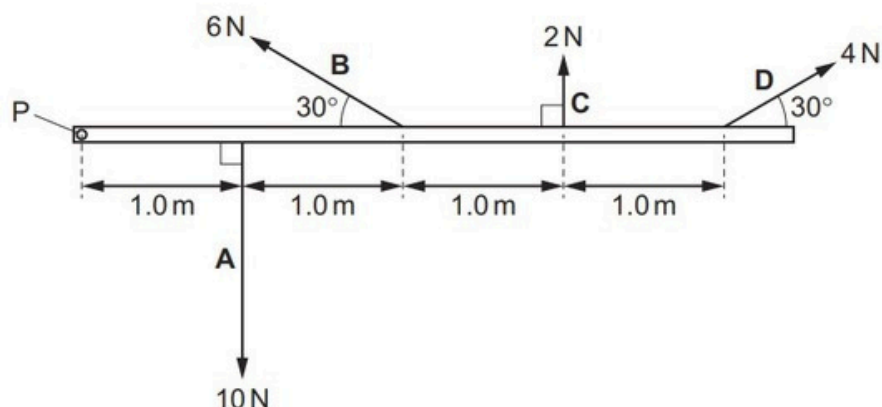
What is a couple?

- A** a pair of equal forces acting in the same direction but along different lines
B a pair of forces that are equal and opposite but acting along different lines
C a pair of forces that produce no resultant force and no resultant moment
D a pair of unequal forces acting in opposite directions but along the same line

284. 9702/13/M/J/21/Q13

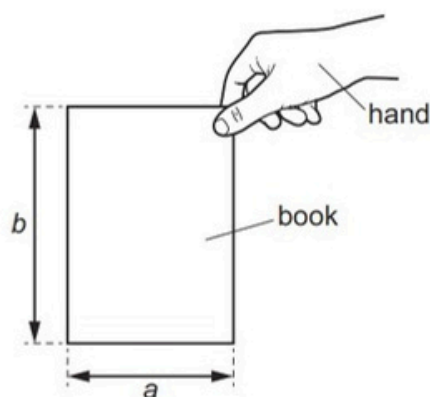
Four coplanar forces act on a rigid rod, as shown. The rod is hinged at P.

Which force produces the greatest moment about point P?



285. 9702/13/M/J/21/Q14

A book of weight W has a rectangular shape and is of uniform thickness. The book is held in a vertical plane so that the longer sides of the book are vertical, as shown.



Which expression gives the approximate torque exerted by the hand on the book?

- A $\frac{Wa}{2}$ clockwise
- B $\frac{Wb}{2}$ anticlockwise
- C Wa clockwise
- D Wb anticlockwise

286. 9702/11/O/N/21/Q14

A bird dives to a depth of 1.50 m below the surface of a lake. Atmospheric pressure is 101 kPa. The density of water is 1000 kg m^{-3} .

What is the pressure at this depth?

- A 14.7 kPa
- B 86.3 kPa
- C 103 kPa
- D 116 kPa

287. 9702/13/M/J/21/Q15

The derivation of the pressure equation $\Delta p = \rho g \Delta h$ uses a number of relationships between quantities.

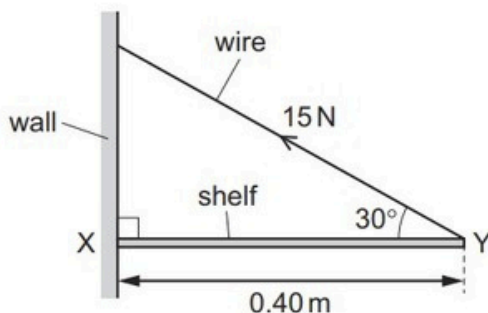
Which relationship is **not** used in the derivation of this equation?

- A density = $\frac{\text{mass}}{\text{volume}}$
- B potential energy = mass $\times$ acceleration of free fall $\times$ height
- C pressure = $\frac{\text{force}}{\text{area}}$
- D weight = mass $\times$ acceleration of free fall

288. 9702/11/O/N/21/Q12

A shelf XY is 0.40 m long and is attached to a wall at end X.

It is kept horizontal by a wire attached to Y and to the wall, as shown.



The tension force in the wire is 15 N at an angle of 30° to the horizontal.

What is the moment of this force about point X?

- A 3.0 Nm
- B 5.2 Nm
- C 6.9 Nm
- D 12 Nm

289. 9702/11/O/N/21/Q13

A statement about the principle of moments with some words omitted is shown.

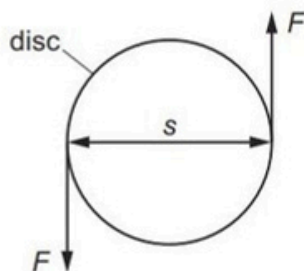
'For an object in a state of rotational equilibrium, the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about '

Which words correctly complete the statement?

- A any point
- B the centre of gravity of the object
- C the pivot
- D the same point

290. 9702/12/O/N/21/Q12

Two forces, each of magnitude F , act on a disc of diameter s , as shown.

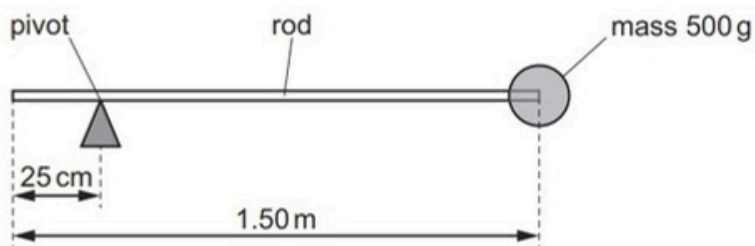


What is the torque exerted on the disc?

- A** zero **B** $\frac{1}{2}Fs$ **C** Fs **D** $2Fs$

291. 9702/12/O/N/21/Q13

A mass of 500 g is attached at one end of a rod of length 1.50 m. The rod is pivoted at a distance of 25 cm from the other end. The rod is horizontal.

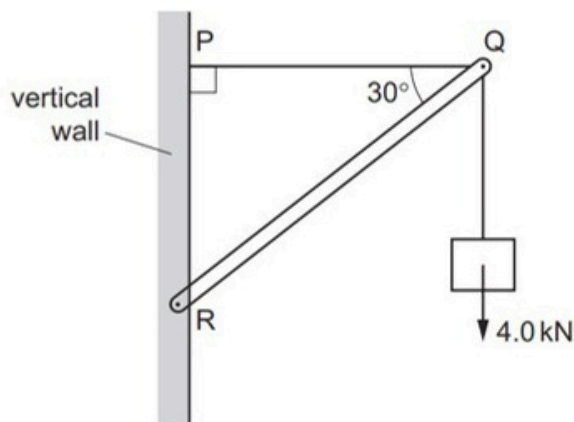


What is the moment about the pivot due to the mass?

- A** 0.63 Nm **B** 1.2 Nm **C** 6.1 Nm **D** 7.4 Nm

292. 9702/12/O/N/21/Q14

A beam QR is held in position by a wire PQ. The structure is used to form a crane supporting a stationary load of 4.0 kN, as shown.

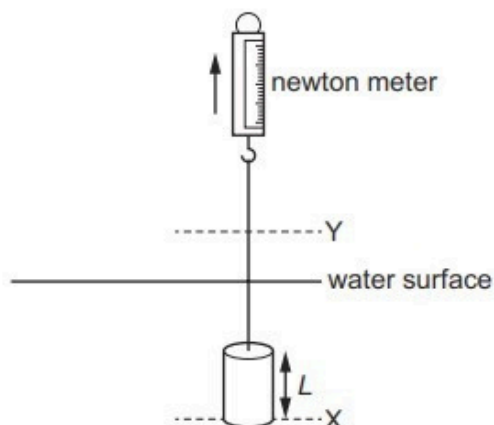


What is the force exerted by the beam QR on point Q?

- A** 4.0 kN **B** 4.6 kN **C** 6.9 kN **D** 8.0 kN

293. 9702/12/O/N/21/Q15

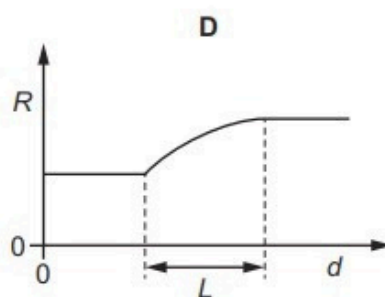
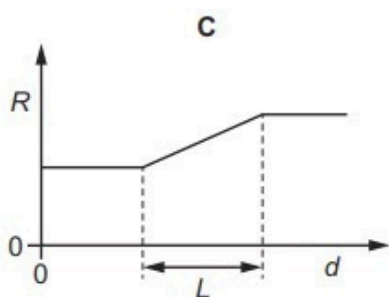
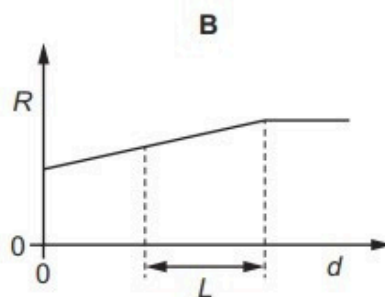
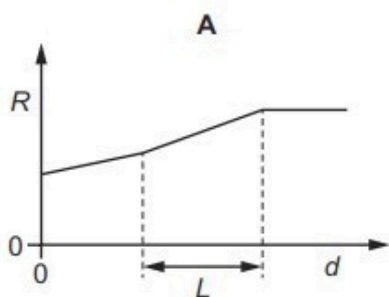
A metal cylinder, totally immersed in water, is hung from a newton meter.



The cylinder, of height L , is slowly raised vertically by lifting the newton meter.

As the base of the cylinder moves from line X in the water to line Y above the surface of the water, the reading R on the newton meter is recorded. The velocity of the cylinder is constant.

Which graph best shows the variation of R with the distance d of the base of the cylinder from line X ?



294. 9702/13/O/N/21/Q12

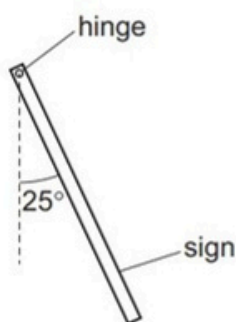
Two forces act as a couple.

Which statement about the two forces must **not** be correct?

- A They act along the same straight line.
- B They act in opposite directions.
- C They are the same type of force.
- D They have the same magnitude.

295. 9702/13/O/N/21/Q13

A sign outside a shop is suspended from a rusty horizontal hinge at its top end. The sign hangs in equilibrium at an angle of 25° to the vertical, as shown.



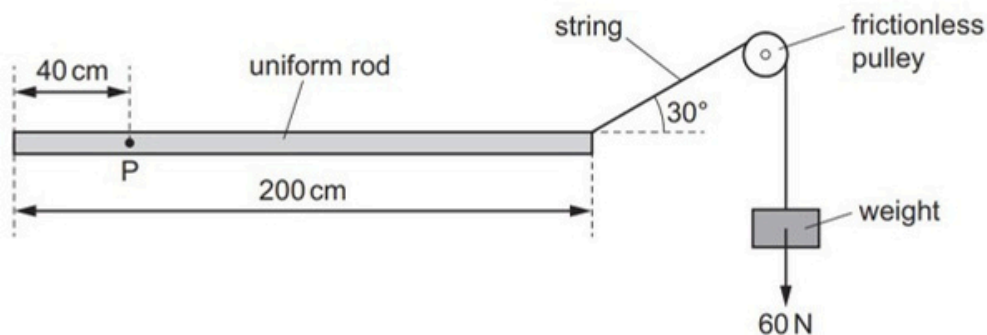
The sign is a square of side length 52 cm and uniform thickness. It has a mass of 36 kg.

What is the moment of the weight of the sign about the hinge?

- A 39 Nm B 78 Nm C 83 Nm D 92 Nm

296. 9702/13/O/N/21/Q14

A uniform rod of length 200 cm is freely pivoted at point P. The rod is held horizontally in equilibrium by a 60 N weight that is attached to the rod by a string passing over a frictionless pulley.

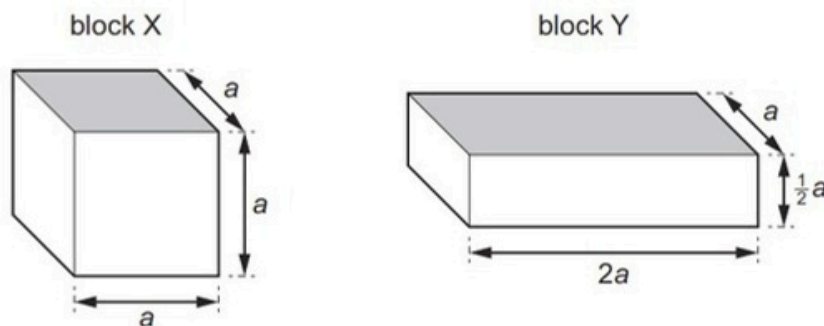


What is the weight of the rod?

- A 30 N B 60 N C 80 N D 140 N

297. 9702/13/O/N/21/Q15

The diagram shows two blocks X and Y.



Block X has sides of length a . When block X is placed in a liquid of density ρ with the shaded face level with the liquid surface, it experiences an upthrust U .

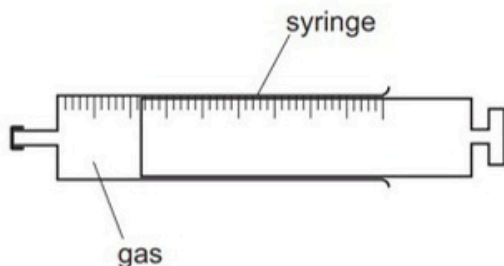
Block Y has horizontal sides of length a and $2a$ and height $\frac{1}{2}a$. Block Y is placed in a liquid of density 2ρ , also with the shaded face level with the liquid surface.

What is the upthrust on block Y?

- A** $\frac{1}{2}U$ **B** U **C** $2U$ **D** $4U$

298. 9702/13/O/N/21/Q16

A gas is contained inside a syringe, as shown.



The initial volume of the gas is 2.00 cm^3 .

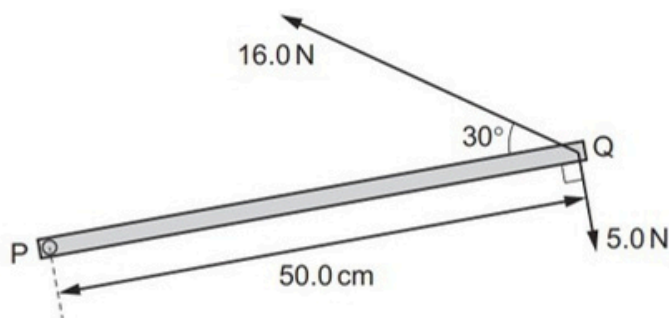
Atmospheric pressure is 101 kPa .

What is the work done by the gas on the atmosphere when the gas is heated and expands to a volume of 6.00 cm^3 ?

- A** $404 \mu\text{J}$ **B** 404 mJ **C** 404 J **D** 404 kJ

299. 9702/12/F/M/22/Q11

A horizontal metal bar PQ of length 50.0 cm is hinged at end P. The diagram shows the metal bar viewed from above.



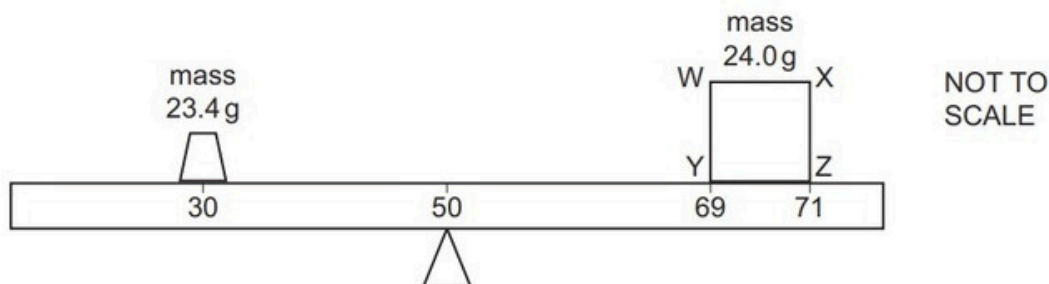
Two forces of 16.0 N and 5.0 N are in the horizontal plane and act on end Q, as shown.

What is the resultant moment about P due to the two forces?

- A** 1.5 Nm **B** 4.4 Nm **C** 6.5 Nm **D** 9.4 Nm

300. 9702/12/F/M/22/Q12

A cube WXZY has sides of length 2.0 cm and mass 24.0 g. The cube rests on a metre rule of negligible mass. The geometrical centre of the cube is vertically above the 70.0 cm mark on the scale of the rule.



The cube has a non-uniform density so that its centre of gravity is **not** at its geometrical centre. The centre of gravity of the cube is in the plane of the diagram.

The rule rests on a pivot at the 50.0 cm mark. A mass of 23.4 g is placed vertically above the 30.0 cm mark. The rule is horizontal and in equilibrium.

What can be determined about the position of the centre of gravity of the cube?

- A** It must be somewhere along a horizontal line that is 0.5 cm from line WX.
B It must be somewhere along a horizontal line that is 0.5 cm from line YZ.
C It must be somewhere along a vertical line that is 0.5 cm from line WY.
D It must be somewhere along a vertical line that is 0.5 cm from line XZ.

301. 9702/12/F/M/22/Q13

A rigid sphere is held at rest on the sea bed. When the sphere is released, it rises to the surface of the sea. The seawater has a uniform density.

Which statement about the sphere, from its release until it reaches the surface, is correct?

- A The sphere always moves with constant acceleration.
- B The sphere always moves with constant velocity.
- C The upthrust on the sphere always decreases.
- D The upthrust on the sphere is always constant.

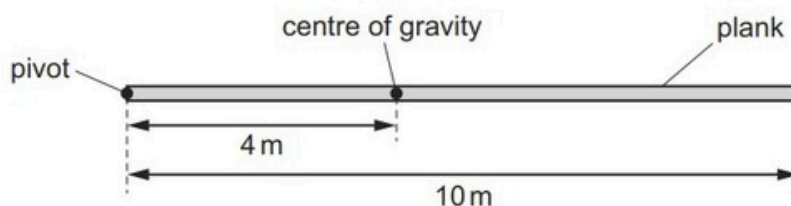
302. 9702/12/F/M/22/Q14

What is a unit for density?

- A Nm^{-3} B gmm^{-1} C kg cm^{-2} D $\mu\text{g mm}^{-3}$

303. 9702/11/M/J/22/Q11

A horizontal wooden plank is pivoted at one end, as shown.



The plank has a mass of 100 kg and a length of 10 m. The centre of gravity of the plank is a distance of 4 m from the pivot.

What is the moment of the weight of the plank about the pivot?

- A $4 \times 10^2 \text{ Nm}$ B $5 \times 10^2 \text{ Nm}$ C $4 \times 10^3 \text{ Nm}$ D $5 \times 10^3 \text{ Nm}$

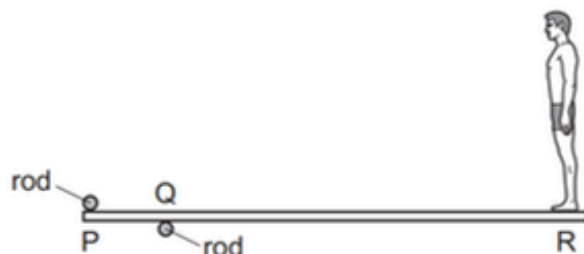
304. 9702/11/M/J/22/Q12

When **must** an object be in equilibrium?

- A when no resultant force acts on the object
- B when no resultant force and no resultant torque act on the object
- C when no resultant torque acts on the object
- D when the upward force on the object is equal and opposite to its weight

305. 9702/11/M/J/22/Q13

A uniform diving board is held by two fixed rods at points P and Q. A person stands at end R of the diving board, as shown.



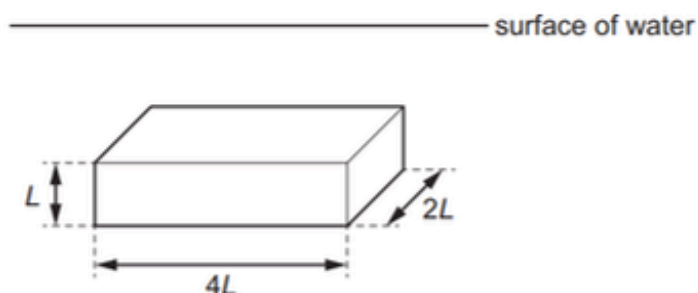
The forces exerted by the rods on the board are vertical. The board remains in equilibrium as the person slowly moves towards point Q from end R.

Which row describes the changes to the magnitudes of the forces exerted by the rods on the board?

	force at P	force at Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

306. 9702/11/M/J/22/Q14

A solid block has sides of length L , $2L$ and $4L$. The block is submerged in water of uniform density so that the faces with the largest area are horizontal, as shown.



The upthrust acting on the block is U .

The block is now rotated to a new position so that the faces with the smallest area are horizontal. The block remains fully submerged in the water.

What is the upthrust acting on the block in its new position?

- A** $\frac{U}{4}$ **B** U **C** $2U$ **D** $4U$

307. 9702/11/M/J/22/Q15

In a large container in an oil refinery, three oils of different densities are mixed. No chemical activity occurs.

The mixture consists of:

1200 kg of oil of density 1100 kg m^{-3}

1500 kg of oil of density 860 kg m^{-3}

4000 kg of oil of density 910 kg m^{-3} .

What is the density of the mixture?

- A** 927 kg m^{-3} **B** 933 kg m^{-3} **C** 957 kg m^{-3} **D** 1045 kg m^{-3}

308. 9702/12/M/J/22/Q11

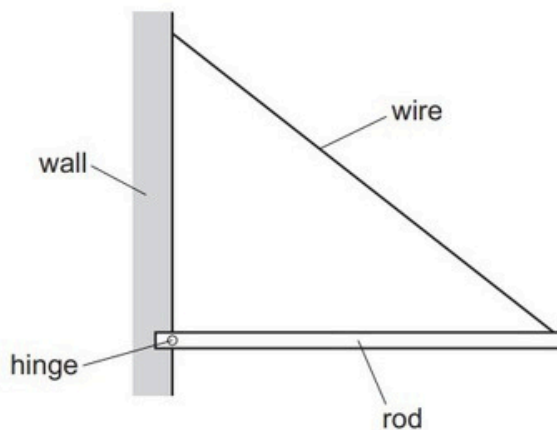
Two forces form a couple.

Which statement describes the two forces?

- A** They are in the same direction.
B They are perpendicular to each other.
C They have the same magnitude.
D They pass through the same point.

309. 9702/12/M/J/22/Q12

A uniform rod is attached by a hinge at one end to a wall. The other end of the rod is supported by a wire so that the rod is horizontal and in equilibrium.



Which arrow shows the direction of the force on the rod from the hinge?

A



B



C

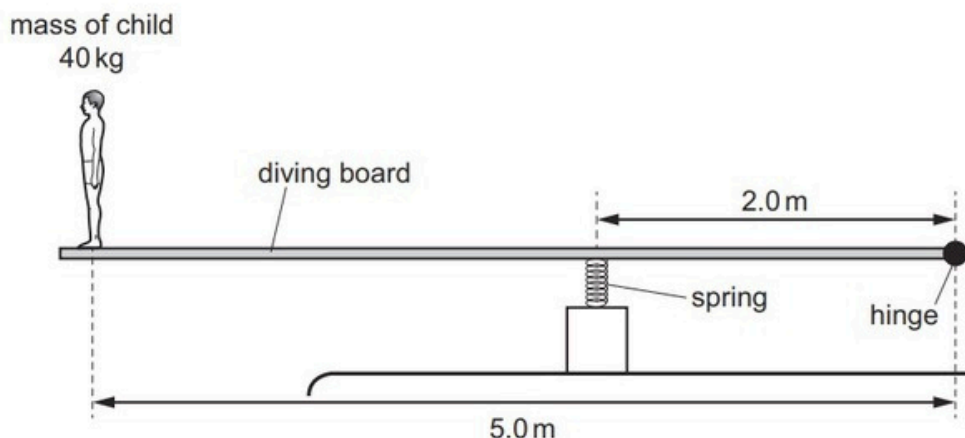


D



310. 9702/12/M/J/22/Q13

A diving board of length 5.0 m is hinged at one end and supported 2.0 m from this end by a spring of spring constant 10 kN m^{-1} . A child of mass 40 kg stands at the far end of the board.



What is the extra compression of the spring caused by the child standing on the end of the board?

- A** 1.0 cm **B** 1.6 cm **C** 9.8 cm **D** 16 cm

311. 9702/12/M/J/22/Q14

A granite rock at the surface of the Earth has density ρ . The rock is transported to the surface of another planet.

The acceleration of free fall on the surface of the other planet is twice that on the surface of the Earth.

What is the density of the rock on the other planet?

- A** 0.5ρ **B** ρ **C** 2ρ **D** 4ρ

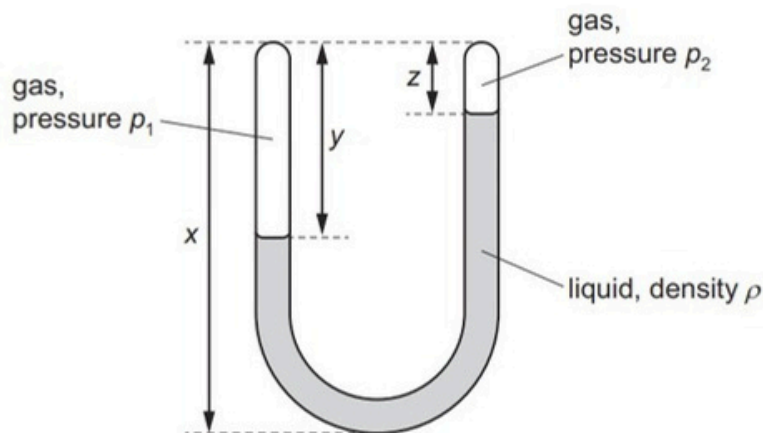
312. 9702/13/M/J/22/Q12

Which statement describes the two forces in a couple?

- A** They act in the same direction.
B They act through the same point.
C They produce zero resultant force.
D They produce zero resultant moment.

313. 9702/12/M/J/22/Q15

A closed U-shaped tube contains a stationary liquid of density ρ . One side of the tube contains a gas at pressure p_1 and the other side contains a gas at pressure p_2 , as shown.



The acceleration of free fall is g .

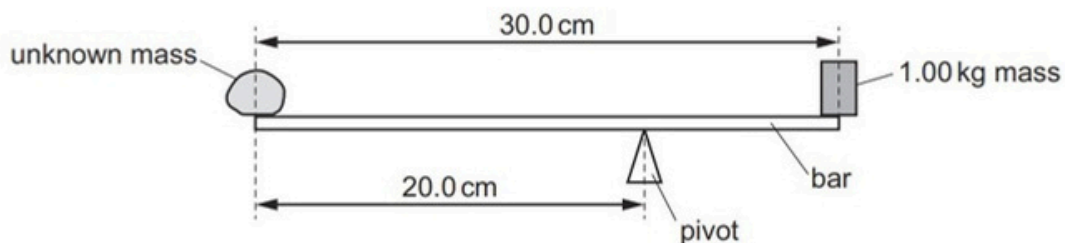
Which equation is correct?

- A $p_1 = \rho g y$
- B $p_2 = \rho g(x - z)$
- C $p_1 - p_2 = \rho g(y - z)$
- D $p_1 + p_2 = \rho g x$

314. 9702/13/M/J/22/Q14

An unknown mass and a 1.00 kg mass are fixed at opposite ends of a bar. The bar has negligible mass and a length of 30.0 cm.

The bar balances when supported by a pivot placed 20.0 cm from the unknown mass, as shown.

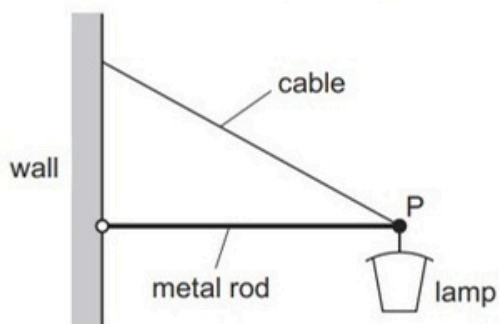


What is the unknown mass?

- A 333 g
- B 500 g
- C 667 g
- D 1000 g

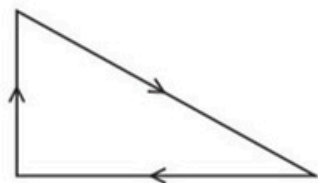
315. 9702/13/M/J/22/Q13

A street lamp is fixed to a wall by a metal rod and a cable.

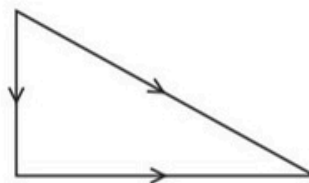


Which vector triangle could represent the forces acting on the end of the rod at point P?

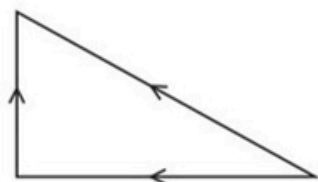
A



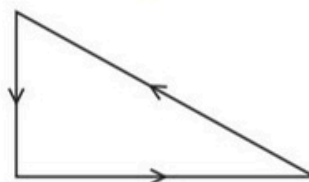
B



C



D



316. 9702/13/M/J/22/Q16

A technical article about diesel engines expresses the energy available from diesel fuel both as 41.8 MJ kg^{-1} and as 34.9 GJ m^{-3} .

What is the density of diesel fuel?

A $8.35 \times 10^2 \text{ kg m}^{-3}$

B $1.20 \times 10^3 \text{ kg m}^{-3}$

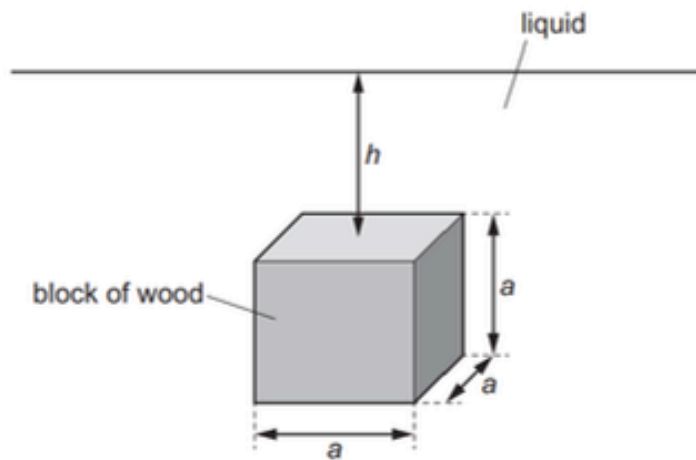
C $8.35 \times 10^5 \text{ kg m}^{-3}$

D $1.20 \times 10^6 \text{ kg m}^{-3}$

317. 9702/13/M/J/22/Q15

A block of wood of density ρ_w has sides of length a .

The block is immersed in a liquid of density ρ_L . The top surface of the block is at a depth h below the surface of the liquid.



The acceleration of free fall is g .

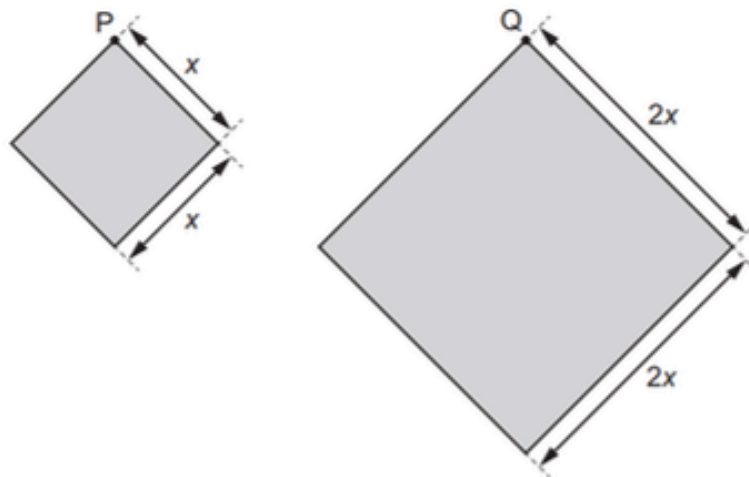
What is the upthrust acting on the block from the liquid?

- A $\rho_L a^3 g$ B $\rho_w a^3 g$ C $\rho_L h g$ D $\rho_L a g$

318. 9702/11/O/N/22/Q11

A square board, of side length x , hangs freely from a nail P, as shown.

The board has uniform thickness and is made from material of uniform density.



A second square board, of side length $2x$, is made of the same material and has the same thickness as the original board. This second board is then hung from a nail Q. Nails P and Q are at the same height.

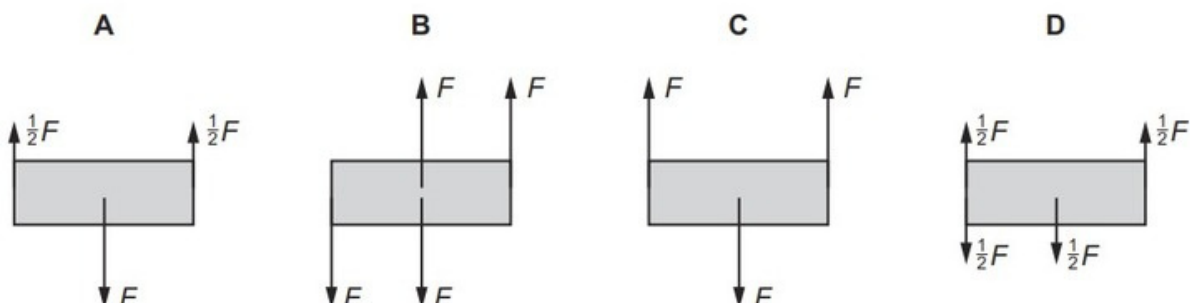
What is the vertical distance between the positions of the centres of gravity of the two boards?

- A 0 B $\frac{x}{\sqrt{2}}$ C x D $x\sqrt{2}$

319. 9702/11/O/N/22/Q12

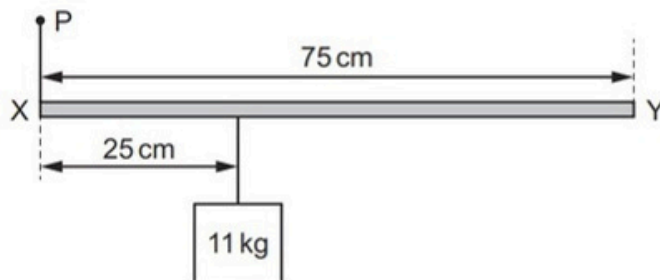
Forces are applied to a rigid object. The forces all act in the same plane.

In which diagram is the object in equilibrium?



320. 9702/11/O/N/22/Q13

A rigid rod XY has negligible mass and length 75 cm. The rod is suspended from a fixed point P by a string attached to end X. An object of mass 11 kg is suspended by a string that is attached to the rod at a distance of 25 cm from end X, as shown.



Which vertically upward force acting on end Y of the rod would hold the rod horizontally in equilibrium?

- A** 3.7 N **B** 33 N **C** 36 N **D** 320 N

321. 9702/11/O/N/22/Q14

The density of water is 1.0 g cm^{-3} and the density of glycerine is 1.3 g cm^{-3} .

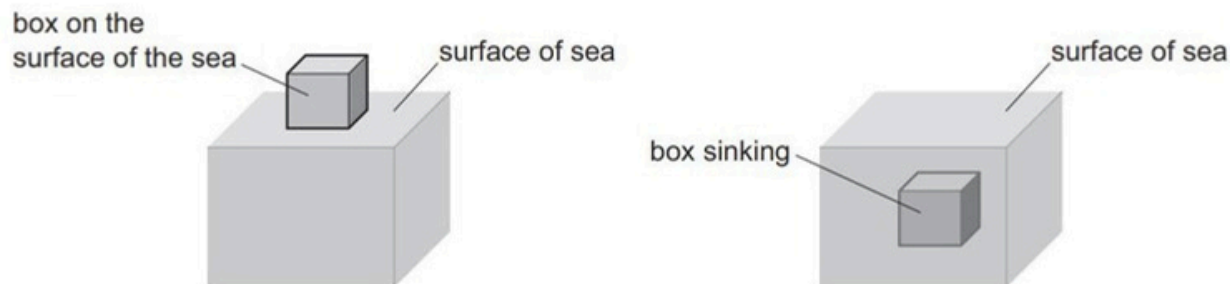
Water is added to a measuring cylinder containing 40 cm^3 of glycerine so that the density of the mixture is 1.1 g cm^{-3} . Assume that the mixing process does not change the total volume of the liquid.

What is the volume of water added?

- A** 40 cm^3 **B** 44 cm^3 **C** 52 cm^3 **D** 80 cm^3

322. 9702/11/O/N/22/Q15

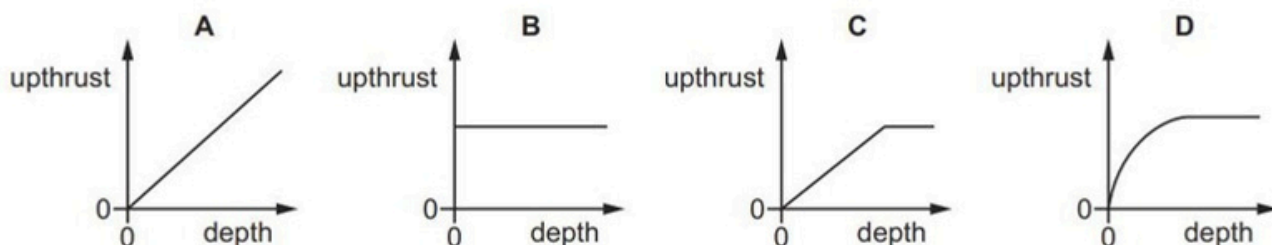
A box, in the shape of a cube, falls from a ship into the sea. The box lands with its lower face level with the surface of the sea.



The box begins to sink, becomes totally submerged and then sinks deeper into the sea.

As the box sinks, its lower face is always parallel to the surface of the sea.

Which graph best represents the variation of the upthrust acting on the box with the depth of its lower face below the surface of the sea?



323. 9702/12/O/N/22/Q12

What is **not** a necessary requirement of the forces in a couple?

- A They act in opposite directions.
- B They act along different lines.
- C They have the same magnitude.
- D They produce a resultant force.

324. 9702/12/O/N/22/Q15

A solid cube is floating in equilibrium in liquid mercury. The cube is made of iron of density 7900 kg m^{-3} .

The cube floats with 42% of its volume above the surface of the mercury.

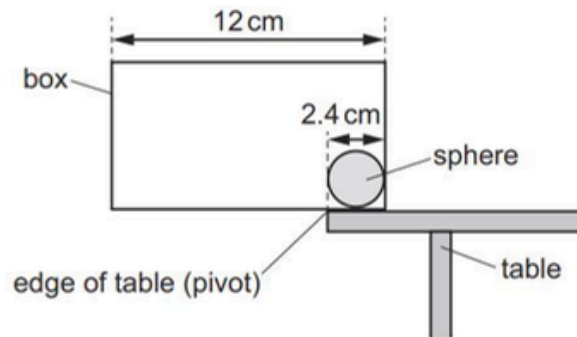
What is the density of the mercury?

- A 3300 kg m^{-3}
- B 4600 kg m^{-3}
- C $14\,000 \text{ kg m}^{-3}$
- D $19\,000 \text{ kg m}^{-3}$

325. 9702/12/O/N/22/Q13

A box of length 12 cm and weight 0.43 N is placed on a horizontal table, with the greater part of its length overhanging the edge of the table. The edge of the table acts as a pivot. The centre of gravity of the box is at its geometric centre.

To balance the box, a uniform sphere of diameter 2.4 cm is placed inside the box, touching one end, as shown.



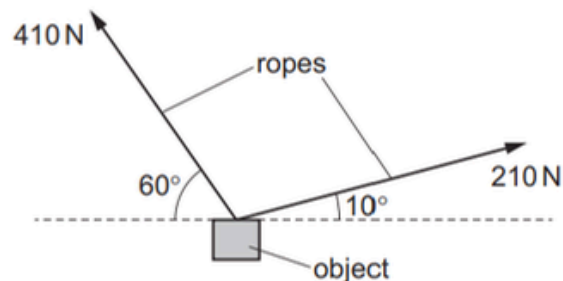
Assume that the forces acting on the box are in the plane of the diagram.

What is the minimum mass of the sphere that is needed to maintain the system in equilibrium?

- A** 0.066 kg **B** 0.13 kg **C** 0.22 kg **D** 1.3 kg

326. 9702/12/O/N/22/Q14

An object is suspended by two ropes. One rope has a tension of 410 N at an angle of 60° to the horizontal. The other rope has a tension of 210 N at an angle of 10° to the horizontal.



The object is in equilibrium.

What is the mass of the object?

- A** 40 kg **B** 42 kg **C** 390 kg **D** 410 kg

327. 9702/12/O/N/22/Q16

The diagram shows two vessels, P and Q, both with sides inclined at 45° to the horizontal.



Vessel P tapers outwards and vessel Q tapers inwards, as shown.

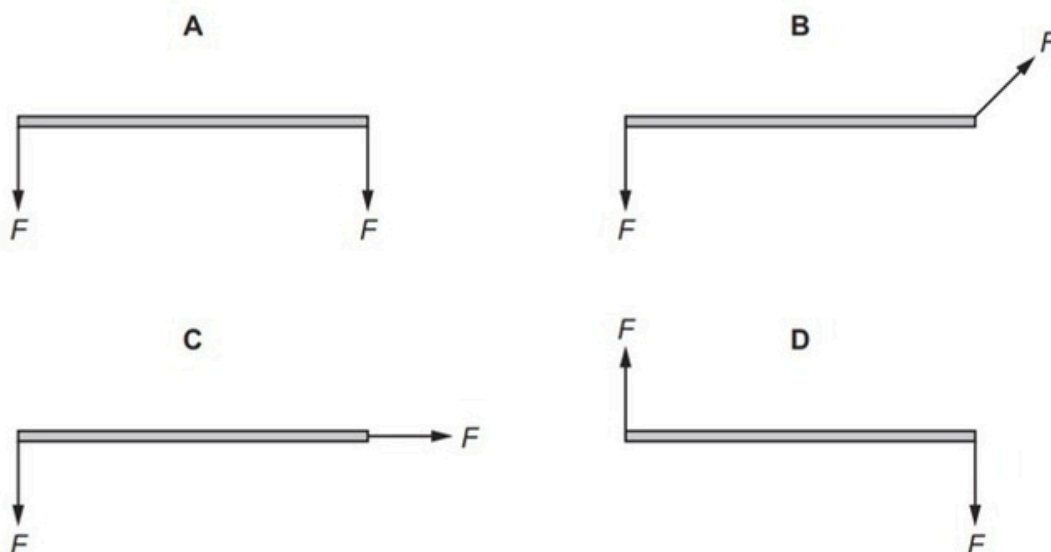
Both vessels contain a liquid. The depth of the liquid in the vessels is the same. The liquid in vessel P is twice as dense as the liquid in vessel Q.

What is the ratio $\frac{\text{pressure due to the liquid on the base of P}}{\text{pressure due to the liquid on the base of Q}}$?

- A $\frac{2}{1}$ B $\frac{\sqrt{2}}{1}$ C $\frac{1}{\sqrt{2}}$ D $\frac{1}{2}$

328. 9702/13/O/N/22/Q11

Which diagram shows a couple formed by two forces, each of magnitude F , acting on a rod?



329. 9702/13/O/N/22/Q12

A student states that:

'If an object is in equilibrium, the sum of the clockwise moments about a point X is equal to the sum of the anticlockwise moments about a point Y.'

Which condition would make the student's statement correct?

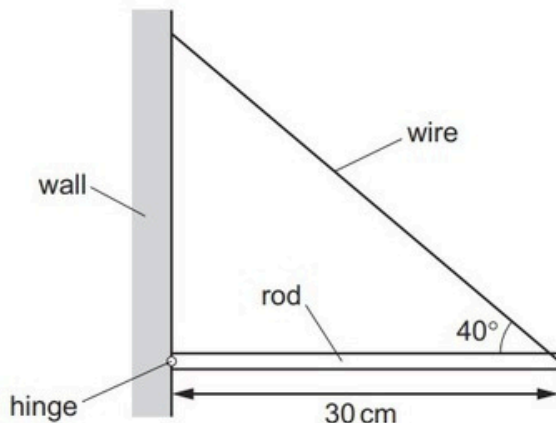
- A Either X or Y is the centre of gravity of the object.
B Either X or Y is the pivot of the object.
C X and Y are at opposite ends of the object.
D X and Y are the same point on the object.

330. 9702/13/O/N/22/Q13

A uniform rod of length 30 cm and weight 5.2 N is attached to a wall by a hinge at one end.

The other end of the rod is supported by a wire so that the rod is horizontal and in equilibrium.

The wire is at an angle of 40° to the horizontal.



What is the tension in the wire?

- A** 3.4 N **B** 4.0 N **C** 6.8 N **D** 8.1 N

331. 9702/13/O/N/22/Q14

Water is pumped through a nozzle at the end of a hose. The nozzle has a circular cross-section of diameter 50 mm. A mass of 100 kg of water takes a time of 2.0 s to move through the nozzle. The density of water is 1000 kg m^{-3} .

What is the speed of the water in the nozzle?

- A** 6.4 ms^{-1} **B** 13 ms^{-1} **C** 25 ms^{-1} **D** 51 ms^{-1}

332. 9702/13/O/N/22/Q15

A solid wooden cube rests on a horizontal surface.

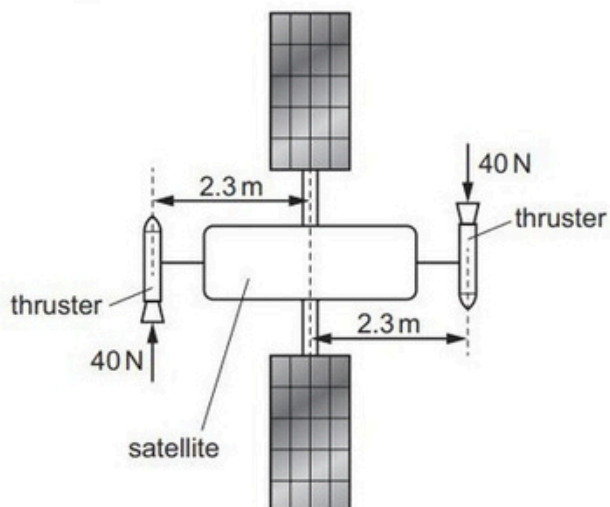
What gives the pressure exerted by the weight of the cube on the horizontal surface?

- A** the product of the acceleration of free fall and the density of the cube
B the product of the acceleration of free fall, the density and the side length of the cube
C the product of the acceleration of free fall, the density and the area of one face of the cube
D the product of the acceleration of free fall, the density and the volume of the cube

333. 9702/12/F/M/23/Q11

A satellite uses two thrusters to adjust its motion in space.

Each thruster exerts a force of 40 N on the satellite. The line of action of each force is a perpendicular distance of 2.3 m from the centre of gravity of the satellite. These two parallel forces act in opposite directions.

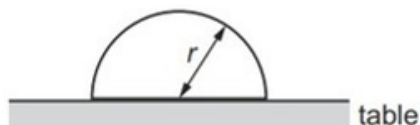


What are the magnitudes of the torque and the resultant force acting on the satellite due to the two thrust forces?

	torque / N m	resultant force / N
A	92	0
B	92	80
C	180	0
D	180	80

334. 9702/12/F/M/23/Q13

An object shaped as a hemisphere rests with its flat surface on a table. The object has radius r and density ρ .



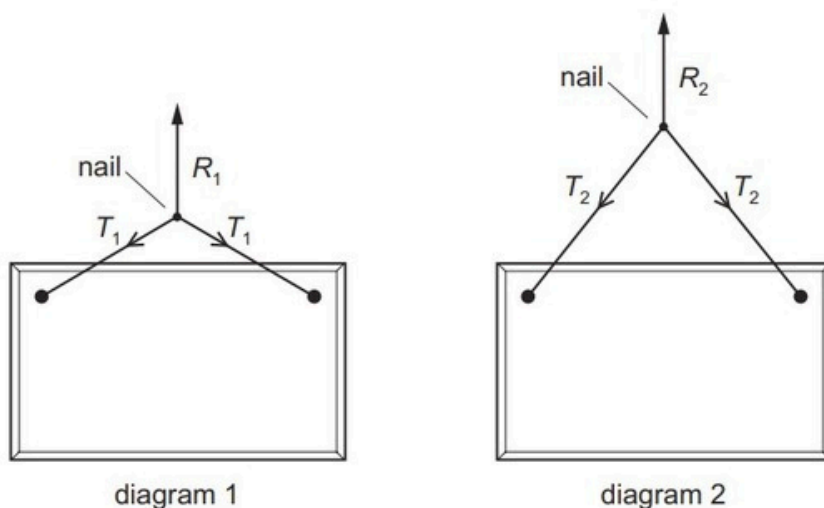
The volume of a sphere is $\frac{4}{3} \pi r^3$.

Which average pressure does the object exert on the table?

- A** $\frac{1}{3} \rho r^2$ **B** $\frac{1}{3} \rho r^2 g$ **C** $\frac{2}{3} \rho r$ **D** $\frac{2}{3} \rho r g$

335. 9702/12/F/M/23/Q12

The diagrams show two ways of hanging the same picture.



In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

- A $R_1 = R_2$ $T_1 = T_2$
- B $R_1 = R_2$ $T_1 > T_2$
- C $R_1 > R_2$ $T_1 < T_2$
- D $R_1 < R_2$ $T_1 = T_2$

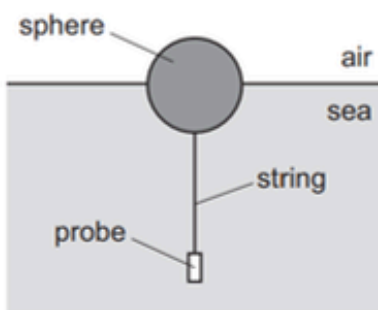
336. 9702/12/F/M/23/Q15

What is the centre of gravity of an object?

- A the geometrical centre of the object
- B the point at which the weight of the object may be considered to act
- C the point on the object about which there is a zero net torque
- D the point where gravity acts on the object

337. 9702/12/F/M/23/Q14

A probe is used to monitor the quality of the water in the sea. The probe is suspended by a vertical string which is attached to a sphere. The stationary sphere floats in equilibrium on the surface of the sea, as shown.



The sphere has a weight of 5.00 N. The probe and string have a combined weight of 2.00 N.

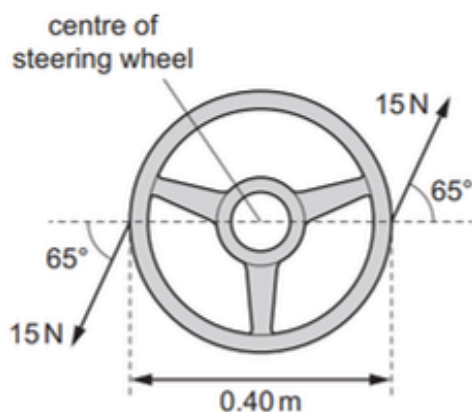
The density of the seawater is $1.03 \times 10^3 \text{ kg m}^{-3}$. The upthrust acting on the probe and thread is negligible.

What is the volume of the sphere below the surface of the sea?

- A $1.98 \times 10^{-4} \text{ m}^3$
- B $2.97 \times 10^{-4} \text{ m}^3$
- C $4.95 \times 10^{-4} \text{ m}^3$
- D $6.93 \times 10^{-4} \text{ m}^3$

338. 9702/11/M/J/23/Q11

The driver of a car applies two parallel forces to a steering wheel, as shown.



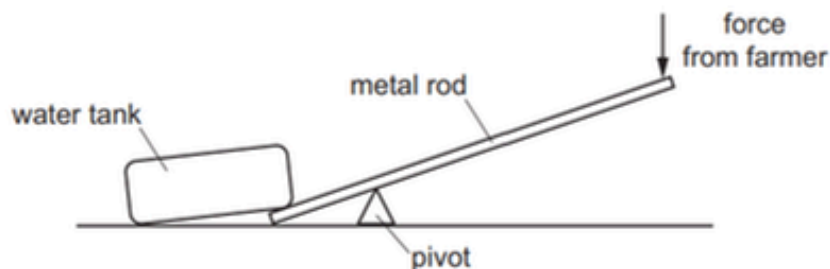
Each force has a magnitude of 15 N and acts in the direction shown. The steering wheel has a diameter of 0.40 m.

What is the torque exerted on the steering wheel?

- A 1.3 Nm
- B 2.5 Nm
- C 2.7 Nm
- D 5.4 Nm

339. 9702/11/M/J/23/Q12

A farmer is trying to lift the corner of a large water tank. She uses a metal rod as a lever.



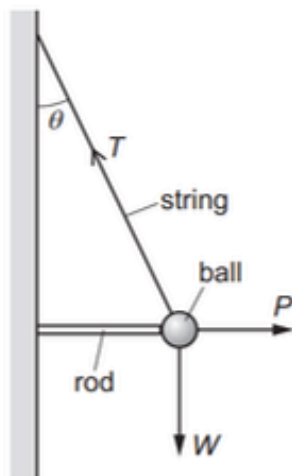
The vertical force from the farmer is constant and is always applied to the end of the rod.

Which change **must** increase the upward force on the water tank?

- A using a longer rod and moving the pivot closer to the tank
- B using a longer rod and moving the pivot further away from the tank
- C using a shorter rod and moving the pivot closer to the tank
- D using a shorter rod and moving the pivot further away from the tank

340. 9702/11/M/J/23/Q13

The diagram shows a ball of weight W hanging in equilibrium from a string.



The string is at an angle θ to the vertical. The tension in the string is T . The ball is held away from the wall by a horizontal force P from a metal rod.

Which relationship between the magnitudes of T , P and W is correct?

- A $P = T \cos \theta$ and $W = T \sin \theta$
- B $T = P + W$
- C $T^2 = P^2 + W^2$
- D $W = P \tan \theta$ and $W = T \cos \theta$

341. 9702/11/M/J/23/Q14

Which expression for pressure is correct?

- A** force per unit area
B force per unit volume
C mass per unit area
D mass per unit volume

342. 9702/11/M/J/23/Q15

A ball has a mass of 0.50 kg and a volume of $1.3 \times 10^{-3}\text{ m}^3$. The ball is floating in equilibrium on still water. The two forces that act on the ball are its weight and the upthrust due to the water.

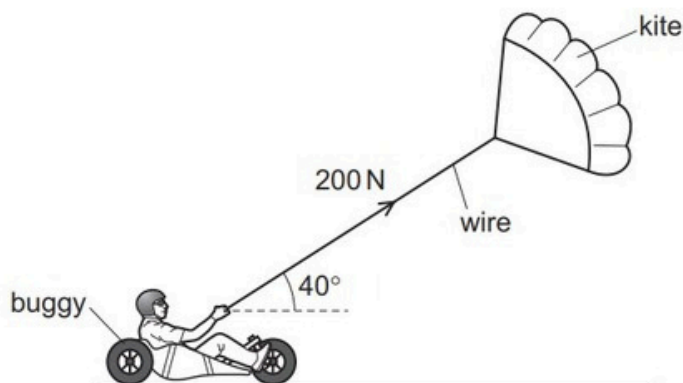
The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

What is the percentage of the volume of the ball above the surface of the water?

- A** 3.9% **B** 38% **C** 62% **D** 96%

343. 9702/11/M/J/23/Q16

A man sits on a buggy that is pulled along by a wire attached to a kite. The wire is at an angle of 40° to the horizontal and has a constant tension of 200 N. The man and buggy travel a distance of 20 m along a straight horizontal path. The wire and the path of the buggy are in the same vertical plane.



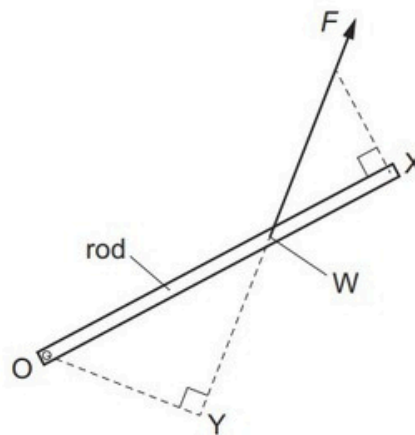
What is the work done by the tension force on the man and buggy?

- A** 2.6kJ **B** 3.1kJ **C** 3.4kJ **D** 4.0kJ

344. 9702/12/M/J/23/Q11

A rod is pivoted at point O.

A force F is applied to the rod at point W, as shown.

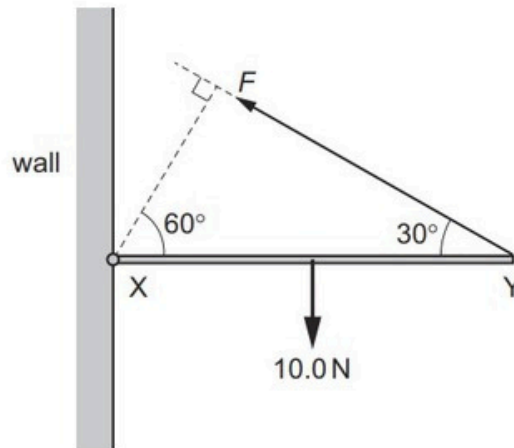


What is the moment of the force F about O?

- A $F \times \text{distance OX}$
- B $F \times \text{distance OY}$
- C $F \times \text{distance WO}$
- D $F \times \text{distance WX}$

345. 9702/12/M/J/23/Q12

A uniform rod XY of weight 10.0 N is freely hinged to a wall at X. It is held horizontal by a force F acting from Y at an angle of 30° to the horizontal, as shown.



What is the value of F ?

- A 5.0 N
- B 8.7 N
- C 10 N
- D 20 N

346. 9702/12/M/J/23/Q13

Two solid cubes X and Y are made of material of the same density. Cube X has twice the mass of cube Y.

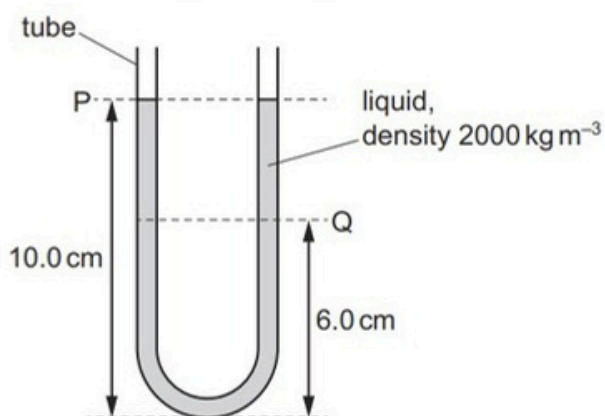
Cube X has sides of length x . Cube Y has sides of length y .

What is the ratio $\frac{x}{y}$?

- A** 1.26 **B** 1.41 **C** 2.00 **D** 8.00

347. 9702/12/M/J/23/Q14

A U-shaped glass tube contains liquid of density 2000 kg m^{-3} , as shown.

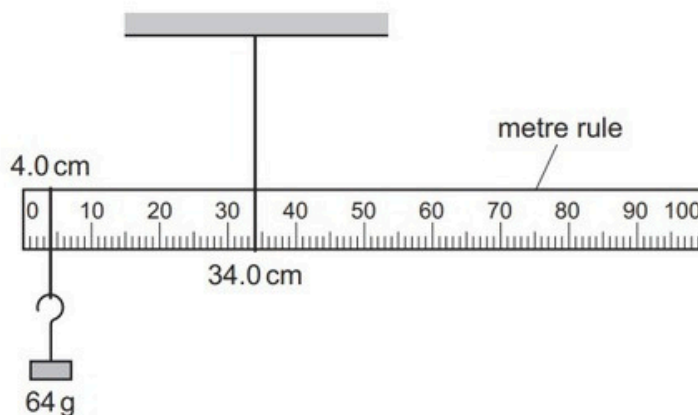


What is the difference in pressure due to the liquid between levels P and Q?

- A** 780 Pa **B** 1200 Pa **C** 1600 Pa **D** 2000 Pa

348. 9702/13/M/J/23/Q12

A uniform metre rule is pivoted at the 34.0 cm mark, as shown.



The rule balances when a 64 g mass is hung from the 4.0 cm mark.

What is the mass of the metre rule?

- A** 38 g **B** 44 g **C** 120 g **D** 136 g

349. 9702/13/M/J/23/Q13

A volume of 1.5 m^3 of water is mixed with 0.50 m^3 of alcohol. The density of water is 1000 kg m^{-3} and the density of alcohol is 800 kg m^{-3} .

The volume of the mixture is 2.0 m^3 .

What is the density of the mixture?

- A** 850 kg m^{-3} **B** 900 kg m^{-3} **C** 940 kg m^{-3} **D** 950 kg m^{-3}

350. 9702/13/M/J/23/Q14

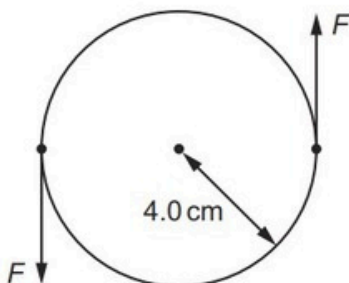
An object is falling at a constant speed through a viscous liquid. F_U is the upthrust on the object due to the liquid. W_L is the weight of the liquid displaced by the object. W_O is the weight of the object.

Which equation **must** be correct?

- A** $F_U = W_L$
B $F_U = W_O - W_L$
C $F_U = W_O$
D $F_U = W_O + W_L$

351. 9702/11/O/N/23/Q11

A minimum torque of 20 N m must be applied to the lid of a jar for it to open. The radius of the lid is 4.0 cm .

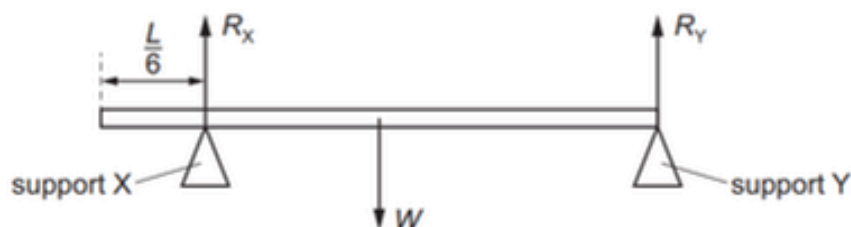


What is the minimum force F that must act on each side of the lid in order to open it?

- A** 2.5 N **B** 5.0 N **C** 250 N **D** 500 N

352. 9702/11/O/N/23/Q12

A uniform bar of length L and weight W rests horizontally on two supports X and Y.



Support X exerts a vertical force R_X at a distance of $\frac{L}{6}$ from one end of the bar.

Support Y exerts a vertical force R_Y at the other end of the bar.

The bar is in equilibrium.

What is the ratio $\frac{R_X}{R_Y}$?

A $\frac{3}{2}$

B $\frac{2}{3}$

C $\frac{3}{5}$

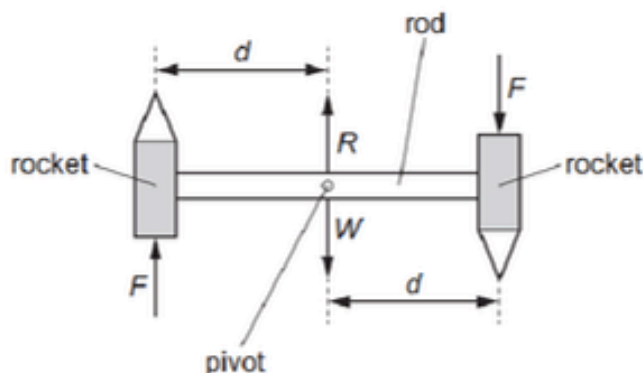
D $\frac{2}{5}$

353. 9702/11/O/N/23/Q13

A type of firework is made by connecting two rockets, facing in opposite directions, to a rod, as shown.

The rod is attached to a frictionless pivot so that the firework can rotate in a vertical plane.

The firework has weight W . The pivot exerts a force R on the rod that is equal and opposite to W .



Each rocket exerts a force of magnitude F on the rod at a perpendicular distance d from the pivot. The forces exerted by the rockets are always in opposite directions.

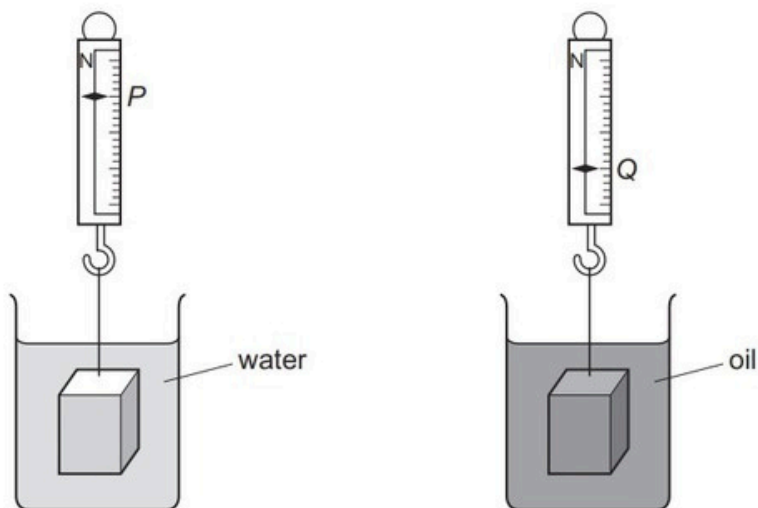
Air resistance is negligible.

Which statement is correct?

- A The firework is in equilibrium because the resultant force acting on it is zero.
- B The firework is in equilibrium because the resultant torque acting on it is zero.
- C The firework is not in equilibrium because the resultant force acting on it is not zero.
- D The firework is not in equilibrium because the resultant torque acting on it is not zero.

354. 9702/11/O/N/23/Q14

An object of weight W is suspended from a newton meter. When the object is completely immersed in water, the newton meter reads P . When the object is completely immersed in oil, the newton meter reads Q .

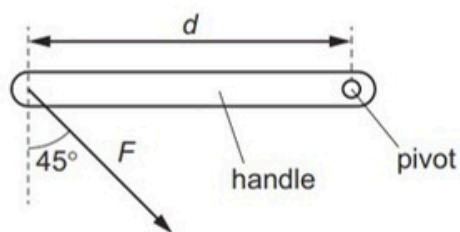


What is the ratio $\frac{\text{density of oil}}{\text{density of water}}$?

- A $\frac{W-P}{Q-P}$ B $\frac{Q-P}{W-P}$ C $\frac{W-P}{W-Q}$ D $\frac{W-Q}{W-P}$

355. 9702/12/O/N/23/Q11

A force F is applied at an angle of 45° to a door handle at a distance d from the pivot of the handle, as shown.



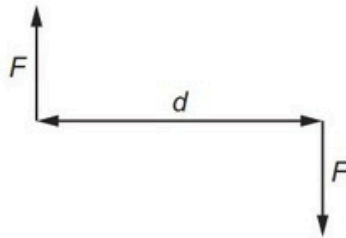
What is the moment of the force about the pivot?

- A $\frac{Fd}{\sqrt{2}}$ B Fd C $Fd\sqrt{2}$ D $2Fd$

356. 9702/12/O/N/23/Q12

A couple consists of two forces, each of magnitude F , that act in opposite directions in the same plane.

The perpendicular distance between the two forces is d .

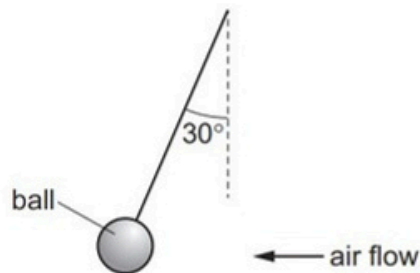


What is the torque of the couple?

- A** $\frac{Fd}{2}$ **B** $\frac{F}{d}$ **C** Fd **D** $2Fd$

357. 9702/12/O/N/23/Q13

The diagram shows an experiment to determine the force exerted on a ball by a horizontal air flow.



The ball is suspended by a light string and weighs 0.15 N .

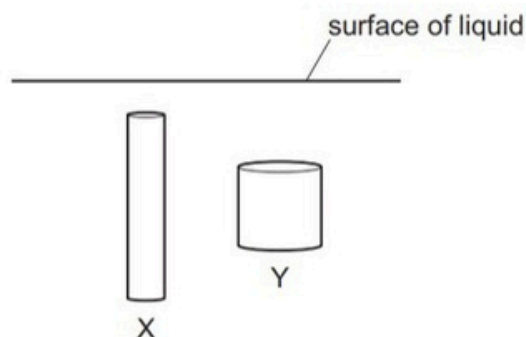
The deflection of the string from vertical is 30° . The ball is in equilibrium.

What is the force on the ball from the air flow?

- A** 0.075 N **B** 0.087 N **C** 0.26 N **D** 0.30 N

358. 9702/12/O/N/23/Q14

Two solid cylindrical objects X and Y are held fully submerged in a liquid, as shown.



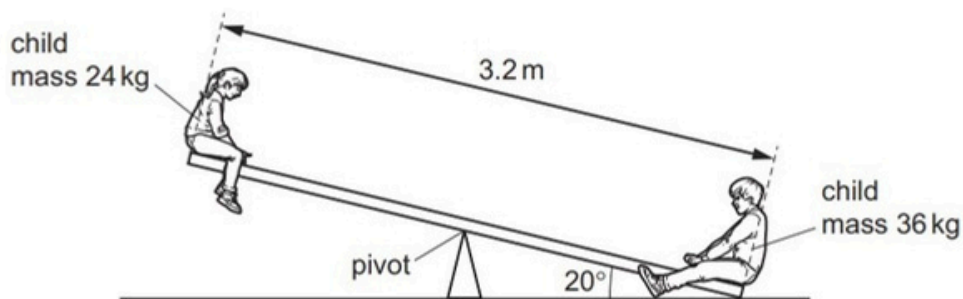
The objects have the same volume. The density of the material of Y is twice the density of the material of X. Both objects are stationary.

Which statement is correct?

- A The force due to the liquid acting on the top surface of X is greater than that acting on the top surface of Y.
- B The pressure difference due to the liquid between the top and bottom surfaces of X is the same as that for Y.
- C The upthrust acting on X is the same as the upthrust acting on Y.
- D The weight of X is the same as the weight of Y.

359. 9702/13/O/N/23/Q11

A uniform rigid beam of length 3.2 m is pivoted at its centre. Two children sit at the opposite ends of the beam, as shown.



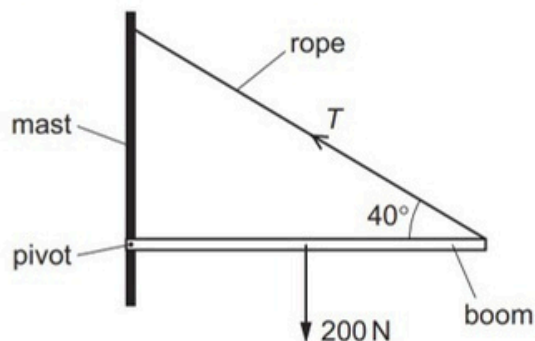
One child has a mass of 24 kg. The other child has a mass of 36 kg. The heavier child causes one end of the beam to permanently rest on the ground, so that the beam makes an angle of 20° to the horizontal ground.

What is the moment of the weight of the 24 kg child about the pivot?

- A 72 Nm
- B 130 Nm
- C 350 Nm
- D 380 Nm

360. 9702/13/O/N/23/Q12

Two parts of a sailing boat are the mast and the boom. The mast is a vertical rigid beam and the boom is a horizontal rigid beam. One end of the boom is attached to the mast by a pivot. The other end of the boom is connected to the mast by a rope, as shown.



The rope is at an angle of 40° to the horizontal and exerts a tension force T on the boom. The weight of the boom is 200 N. The mass of the boom is uniformly distributed along its length. The boom is in equilibrium.

What is the magnitude of T ?

- A** 130 N **B** 160 N **C** 260 N **D** 310 N

361. 9702/13/O/N/23/Q13

Full-fat milk is made up of fat-free milk mixed with fat.

A volume of $1.000 \times 10^{-3} \text{ m}^3$ of full-fat milk has a mass of 1.035 kg. It contains 4.00% fat by volume.

The density of fat-free milk is $1.040 \times 10^3 \text{ kg m}^{-3}$.

What is the density of fat?

- A** $1.25 \times 10^2 \text{ kg m}^{-3}$
B $9.15 \times 10^2 \text{ kg m}^{-3}$
C $9.28 \times 10^2 \text{ kg m}^{-3}$
D $1.16 \times 10^3 \text{ kg m}^{-3}$

362. 9702/13/O/N/23/Q14

Which expression for pressure is correct?

- A** force per unit area
B mass per unit area
C mass per unit volume
D weight per unit volume